

# ***Classification Of GAIT Data Using Machine Learning Techniques To Categories Human Locomotion***

**Submitted in partial fulfillment of the requirements  
for the award of the degree of**

**Master of Technology  
In  
Computer Science & Engineering**

Supervisor  
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**Vaish College Of Engineering**

**Maharishi Dayanand University, Rohtak**

**JUNE, 2016**

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**VAISH COLLEGE OF ENGINEERING  
Maharishi Dayanand University, Rohtak  
JUNE, 2016**

## **CANDIDATE'S DECLARATION**

I hereby declare that the work which is being presented in the thesis entitled “**CLASSIFICATION OF GAIT DATA USING MACHINE LEARNING TECHNIQUES TO CATEGORISE HUMAN LOCOMOTION**” in partial fulfillment of requirements for the award of degree of M.Tech. (CSE) submitted in the Department of COMPUTER SCIENCE AND ENGINEERING at VAISH COLLEGE OF ENGINEERING under MAHARSHI DAYANAND UNIVERSITY, ROHTAK, is an authentic record of my own work carried out during a period from 15.01.2016 to 15.06.2016 under the supervision of Mr. DEEPAK GOYAL. The matter presented in this thesis has not been submitted by me in any other University / Institute for the award of M.Tech. Degree.

**Date:**

**Place:**

**Signature of the Student**

## CERTIFICATE

This is to certify that the work which is being presented in the thesis entitled “**CLASSIFICATION OF GAIT DATA USING MACHINE LEARNING TECHNIQUES TO CATEGORISE HUMAN LOCOMOTION**” by “**ANUBHA PARASHAR**” in partial fulfillment of requirements for the award of degree of M.Tech. (CSE) submitted in the Department of COMPUTER SCIENCE AND ENGINEERING at VAISH COLLEGE OF ENGINEERING under MAHARSHI DAYANAND UNIVERSITY, ROHTAK, is an authentic record of her own work carried out during a period from 15.01.2016 to 15.06.2016 under the supervision of Mr. DEEPAK GOYAL. The matter presented in this thesis has not been submitted by her in any other University / Institute for the award of M.Tech. Degree.

This is to certify that the above statement made by the candidate is correct to the best of my/our knowledge.

Signature of the Supervisor  
Mr. Deepak Goyal

The M.Tech Viva–Voce Examination of **ANUBHA PARASHAR** has been held on \_\_\_\_\_ and accepted.

Signature of Supervisor  
Mr. Deepak Goyal

Signature of External Examiner

## **DEDICATION**

**I dedicate this thesis to my beloved grandmother Smt. Santosh Sharma and grandfather  
Dr. D.P Sharma.**

## **PREFACE**

This thesis includes a number of figures and tables previously published in my four conference papers two journal paper. All these figures and tables have been produced by the thesis author on the basis of this thesis research.

## **ACKNOWLEDGEMENTS**

I would like to express my sincere gratitude to Mr. Deepak Goyal for giving me the chance to conduct my studies under his supervision. His continuous encouragement and his generous knowledge sharing were essential during my master studies. Furthermore, the care with which he reviewed the chapters of this thesis really improved the quality of the work. His commitment to the work will be an example and an inspiration for my future work.

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I am always thankful for my parents for always motivating me and being my inspiration. Finally, I wish to express my most heartfelt gratitude to my beloved and missed grandmother Smt. Santosh Sharma who taught me the most important lesson of my life, the only one that I am sure I will never forget. Thank you because you made me the person I am today.

## ABSTRACT

Human Gait is a model of movement while locomotion and it is known as behavioural characteristic of human. Gait recognition is a behavioural biometric method that includes human identification at a distance i.e. in absence of human and this is achieved by analysing the behaviour of their walking. Main aim to make GAIT recognition system is to impart a best method to recognize risks in places where high security is required like in banking, parking, airports etc. or to detect disease like Parkinson's (Classification is utilized to see Gait pattern of different person and any upcoming disease can be detected earlier). The Classification of Humanoid locomotion is a troublesome exercise because of non linearity associate with gait. The high dimension feature vector requires a high computational cost. The classification using the different machine learning technique leads for over fitting and under fitting. To select the correct feature is also the difficult task. The hand craft feature selection machine learning techniques performed poor. We have used the deep learning technique to get the trained feature and then classification we have used deep belief network based deep learning.

In the thesis I have shown the importance of gait recognition in order to detect whether a human GAIT is normal gait or pathological gait. So we first selected the feature and identify the principle feature then we classify gait data and use different machine learning technique (K-Means, KNN and Back Propagation) and performance comparison is shown. Experimental result on real time datasets propose method is better than previous method as far as humanoid locomotion classification is concerned.

To test the proposed system the database contains 204 gaits and 16 different features were recognised, in which 3 datasets are crouch gait dataset and one normal gait dataset of 4 different humans is collected. We have obtained 100% of classification accuracy for a training data. Testing results shows 43% accuracy overall accuracy of system is 69.825%. Moreover, 10 fold cross-validation test also conducted to validate the statistical significance of the results.



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## LIST OF PUBLICATIONS

### ❖ International Conferences

1. Anubha Parashar, Apoorva Parashar, Somya Goyal, Bharat Sahjolan, "Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Means.", [\*Second International Conference on Information and Communication Technology for Competitive Strategies\*](#) (ICTCS-2016)(Conference Proceedings by **ACM – ICPS** ISBN No 978-1-4503-3962-9). (see in *Appendix –I*)
2. Anubha Parashar, Apoorva Parashar, Somya Goyal, " Classifying Gait Data using different machine learning techniques and finding the optimum technique of classification.", [\*International Conference on ICT for Sustainable Development\*](#) (ICT4SD-2016)(Conference Proceedings by **Springer** - *Lecture Notes in Networks and Systems* (LNNS Series)).(see in *Appendix – II*)
3. Anubha Parashar, Deepak Goyal, "Identification of GAIT data using machine learning technique to categorise human locomotion", *IEEE Conference at IIT KGP*. (see in *Appendix –IV*)
4. Anubha Parashar, Deepak Goyal, "Clustering Gait Data using different machine learning techniques and finding the best technique ", *International Conference on Smart Trends for Information Technology and Computer Communications* (SMARTCOM 2016) Conference Proceedings by **Springer CCIS Series** (ISSN No. -1865-0929). (see in *Appendix –III*)

### ❖ Journals

- 1 Anubha Parashar, Somya Goyal, Apoorva Parashar "Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Means." in *International Journal of Interactive Multimedia and Artificial Intelligence* (**IJIMAI**2016).
2. VS Bhaskar, AK Singh, J Dhruw, A Parashar, M Sharma, "Business and Social Behaviour Intelligence Analysis Using PSO." in *International Journal of Interactive Multimedia and Artificial Intelligence*, (**IJIMAI**) Vol.2 (6), Page No. [69-74](#).

## ❖ E-Prints

1. A Parashar, S Kumar, VS Bhaskar, R Chinia, "Noisy Distance Measurements Using 3-D Localization with Rb-Rf Methods", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.2125.pdf>
2. A Parashar, V Bijalwan, "A Text to Speech Conversion Engine", available from [https://www.researchgate.net/profile/Vishwanath\\_Bijalwan3/publication/274195825\\_A\\_Text\\_to\\_Speech\\_Conversion\\_Engine/links/5518e2e70cf2d70ee27b752d.pdf](https://www.researchgate.net/profile/Vishwanath_Bijalwan3/publication/274195825_A_Text_to_Speech_Conversion_Engine/links/5518e2e70cf2d70ee27b752d.pdf)
3. M Rani, A Parashar, J Chaturvedi, A Malviya, "Search Space Engine Optimize Search Using FCC\_STF Algorithm in Fuzzy Co-Clustering", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.6952.pdf>
4. J Chaturvedi, A Parashar, AA Manjrekar, VS Bhaskar, "Social and Business Intelligence Analysis Using PSO", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.6090.pdf>
5. K Vaish, SM Rajesh, K Pasupatheeswaran, A Parashar, J Chaturvedi, "Design and Autonomous Control of the Active Adaptive Suspension System Rudra Mars Rover", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.5197.pdf>

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## LIST OF SYMBOLS AND ABBREVIATIONS

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| $\Phi_1$     | : orientation of horizontal ellipses                                                 |
| $\Phi_2$     | : orientation of vertical ellipses                                                   |
| KNN          | : K Nearest Neighbor                                                                 |
| BP           | : Back Propagation                                                                   |
| NN           | : Neural Network                                                                     |
| HMDC         | : Human Motion Capture Device                                                        |
| PCA          | : Principal Component Analysis                                                       |
| $X^T$        | : Data matrix                                                                        |
| $W$          | : the matrix of eigenvectors of the covariance matrix $XX^T$                         |
| $\Sigma$     | : $m \times n$ rectangular diagonal matrix with nonnegative real numbers on diagonal |
| $V^T$        | : matrix of eigenvectors of $X^T X$                                                  |
| $Y^T$        | : a linear transformation of the corresponding row of $X^T$                          |
| $\tau$       | : torque                                                                             |
| $M(\theta)$  | : inertial torque                                                                    |
| $C(\theta')$ | : centripetal and coriolis forces                                                    |
| $G(\theta)$  | : gravitational force                                                                |

# CHAPTER-1

## INTRODUCTION

To verify/validate people identity is called as authentication. And now day's authentication is becoming very popular to check whether a person is authentic or not. Authentication is useful at various places and previously authentication is done with secret passwords or PIN codes. The main disadvantage to such authentication systems are anyone can crack the password or PIN codes so this method fails where very high security is needed.

So, biometric (in fig.1.) technology came to existence where authentication is done using a particular body part like fingers, iris etc. Lately biometric techniques are used to verify and validating the identity of some person. For authentication purposes biometrics are increasingly being used like fingerprint scanner, iris scanner etc. and one can say these are reliable too. But now days these biometrics are not user friendly as people dislike accessing finger print machine or avoid iris scanner as it may weaken or damage their vision. Therefore it is a good technique but consumes time as people are required to give their details to the authentication system.

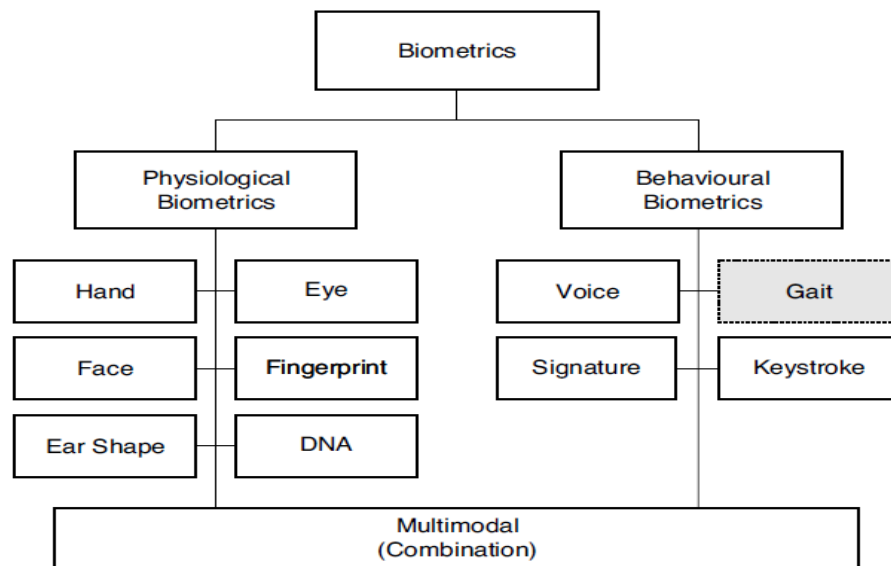


Fig. 1. Biometrics types (in this thesis GAIT biometric classification is used)

Human gait is supposed as unique biometric identification like thumb print. So in GAIT analysis to examine the human walking moment we collect and analyze the data [11]. Human gait can be utilized to identify people for diverse security reason and exercised before to detect many abnormalities. For disable person the classified data is used to implement the prosthetic leg.

The GAIT data can be useful to help in walking of old people and is used in pattern analysis and classification of different walk. By seeing the abnormality in GAIT cycle, it is worth to do research for prognostication for many upcoming diseases. Differently abled and elderly people can be helped by any research regarding GAIT [20]. During human movement, top and bottom limb of human body yield the periodic motion. Due to this particular motion human beings get a particular motion and a unique walking pattern. The biometric identity like walk, jump, jog and run based on human locomotion can be obtained by the use of GAIT analysis shown in figure 2.

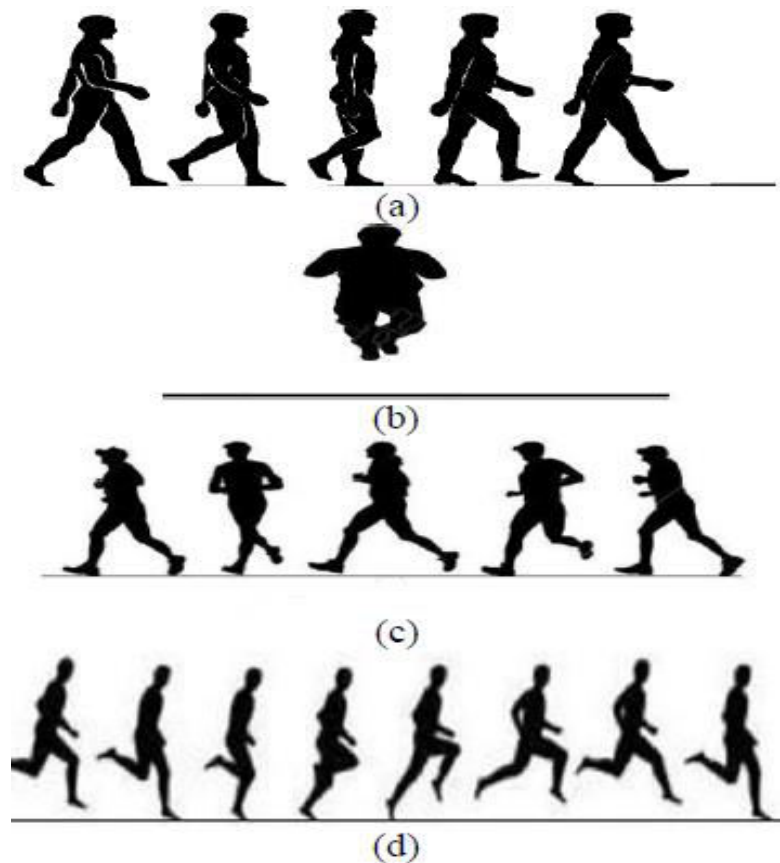


Fig. 2. Biomechanics of Locomotion: (a) Walk (b) Jump (c) Jog (d) Run

A lot of precursory research work about various machine learning techniques is gathered from the paper given [2-6] below. The classification of GAIT pattern is very accurate and unique due to rhythmic pattern. The new arena of biometric research is different category of GAIT classification. The GAIT can be helpful for security reason as it can identify from distance. In comparison with other biometric identification technique such as face reorganization i.e. iris, finger print which is considered as first generation biometric, the GAIT is divergent as it does not require any subject contact other than walking. This can be considered as behavioral authentication technique. Therefore identification using gait is becoming very popular in security. The development of highly sophisticated and more accurate humanoid can be made easy with the study of human GAIT data. So the foremost requirement is the feature reduction [21].

So we can say using gait as biometric where a person don't have to be interactive will not cause any problem as it asks nothing other than walking by. Therefore this system has more advantage over the previous one and it is very user friendly and has large user acceptance. The reasons for large user acceptance are as people do not need to interact or give time to the sensors which are applied to verify a person, so authentication becomes so natural. Secondly gait detection is a behavioural identity of a person which is used as biometric so it cannot be lost/stolen. Thirdly people don't have to waste time to give any details about them, as their gait identity is used in biometric verification [7-10].

So with this discussion we can say that all people walk so we can take gait as biometric identity in verification. As the walking of two persons always varies, biometric identification becomes very easier. So when the dataset of GAIT is collected one can classify into separate groups to know the normal behaviour of walking.

## **1.1 Aim**

The aim of this masters thesis is to evaluate and implement a sufficiently robust and accurate method for human to identify whether it's a **normal gait** or **pathological gait** and to improve the speed and accuracy of classification in the presence of category hierarchies.

## **1.2 Problem description**

Identifying normal or pathological human gait poses many challenges. Therefore a robust method for detecting and capturing the body, face and/or gait of each individual person has to be found. Traits or features have to be extracted from each detection. How many and which features are needed for classification? Which classification method gives the most accurate results? How high will the classification rate be?

## **1.3 Limitations of previous works**

As has been formalised in the preceding section, identity verification, i.e. authentication, is a technology with increasing importance. With today's systems, individuals can authenticate their identity by one or an arbitrary combination of the following three means:

- Knowledge: The specific knowledge of a secret, such as a password, passphrase, or PIN code.
- Possession: The possession of a specific item or token, for example a key, smart card, or identity card.
- Biometric: With a specific characteristic of the individual's body, such as the fingerprint, iris pattern, retina pattern, genetic fingerprint, voice features, facial properties, signature, or knuckle profile.

The combination, i.e. fusion, of two or three of the aforementioned attributes can be used to further increase the security level. All of these attributes have their specific advantages and disadvantages; they will be discussed in the next three sections [22-27].

### **1.3.1 Authentication by Knowledge**

The most common method of authentication obtained through knowledge is the use of passwords and PIN codes. Today's modern computer systems prompt the user for his identification (username, ID) and the appropriate password and compare them against the previously stored and eventually encrypted password of that particular user. Another ubiquitous example is cash dispensers, where the user has to insert his card and enter the numerical PIN code in order to withdraw money. Authentication by knowledge is

intuitive, cheap and very simple to implement. However, there are some important security considerations related to this method.

**Password/PIN Selection:** The selection and administration of passwords or PINs is crucial for the success of the method. A good password should be easy to remember, but nevertheless hard to guess. Thus, passwords should not be based on one's name, date of birth, or children names nor should they contain sequences such as 654321. Such passwords and PIN codes would be rather easy to guess for outsiders.

Although many security experts still recommend frequent password changes this has proven to be a suboptimal strategy. Users are generally lazy and do not want to be bothered with remembering new passwords. If they are forced to, they tend to pick easy to remember passwords or even worse, they write them down somewhere (on a post-it under the keyboard probably). Additionally, it takes some adaptation time until they key in the new password in a fluently and hard to track manner. Until that stage, they are vulnerable to password theft by observation. It is best to choose a secure and hard to guess password in the first place and not change it for a longer period of time. Once users have learned them by heart, it is almost impossible to copy the code by merely looking at them typing.

**Theft:** The basic philosophy behind knowledge-based authentication is that under no circumstances should the secret be disclosed to an undeserving party. However, in extreme cases, users could be forced to reveal their secret. Additionally, the user can be induced into divulging his password unintentionally, as a result of deception (e.g. with phoney or tampered cash dispensers).

**Interception:** An alternative method to attack knowledge-based authentication is to intercept the secret information either during transmission from the input device to the terminal or during transmission over the network. Sophisticated cryptographic methods such as Zero Knowledge Protocols are needed to prevent this type of attack. Although the cryptographic theory behind secure protocols is well understood, they are often not used for the sake of simplicity and thus leave the door wide open for attacks.

A different method to overcome this shortcoming is the application of one-time-passwords, where a single password is only used once. This practice is very common in banking applications.

### **1.3.2 Authentication through Possession**

Authentication through possession is solely based on the fact that a user possesses a certain token or device such as a magnetic card, smartcard or key. This authentication method can also be linked to the particular knowledge held by the user to augment overall security. However intuitive this authentication method is, it needs additional hardware that all users have to bear with them as well as the card readers.

Common to this class of authentication is the possibility to pass the authentication token onto other people. But it can only be possessed by one single person at a time.

Key: Authentication through the possession of a key is a method already known for several millennia. Its origins lie in the near east, where the oldest known example was found; possibly more than 4'000 years old. In most cases, keys identify a specific user group (namely all the owners of the same key) rather than an individual user.

If a key is lost, security is immediately endangered and all locks and keys need to be changed in an expensive and time consuming procedure. Today, modern keys with an integrated Smart Token, i.e. microchip, exist to overcome the hassles when a key is lost. The lost key is simply blocked from the list of accepted key identifiers. Additionally, combined with a real time clock, those systems allow access restrictions for the individual keys to be limited to specific work hours and days, since every key bears a unique identifier.

Memory Tokens: Memory Tokens are a more advanced variant of the well known keys mentioned above. They do not perform any information processing, but they merely make information available, such as a unique identifier. Possible examples of memory tokens are the widely used magnetic strip cards or RFID-tags<sup>4</sup> respectively. The information is accessed by read/write devices such as magnetic card readers or radio transmitters. Of course, memory tokens can additionally be combined with the entry of a PIN or password to augment the level of security.

Smart Tokens: The Smart Token, i.e. Chip Card, allows more elaborate functions than a simple memory card. Thanks to the integrated circuits, they provide additional functions to increase the level of security. In particular, smart tokens can (in contrast to memory cards) protect their stored contents with the requirement of entering a PIN code or

password in order to decrypt the stored data. This further reduces the security risk imposed by stolen, lost or forged cards.

### **1.3.3 Authentication with Biometrics**

Biometric authentication systems make use of unique physiological or behavioural characteristics of the user. As of today biometrics is the most secure way to establish authentication. However, hardware requirements are higher compared to the previously mentioned authentication methods but with large scale production it is possible to limit system costs.

There is a fundamental and very important difference between authentication methods based on biometric traits and the methods based on knowledge or possession. The biometric methods check for the physical presence of the specific user, whereas the knowledge/possession based methods are only capable of verifying the presence of the secret or token respectively. The biometric methods can therefore establish a direct relation between the presence of the biometric trait and the person itself. It is thus not possible to hand over a biometric trait to someone else as it is with passwords or tokens. It is therefore very difficult to lose, forget, or steal someone's biometric signature. For most application fields this is a highly valued quality. However, there are certain applications where this close relation between the person and its biometric signature is undesirable.

A possible example is a home owner, which goes on holiday and asks his neighbouring couple to water his plants in the meantime. In this scenario it is more convenient to simply pass the door key to them, rather than adding the couple to the access database of the biometric system.

An additional peculiarity of all biometric methods is that they cannot rely on perfect agreement between the stored template and the newly acquired data. They have to tolerate a certain amount of variation in order to make allowances for natural fluctuations of the biometric trait. This is in stark contrast to knowledge and possession based methods where a perfect match to the value stored in the system database is mandatory. Therefore, all biometric systems can only produce probability estimation to what extent the new and the stored template correspond to each other.



**Complex Technology:** The relatively complex analysis of the behavioural and physical attributes with all their possible influences and natural variability of the biometric traits as well as ageing, make the biometric technology elusive. Complicated schemes are necessary to compensate for all those effects.

**Feature Selection:** The choice of the biometric methodology, i.e. feature (fingerprint, retina, iris), is crucial. However, there is no general rule to answer this question because many different factors have to be taken into account. Furthermore, there has to be some kind of backup authentication because not every person has every biometric trait. There are people, for example, without a prominent fingerprint, retina pattern and so on.

**User Acceptance:** User acceptance is another crucially important area which significantly affects the realised overall performance of the biometric system. The importance should not be underestimated because the introduction of any biometric system that lacks broad user acceptance is doomed to fail.

In order to attain acceptance the users have to feel comfortable with the usage of the biometric system. Thus, each user needs to be given instructions in such domains as system handling, working principle, data security, and privacy of stored information as well as the opportunity to ask whatever questions might occur to them in this context.

**Identification vs. Authentication:** In contrast to the other authentication systems, biometric systems can be operated in an additional so called identification mode. Thus, the biometric systems not only allow verification of a claimed user's identity but they can also produce a probability estimate of the previously unknown identity of the client from the database of known users. The computational requirements in terms of computing power are considerably higher in identification than in verification mode because the feature has to be compared to every single template stored in the database and not just the one of the claimed identity.

#### **1.3.4 Comparison**

In the previous sections all three authentication methods were shortly described with their respective advantages and drawbacks.

Table 1 briefly summarises the most important findings:

| Criterion             | Knowledge | Possession | Biometrics |
|-----------------------|-----------|------------|------------|
| Technology            | trivial   | moderate   | difficult  |
| User friendly         | yes       | yes        | depends    |
| Can be stolen         | yes       | yes        | no         |
| Can be lost           | no        | yes        | no         |
| Can be forgotten      | yes       | yes        | no         |
| System price          | marginal  | moderate   | high       |
| Hygienic reservations | no        | no         | yes        |

### **1.4 Main Contribution**

In the thesis (fig.4.) we collect the human gait data from Human Motion Capture Device (HMCD). Then training and testing of GAIT data is done to get the maximum possible accuracy of the system. Then features of GAIT data is extracted and classified. Classification is done using Back Propagation technique and KNN; both are supervised learning technique and K-Means technique. Comparison is done in three techniques and best technique is used to train and test the system [28-31].

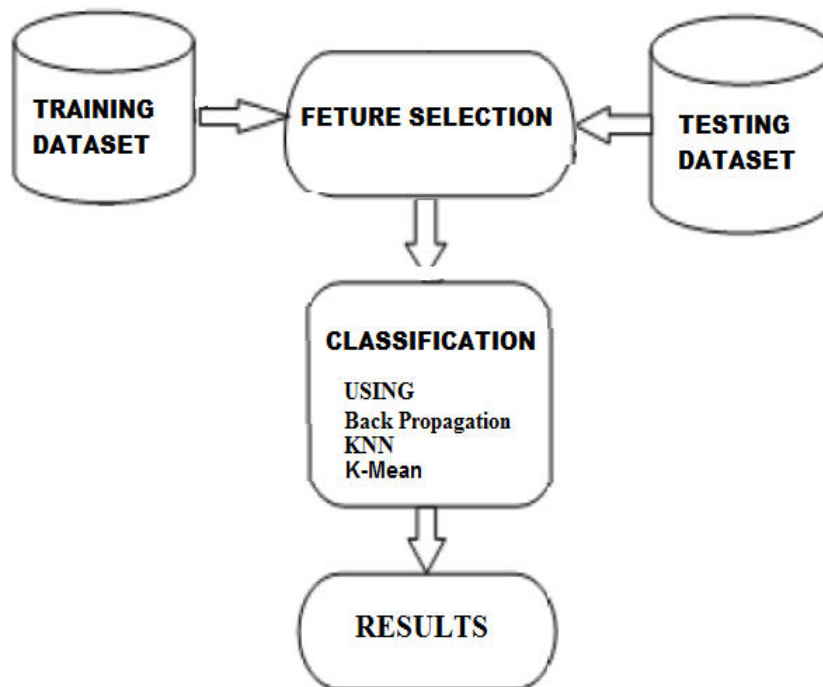


Fig. 3. Block diagram of classification system

## **1.5 Objectives**

The objectives include the following research and experimental tasks:

1. Conduct a literature review on the currently available methods of classifier combination.
2. Research various classifier methods to improve classification.
3. Propose architecture for improved learning using the various machine learning techniques.
4. Evaluate the performance of the proposed architecture using various performance methods.

## **1.6 Outline of the Thesis**

In this thesis we collect the human gait data from Human Motion Capture Device (HMCD). Then training and testing of GAIT data is done to get the maximum possible accuracy of the system. Then features of GAIT data is extracted and classified. Classification is done using Back Propagation technique, K-Means and KNN learning techniques. Comparison is done in all techniques and best technique is used to train and test the system. The thesis concludes with a chapter that contains the main contributions and open problems as well as an outlook on potential topics for further research.

- Chapter 1 introduces the subject of this thesis.
- Chapter 2 gives an introduction to the fundamentals of biometric technology as well as explains some of the terminology extensively used throughout this thesis.
- Chapter 3 describes the literature review of this thesis.
- Chapter 4 describes the design and implementation of the laboratory prototype biometric system built and used during the course of this thesis. This chapter

details the computationally efficient methods for extracting the individual features and also explains the experimental methodology and reasoning for system design decisions. Additionally, a novel computationally efficient variant of the generalised PCA is described.

- Chapter 5 discusses and compares the performance of the learning techniques and describes the results of the experiments, including an analysis of the findings.
- Chapter 6 gives a summary of the described work, discusses the major contributions of this thesis, limitation and gives an outlook for future directions of research.

## CHAPTER-2

### GAIT

#### **2.1 General Overview**

The process of verifying a person's identity, also called authentication plays an important role in various areas of everyday life. Any situation with user interaction where the identity is required needs a means to verify the claimed identity. One of the more obvious and commonly known application areas for identity verifying technologies, i.e. authentication, is the Logical Access Control to computer systems, where authenticity is normally established by confirming a claimed identity with a secret password or PIN code. Cash Dispensers or Computer Login Procedures are two other ubiquitous examples of this application area. On the other hand, authentication mechanisms are also applied to control Physical Access of persons to hazardous, dangerous, or high security areas. Similar or enhanced applications of this area include attendance monitoring of employees and the control of visitors in prisons. Traditional methods of confirming the identity of an unknown person rely either upon some secret knowledge (such as a PIN or password) or upon an object the person possesses (such as a key or card). But testing for secret knowledge or the possession of special objects can only confirm the knowledge or presence, and not, that the rightful owner is present. In fact, both could be stolen [32].

Conversely, biometric technology is capable of establishing a much closer relationship between the user's identity and a particular body, through its unique features or behaviour. All of the above mentioned application areas offer potential for biometric authentication technology, where the user's identity is verified using a physiological or behavioural characteristic such as the iris pattern, a fingerprint, or the voice.

This thesis describes the development of a novel type of comfortable and easy to use biometric system. The system uses human gait as the biometric trait to authenticate people. Gait as a biometric has several important properties that make it an interesting

solution to the authentication problem. First of all, people need not to interact with a sensor in an unnatural way. Second, since gait is a behavioural biometric, it performs implicitly a living person test and can thus neither be stolen nor lost. Finally, users do not need to unveil additional information about them other than already available.

In contrast to the thesis above, the quest of biometrics is to find the particular within the universal. Nearly all people can walk; detecting these universal human traits helps to distinguish a person from a dog, but serves little to distinguish among individual persons. For that task, one needs to find unique aspects in human gait which are more particular than universal. Not only must there be great variability in such features amongst different individuals (else the feature would not be unique), but also there must be little or no variability in those same features for a given person over time and conditions (else they would not be reliable). Everything in the science behind biometric technologies depends upon the relative size of these two variability: the between-person and the within-person variability.

## **2.2 Biometrics**

The word biometric is derived from the Greek roots – *bios* for *life* and *metrikos* for *measurement*.

It consists of methods for recognising a person through their unique distinguishing characteristics. These characteristics can be categorised in two forms; physiological and behavioural characteristics.

*Physiological* – these are biometrics derived from a direct measurement of human body parts. The most prominent and successful of these types of measures, to date, are fingerprints, face recognition, iris-scans and hand scans.

*Behavioural* – characteristics extracted based on the uniqueness of the way an action is performed by an individual and extend over time. Established measures include gait, keystroke-scan and speech patterns.

Furthermore, the quality of the biometric can be described using key characteristics [12] such as:

Universality: present in each and every person.

Distinctiveness: dissimilar for two different persons.

Permanence: is invariant over a period of time.

Circumvention: dissimilar personal space-a protective zone.

In our narrowly interconnected society human recognition gained considerable importance. Establishing the identity of an individual is critical in a variety of scenarios ranging from issuing a driver's license to granting access to highly secured resources. The need for reliable identification (to determine our identities) and authentication (to confirm our claimed identities) systems has increased due to rapid advancements in networking, communication, and mobility. Traditional passwords (knowledge-based schemes) and ID cards (tokenbased schemes) have been used for authentication in many applications (e.g. Internet banking) or facilities (e.g. library), although such mechanisms have several limitations. Passwords can be guessed or divulged to illegitimate users and ID cards can be stolen or forged, resulting in a breach of security. Thus, it is necessary to utilize alternative methods that are not merely based on what you know or what you have but, rather, on what you are.

Biometrics [24], as the science of recognizing an individual based on his inherent biological traits, has gained acceptance as a sound method in various commercial, civilian, and forensic applications, such as access control and visual surveillance. There are numerous human measurable characteristics, which can be used to help to derive an individual's identity. Looking at the nature of the underlying modalities, we distinguish two basic categories of biometric traits, physiological (passive) and behavioral (active).

Physiological biometrics are derived from a direct measurement of a person's physical characteristics which are assumed to be relatively unchanging, such as a part of a human body. We all use a form of physiological biometrics when we recognize our friends and acquaintances, at least by looking at them. The most prominent and successful of these types of measures to date are:

### **Fingerprints**

A fingerprint is formed by friction ridges and valleys of the surface of a fingertip. It has been proven that each single finger generates a different fingerprint. Multiple fingerprints of an individual provide additional information to allow mass identification involving millions of identities. However, fingerprints of a small fraction of the population may be unsuitable for automatic identification because of genetic factors, aging, environmental, or occupational reasons (manual work).

### **DNA**

DNA (deoxyribonucleic acid) contains genetic instructions of all living creatures. DNA recognition instruments are likely to be used in investigation of suspicious objects and in diagnostics. The greatest advantage is its high distinctiveness, where the probability of two people having exactly the same DNA code is one in a hundred billion. On the other hand, DNA matching is not available in real-time because it needs a physical sample, while other biometric systems only use an image or recording.

### **Retina**

Capillary vessels located at the inner most sensitive layer of eye have a unique pattern. Retina scans require the person to remove his glasses, place his eye close to the scanner, stare at a specific point, and remain still while the scan is completed. A retina scan cannot be faked as it is currently impossible to forge a human retina. Furthermore, the retina of a deceased person decays too rapidly to be used to deceive a retinal scan. Although the enrollment and scanning are intrusive and slow, the method is highly accurate.

### **Iris**

Iris is a pigmented portion of eye, in which a pupil is situated centrally. Its visual texture is stabilized during the first two years of life. The complex iris texture carries very distinctive information useful for personal recognition. The hippus movement of the eye may also be used as a measure of liveness for this biometric.

### **Palm vein pattern**



Pattern of blood veins in the back of the hand and the wrist contain differentiating features for personal identification. An individual's vein pattern image is captured by radiating his hand with near-infrared rays. It is a very secure method of authentication because this blood vein pattern lies under the skin. This makes it almost impossible for others to read (without permission) or copy. Furthermore, it will not vary during the person's lifetime.

### **Face**

Applications of this non-intrusive method range from a static and controlled authentication to a dynamic uncontrolled identification in a cluttered background. The most popular approaches to face recognition are based on either the location and shape of facial attributes (eyes, eyebrows, nose, lips, chin) and their spatial relationships, or the overall analysis of the face image as a weighted combination of canonical faces. A facial recognition system should automatically detect whether a face is present in an acquired image, locate the face if there is one, and recognize the face from a general viewpoint [33-35].

### **Hand geometry**

Hand geometry recognition systems are based on a number of measurements taken from the human hand, including its shape, size of palm, and the lengths and widths of its fingers. Since hand geometry information may vary during lifetime and accessories (jewelry) may touch extracting the variables, this biometric is not known to be very distinctive.

By contrast, behavioral biometrics are an indirect measure of the characteristic of the human form. Based on (hopefully) unique ways people do things, we extract features from an action performed by an individual. The main aspect of a behavioral biometric is the use of time as another dimension. From the user's point of view, data can be acquired even without his explicit consent. Established measures include:

### **Signature**

The way a person writes his name has been accepted in government, legal, and commercial transactions as a method of authentication. This is an obtrusive method, since the user's effort and contact with a pen is required. The actual signature changes over time and is influenced by physical and emotional conditions of the signatories. In addition, professional forgers may be able to reproduce signatures that fool the signature verification system.

### **Gait**

Gait refers to the manner in which a person walks, and is one of the few biometric traits that can be used to recognize people at a distance. Therefore, this trait is very appropriate in surveillance scenarios. Most gait recognition algorithms attempt to extract the human silhouette in order to derive the gait variables. Hence, the selection of a good model to represent the human body is pivotal to the efficient functioning of a gait recognition system. However, the gait of an individual is affected by footwear, clothing, affliction of the legs, or walking surface.

### **Keystroke dynamics**

Each person is hypothesized to type on a keyboard in a characteristic way. Keystroke dynamics is a collection of detailed press and release timing information about typing on a keyboard. This biometric is not expected to be unique to each individual but it may offer sufficient discriminatory information to permit identity verification. The keystrokes of a person could be monitored unobtrusively and continuous verification of an individual's identity is available.

### **Voice**

The measurable parameters of a person's voice are tone, pitch, cadence and frequency. However, they change over time due to age, medical conditions (common cold), or emotional state. Being not very distinctive, voice and may not be an appropriate trait for large-scale identification. A text-dependent voice recognition system is based on the utterance of a fixed predetermined phrase. A text-independent voice recognition system offers more protection against fraud, but is more difficult to design. Speaker recognition

is possible in telephone-based applications but the voice signal is typically degraded in quality by the communication channel.

All biometric traits have their own relative merits in various operational scenarios and their choice for a particular application depends on a series of conditions besides its match performance. No single biometric ever meets all the requirements (accuracy, cost, efficiency) imposed by all applications. However, the inherent limitation factors of processing a single biometric can be alleviated by fusing the information obtained from multiple sources. For example, the fingerprints of the right and left index fingers, or multiple images of face from different angles, or a combination of the face and gait traits may be used to resolve the identity of an individual. Consolidating the evidence presented by multiple sources is expected to be more reliable. Furthermore, fusion of biometrics helps users to get successfully enrolled and makes the system more robust against hackers.

Most of the human recognition technologies generally require a subject that permits and enables physical contact, close proximity or views from specific angles, in order to help the recognition procedure. Without a cooperating subject, individuals are difficult to be reliably recognized from arbitrary views when walking at a distance in real-world changing environmental conditions. For optimal performance, we should make use of as much information as can possibly be obtained from the available observations. Gait and face are two traits available to be captured on a video that is acquired from a distance.

Considering biometrics, there is nothing like a unique signature for a particular individual. Despite the best efforts, each time he generates a slightly different sample. Since exact match is simply inapplicable, we are induced to employ similarity search methods, such as  $k$ -NN query. The recognition is done through correlation and thresholding. This does not mean the technology is inherently insecure, as very high success rates of have been achieved. However, technology offering 100% recognition should be greeted with a pinch of salt.

In the world of information technology security we require biometric techniques to be automatically verifiable. Manual feature extraction would be both undesirable and time

consuming due to acquiring and processing of large amount of data. Inability to automatize would render the process infeasible on realistic size data sets. This leads to development of fast and convenient recognition systems (see Figure 1) that automatically evaluate plenty of queries and give instant answers.

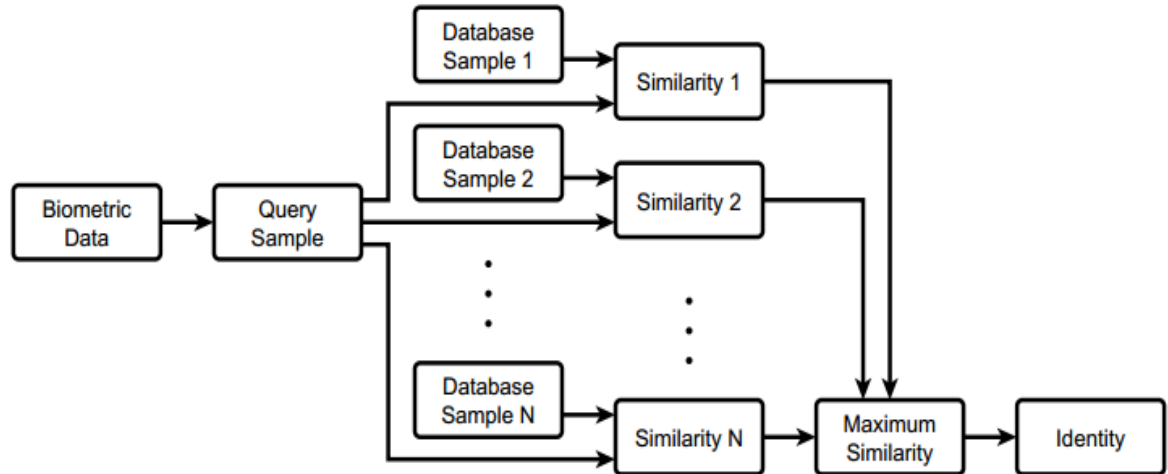


Fig. 1. Biometric identification scheme.

For a biometric to be useful in an application, it must also be easily measurable, have ability to achieve high accuracy of correct recognition rate and must be user acceptable. In combination of all these attributes, there is no such thing as ideal biometric; each has its advantages and disadvantages, which means that a "fit for purpose" compromise to a specific application is the solution.

A biometric system can be deemed as a pattern recognition system which consists of three main parts, namely, a) acquisition of biometric data, b) feature extraction and, c) matching of the acquired features against template set in the database of known subjects.

The operation of the biometric system can be divided in two modes:

Verification mode - the system validates a person's identify by matching the acquired biometric features with the claimed identity template stored in the database. The system verifies if the user is the person who he/she claims to be by conducting one-to-one matching. The purpose of such system is to prevent multiple people to use the same identify and is used typically for positive recognition.

Identification mode – the system recognises a person by matching the acquired biometric feature against the templates of all users in the database. Hence, to identify the person's, the system attempts to establish "whose identity is this?" and conducts one-to-many comparison. The purpose of such system is to prevent a single person from using multiple identities and is used in negative and positive recognition.

Due to intra-variation of the biometric trait the result of the matching the newly acquired data and the stored template will not produce a perfect match. The system will have to be tolerant of the slight variation due to natural fluctuations of the biometric trait. Therefore, biometric systems match is based on the probability estimate to what extent the newly acquired data corresponds to the stored template. A means of user interaction is inevitable in order to verify a claimed identity or where an identity is required. This generally requires that the subject be in close proximity, or even physical contact, with the biometric sensor in order to be effective [13]. Hence, it is rather important that user acceptance is not underestimated. A biometric system that lacks broad user acceptance will most certainly fail in performance. In order to secure user acceptance, the users must feel comfortable and at ease with the use of the biometric system. For instance, finger printing has always been associated with criminal activities and can be very intimidating for a normal person, who gets subjected to these techniques just to ascertain his/hers identification and generally lacks user acceptance. This is where a non-contact biometric such as facial appearance, ear characteristics or gait can be extremely useful. Such alternative biometric could identify a person without much knowledge and the support of the user. The facial and ear characteristics, even though being a non-contact biometric, still require a close view to capture the detailed biometric data. In an application which requires recognition "at a Distance", the facial and ear based recognition becomes impractical. In such application, gait has the advantage of being an unobtrusive biometric that can be observed from a distance with much more information for recognition – making it an attractive feature.

## **2.3 Gait**

Walking is a complex dynamic activity that is fast, animate, irreversible and also rigid. Many components of the body are involved in moving and interact mutually with each

other and with the environment. Gait is defined as a manner of walking. However, human gait is more than that. It is an idiosyncrasy of a person that is determined by, among other things, an individual's weight, bones' length, footwear, and posture combined with characteristic motion. Considering all the features, human gait has been proven peculiar to individuals.

We use the term gait recognition to signify the identification of an individual from a video sequence of the subject walking. This does not mean that gait is limited to walking, it can also be applied to running or any means of movement on foot. With current technology supporting in-depth measurement, we can capture diverse elements of gait, which might be helpful at recognition

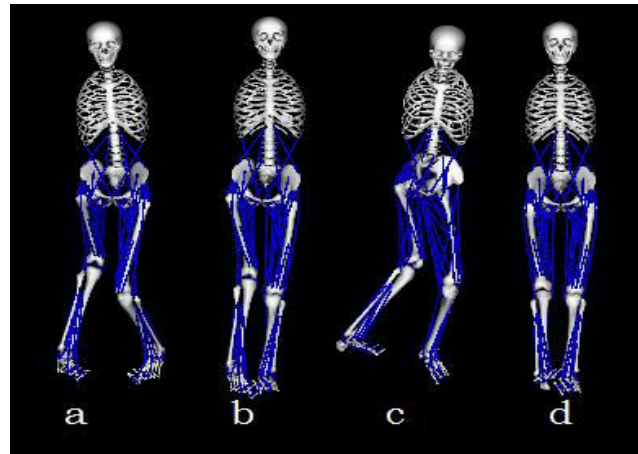


Fig. 2. Contrast enhanced images in gait sequences of Gait Database (Dataset - a (croubh2) b (croubh3) c (croubh4) d (normal) ). Four samples from left to right show the gait of pathological walking (a,b,c) and normal walking (d).

Gait can be seen as advantageous over other traits since the information is captured without usage of any obtrusive body-invasive equipment or subject's cooperation. From a surveillance perspective, walk pattern biometrics is appealing because of its possibility to be performed at a distance, even surreptitiously. Together with high rate of collectability, these are the reasons why the method is preferably employed at human identification rather than authentication. Another motivation is that video footage of suspects is readily available, as surveillance cameras are relatively low cost and installed at most locations requiring presence of security. Unlike face recognition, which can be

easily affected by low resolution, here a video of reduced quality is often sufficient. Moreover, gait is difficult to disguise, and by trying to do so the individual will probably appear even more suspicious, while face can easily be altered or hidden.

On the other hand, slightly modified conditions may introduce a large appearance change, which may lead to a failure in recognition. Gait can be affected by clothing (long cloak, shoes, carrying objects), physical changes (pregnancy, injury, weight gain/loss) or environmental context (walking surface, background). Temporary stimulants (drugs and alcohol) as well as mood can change a person's walking style. The large gait variation of the same person under different conditions intentionally or unintentionally reduces the discriminating power of gait as a biometric. Hence it may not be as unique as fingerprint or iris, although the inherent gait characteristic of an individual still makes it irreplaceable and useful in visual surveillance. The ultimate goal is to detect, classify and identify humans at long distances, under day or night, all-weather conditions, alone or among a group of people. Light deficit and self-occlusion of feature points (overlapping legs) kills the process of gait features extraction from the external factors. The challenges involved in gait recognition include also imperfect foreground segmentation of the walking subject from the background scene and variations in the camera viewing angle with respect to the walking subjects. Large variation of the same individual under different conditions requires more gallery samples collected from different environmental contexts. Despite having samples generally obtained under similar environmental conditions, a reliable gait recognition system with large-scale database seems unreal due to the complexity of real-world situations. Therefore, a reasonable solution is not to ask the system for a single (the best match) identity of an individual, but for a set of most probable identities. The set can be constructed by taking first  $k$  most similar patterns ( $k$ -NN query), or by setting a similarity threshold (range query).

It is important to adjust the values sensitively with respect to database size and similarity function in order to filter out maximum of unrelated identities, but also to keep the related one. The goal of a dramatic reduction of the set of all identities (from billions to hundreds) is now achievable.

## **2.4 Human gait**

Human gait analysis refers to the study of human locomotion through the evaluation of kinematic, kinetic and spatial/temporal parameters. This field of study finds application in a large variety of domains, including entertainment, healthcare and pedestrian navigation. The act of walking involves the coordination of different human body parts, such as the skeleton, the muscle and the neural systems. Various factors can affect the complex interaction between the body parts, for example any pathological nature will require a distinction between — **normal gait** and — **pathological gait**. —Normal gait refers to the naturalistic and general human walking parameters without differentiating the age, sex or individual physical parameters. Conversely, —pathological gait refers to an abnormal gait, for example affected by pathologies such as muscle weakness or skeleton deformities.

In this thesis, we differentiate the normal gait and pathological gait and differentiate between them. Indeed it has been shown that a normal gait is characterized by a fundamental pattern that is not subjected to inter and intra individual variations. Specifically the normal human gait is marked by the periodic repetition of two main phases, the **stance phase**, when the foot is in contact with the ground, and the swing phase, when the foot is in the air. These two phases are represented in Figure 1. The stance phase occurs between two events: the foot strike and the ipsilateral (same foot) foot off. Complementary, the swing phase starts with the foot off event and ends when the second ipsilateral foot strike.

### **2.4.1 Principles of gait**

Gait is the way locomotion is achieved using human limbs. Differences in limb movement patterns, velocity, forces, kinetic energy cycles and changes in the contact with surface characterises different gaits. The methods and mechanics of human ambulation - the bipedal gait - are quite unique in the nature. In relation to human walking, gait refers to the style or manner of walking, rather than the walking process itself [19]. Once gait is learned it becomes almost subconscious. Individual gait is dependent on many variables such as surface properties, confidence, mood, quantity and



quality of sensory information (e.g. light levels, noise, and distractions), any chemical effects on the body (such as alcohol or medication), energy levels, mental alertness and muscle fatigue. Walking mainly consists of the ability to maintain the equilibrium and the ability to initiate and maintain a rhythmic stepping motion – locomotion. In conjunction with these two abilities, there are other many contributing factors involved in walking such as the skeletal system with the joints as well as neuro-muscular system. Although the human gait varies from individual to individual, some basic stages are common for all normal gaits. We divide human gait into gait cycles, which consists of a repeated sequence of events and is the time between two equal events in the walking cycle, such as the heel to heel strike of the right foot, as shown in Figure 3.

#### **2.4.2 Human Gait Biomechanics**

Gait cycle is defined as the periodic cycle/motion of human locomotion/walking. It consists of 8 phases shown in fig.3 [18].

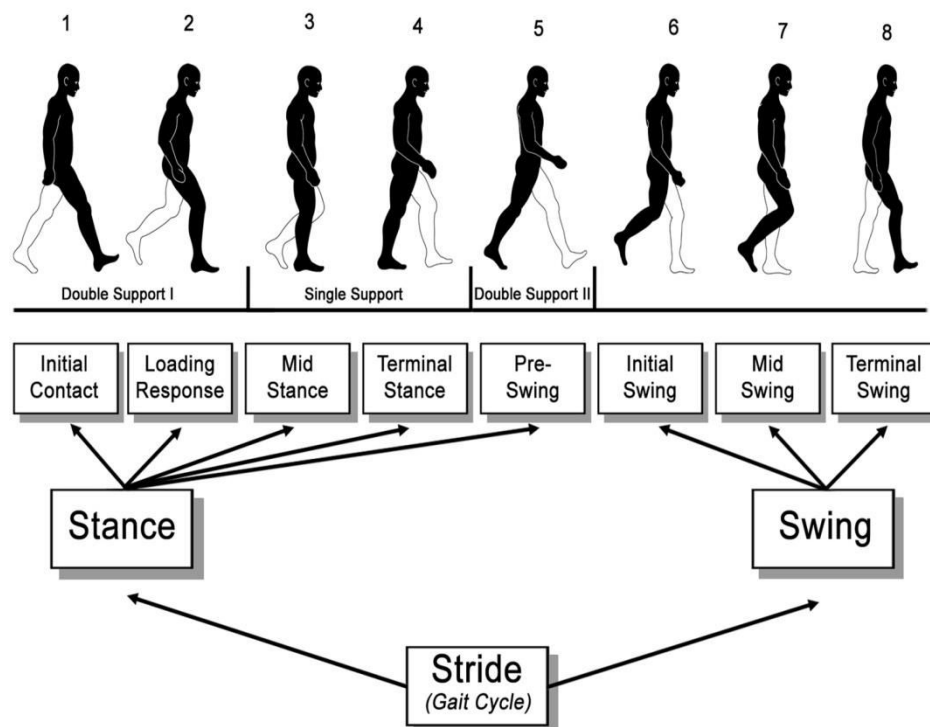


Fig. 3. Eight phases of Gait cycle

In *INITIAL CONTACT* phase foot of human contacts the ground. In *LOADING RESPONSE* phase the weight of body is shifted to leg. In *MID-STANCE* phase the human body moves over the leg. In *TERMINAL STANCE* phase the human body moves in front of the leg. In *PRE-SWING* phase leg pushes off the surface and opposite foot contacts the surface. In *INITIAL SWING* leg lifts off the surface. In *MID-SWING* phase leg moves ahead of the human body. In *TERMINAL SWING* phase leg comes to the ground [15].

As shown in Figure 1, in the case of normal gait, the stance and swing phases can be further divided in eight sub phases as follows:

- Initial contact: refers to the precise instant when the foot touches the ground. The initial contact marks the beginning of the stance phase.
- Loading responses: corresponds to about 10% of a gait cycle. During this phase the foot comes in full contact with the ground and the stance limb supports all the body's weight.
- Mid-stance: starts when the foot leaves the ground and ends when the body advances over the stance limb.
- Terminal stance: starts when the heel leaves the ground and ends when the contralateral foot touches the ground.
- Pre-swing: in this phase the knee flexes before the swing phase and the body's weight is loaded to the opposite limb.
- Initial swing: represents the first third of the swing period. It starts when the foot leaves the ground and ends when the maximum knee flexion occurs.
- Mid swing: is the middle third of the swing period. It starts with the maximum knee flexion and ends when the tibia is vertical.
- Terminal swing: is the third swing period during which the knee fully extends before the heel's contact. The repetition of the same event for the same foot marks the occurrence of the **stride** event. The repetition of the same event for different feet marks **step** events. In general, the period when the foot is flat on the ground is assumed as reference event since it is the most easy to be identified.

The running gait shows some differences with respect to the walking gait. While in the walking case there is at least one foot that touches the ground, in the running case, aerial phases, when both feet are in the air, can be identified. For this reason, in the running mode, the vertical component of the ground reaction assumes bigger values than in the walking mode. More specifically, during running the stance phase decreases by 35% while the peak ground reaction force increases by 50%. These different values of the stance phase and the ground reaction force are reflected in different patterns of the **Centre Of Mass (COM)**, which is the point of the body where the entire mass can be considered concentrated (Rose & Gamble 2006). In walking mode the centre of mass assumes its biggest value at the middle of the stance phase. Conversely, during the running mode, the centre of mass reaches its lowest values at the middle of the stance phase. Indeed the different profile drawn by the centre of mass during motion can be used to distinguish running and walking modes. However, this approach will not be used in this thesis. The COM plays an important role in gait analysis. In Winter (2004), the COM is considered as the intersection of the sagittal, the transverse and the frontal plans cutting the human body as shown in Figure 2. Although the COM varies dynamically according to the performed motion, when the subject is static, it can be considered placed close to the hip. Indeed, since the COM's motion reflects the global motion, it is common to place the sensor used for gait analysis close to the hip. However, this location is not suitable for mobile devices. Indeed mobile devices are generally held in hand. In order to track the global user's motion even if the sensor is held in the hand, it is necessary to analyze how the motion of the body's upper and lower parts are related. Generally, human gait analyses are focused on the motion of the body's lower part. For this reason, gait models are generally designed excluding the subject's arm contribution. A few studies have analyzed the arms' swing in conjunction with the foot's vertical moment.

Specifically, it has been observed that during normal walk, coordination between upper and lower parts of the human body exists. Indeed, the arms' swing has the purpose of decreasing the reaction momentum of the vertical axis of the foot (Park 2008). For example, when the left foot is on the ground (during the stance phase), a positive torque induced by the arm swing allows advancement of the right leg. The same event happens

for the right leg. Consequently, this coordination can be exploited to track the lower part motion even if the sensor is handheld.

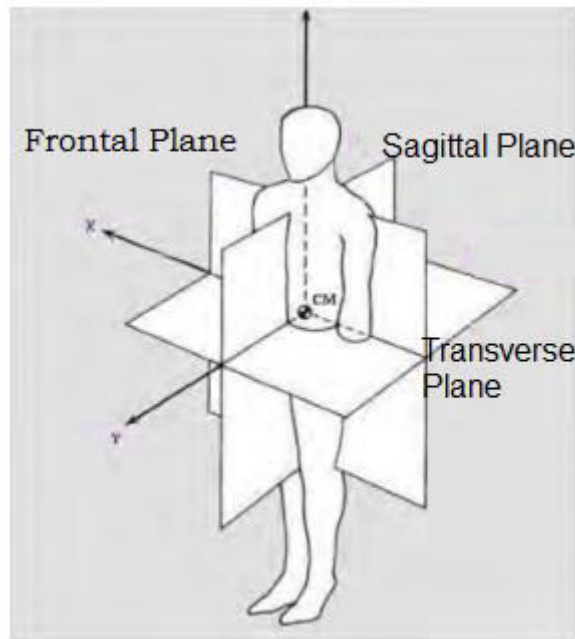


Fig. 4. Representation of the human body and the three plans defined by the body's COM - adapted from Winter (2004)

## **2.5 Gait as a Behavioural Biometric**

The study of human movement known as the *gait* has been a keen interest from a wide variety of disciplines. It has been investigated and researched in various contexts, mainly in the area of medical applications. Medical and psychological studies have indicated the potential of gait in personal identification.

Jay Prakash Gupta [14] conducted experiments with moving light display attached to body parts. He showed that humans' observers can almost instantly recognise biological motion patterns, even when it was presented with only a few of the moving dots. The further studies in [15] confirmed that humans can correctly discriminate between gender of a walker.

Navneet Kaur [16] conducted study on what constituted the walking act through a systematic study that fully characterised the coordinated movement patterns of the various parts of the body, namely head, neck, trunk, and upper and lower limbs. He

obtained the walking patterns by attaching reflective targets to specific anatomical landmarks which was illuminated with a strobe light flashing 20 times per second. His study suggested that the gait is a unique personal characteristic, if all gait movement pattern is considered. This uniqueness feature of gait indicates that it could be used as a feature for biometric.

Navneet Kaur [17] empirically showed that it was relatively simple for friends recognised their friends by their gait; even when known cues such as hairstyle and clothing was not visible. Light sources mounted on joint, know as Moving Light Display (MLD), were sufficient to capture motion pattern for identification. The study indicated that people recognised others not only by static features such as height but through dynamic features such as arm swing, rhythm of the walk, bounciness or length of steps.

The early researches and their results are encouraging and promising, leading towards gait as a biometric feature. Even though these studies have laid the foundation of the gait as a biometric, it is not until recently that use of gait as a biometric has gained momentum in area of surveillance applications. The key challenge in such an application is how to replicate the thought process of humans through a use of a computer so that the identification of people becomes an easy task. What appears to be an easy task for humans does not necessarily apply to computers.

## **CHAPTER-3**

### **LITERATURE REVIEW**

#### **3.1 Introduction**

This chapter presents literature review on gait analysis using different methods. The performance of different techniques used for gait analysis by the previous researchers is presented.

#### **3.2 Literature Review**

Gait recognition can be related to human movement analysis methods i.e. vision methods that detect, track and / or recognize classes of human motion and more specific methods that deal with whole-body movement (without regard to its individual parts). A comprehensive survey of gait recognition techniques can be found in [T INDUMATHI, M.PUSHPARANI, 2014].

##### **3.2.1 Human Movement Analysis**

It has been reported by different researchers that one of the main benefit of gait recognition over other biometrics is its nonintrusive nature [Rong Zhang et al., 2013]. Hence forth, the analysis of the covariate factors becomes essential to the understanding of the uniqueness of gait recognition. Several areas of computer vision are interested in the analysis and modeling of human motion in video, including pedestrian detection / counting, gesture recognition, action / activity recognition, lip reading and person identification. Existing methods can be grouped into structural methods, which recover the structure of the human body prior to recognizing its movement [Davrondzhon Gafurov, 2010] and structure-free methods which directly model the 2D motion patterns generated in the image sequence by any particular type of movement.

##### **3.2.2 Structural Methods**

A 2D or 3D structural model of the human body is assumed and body pose is recovered by extracting image features and mapping them to the structural components of the

model (i.e. body part labeling). Hence, a human is detected in the image, if there exists a labeling that fits the model well enough (based on some measure of goodness of fit). Once a person has been detected and tracked in several images, motion recognition is done based on the temporal trajectories of the body parts, typically by mapping them to some low dimensional feature vector and then applying standard pattern classification techniques [S.M.H. Sithi Shameem Fathima and R.S.D. Wahida Banu, 2015]. Tracking body parts in 3D over long sequences remains an unsolved computer vision problem; the effectiveness and applicability of these methods remain limited.

S.M.H. Sithi Shameem Fathima et al., 2015, proposed methodology for categorizing the walking style as normal or abnormal gait. The block diagram consists of training and testing phases. Initially, a camera with low resolution recorded the gait of the persons in the training phase. A simple background subtraction algorithm extracts the human silhouettes. In order to efficiently classify the gaits as normal and abnormal gait the ellipses are fitted vertically and horizontally to cover the head to toe portion and to the leg portion as shown in Fig. 1. The elliptical features such Major Axis (MA) and minor Axis (mA) lengths as well as the orientation ( $\phi$ ) of both the horizontal and vertical ellipses are computed to serve as feature vectors. These feature vectors differentiates the normal and abnormal gait. In testing phase, the query gait is classified using SRC classifier based on the elliptical features.



Fig. 1. Ellipse fitting model where MA1, MA2 denotes major axes, mA1, mA2 are minor axes and  $\Phi_1$ ,  $\Phi_2$  are the orientation of horizontal and vertical ellipses

**Feature extraction:** The efficient elliptical feature extraction to distinguish the normal and abnormal gait is explained in this sub section. At first the vertical and horizontal ellipses are fitted on head to foot and leg portion respectively using least square method.

**Dataset description:** The abnormal and normal walking patterns of various persons were acquired by a static camera. The persons both female and male having 18 to 35 ages are considered to create the dataset. During video acquisition subjects wear different types of clothes, shoes and slippers. They were asked to walk on the track shown in Fig. 2a and 2b. Totally 105 subjects were considered for creating the dataset. The video has been recorded using SONY Handy cam HDR-CX240. It has CMOS sensor with size of 1/5.8 inch. Resolution of "1920 x 1080", "1440 x 1080", frame rate of 60 pixels. The distance between camera and track was around 25 feet and length of track was 20 feet. Three minutes of continuous walk converted around 1200 frames in JPEG (.jpg) format with frame size of 1920 x 1080 were obtained. These images were converted into silhouettes.

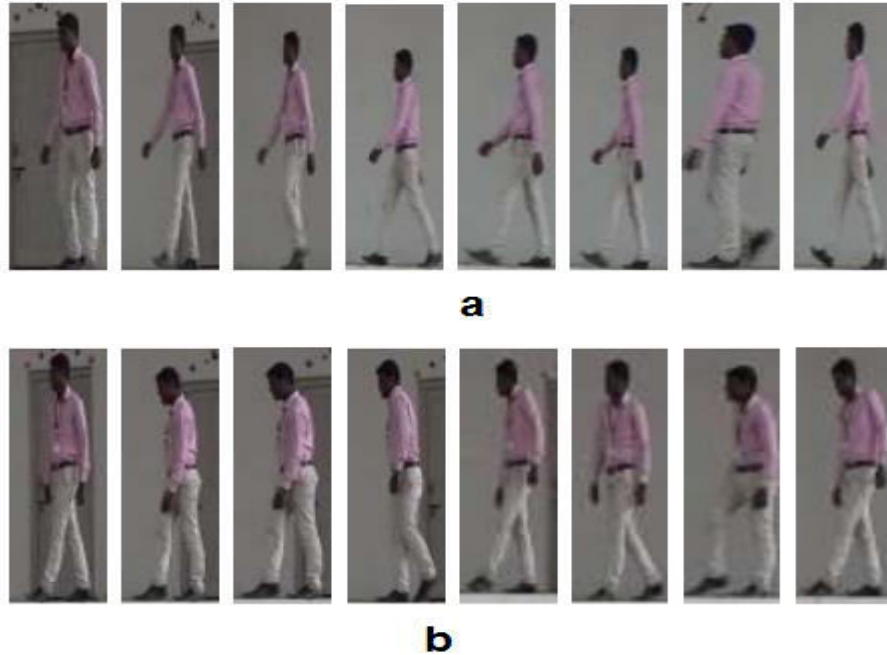


Fig. 2. 2a: The normal gait cycle of a person 2b: Abnormal gait cycle of a person



**CONCLUSION:** They presented a methodology for classifying the walking style as normal or abnormal gait. In order to distinguish the normal and abnormal walking style, this study adopted vertical and horizontal ellipse fitting algorithms on head to foot and leg portion of the silhouettes respectively. From the horizontal and vertical ellipses, the major and minor axis lengths as well as the orientations of both ellipses are calculated. The automatic classification has been done using SRC classifier based on the elliptical features. Furthermore the dataset of normal and abnormal gait is created using 105 subjects. The proposed methodology proved that it is efficient for categorizing the normal and abnormal gait to about 99.52% accuracy from the experimentation.

Adam ŚWITOŃSKI et al., 2011, analyzed the performance of the bottom-up model-based approach which is satisfactory for extracting the most important dynamic gait features for people recognition, derived binary silhouettes of walking subjects which are converted into a one-dimension normalized distance signal by contour unwrapping with respect to the centroid position. Most vision systems for human motion analysis rely primarily on the extraction of markers attached to the joints of moving people. However, most applications such as visual surveillance require the deployment of an automated marker-less vision system to recover the joint trajectories. It is claimed that automated extraction of the joints position from videos of walking subjects is an unsolved problem. To explore the most individual joints movements, aggregated feature rankings are calculated. To examine the proposed method, the database containing 353 gaits of 25 different actors is collected in the motion capture laboratory.

Romualda MUCHA et al., 2011, showed that the proposed method consists of the following steps: kinematical data reduction by Principal Component Analysis, determination of the Fourier component for the 3D PCA trajectories and supervised learning. To examine proposed approach, they have collected database of gaits containing data of coxarthrosis patients. In these publications measurements of gait indexes are further combined with several machine learning techniques in order to extract appropriate information. Some of the publications and methodologies are overviewed below. In the paper four types of health problems of the elderly are detected:

Parkinson's disease, hemiplegia, pain in the leg and pain in the back. The proposed method is based on the feature extraction approach and supervised learning. The 13 different features, calculated directly from the motion capture sequences, are prepared by the medical expert. For instance, one of them represents the quotient between maximal angle of the left knee and maximal angle of the right knee. Obtained feature's vectors are classified by the decision tree and neural network classifiers.

A. Hayder et al., 2011, observed that Biometric gait recognition refers to verifying or identifying persons using their walking style. Human recognition based on gait is relatively recent compared to other biometric approaches such as fingerprint, iris, and facial etc. The Radon Transform is used to detect features within an image and PCA is used to reduce the dimension of the images without much loss of information. The side view of slow walk; fast walk and carrying a ball walk have been selected from the CMU MoBo database for experimental purposes. The two techniques experimental result achieved equal recognition rates (EER) of 85.40%, 78.07% and 90.05% for RT with PCA and 85.18%, 80%, and 89.90% for PCA only for slow walk, fast walk and carrying a ball walk respectively.

Hoang Minh Thang et al., 2013, proposed two approaches for identification based on biometric gait using acceleration sensor - called accelerometer. In contrast to preceding works, acceleration data are acquired from built-in sensor in mobile phone placed at the trouser pocket position. Data are then analyzed in both time domain and frequency domain. In time domain, gait templates are extracted and Dynamic Time Warping (DTW) is used to evaluate the similarity score. On the other hand, extracted features in frequency domain are classified using Support Vector Machine (SVM). With the participation of total 11 volunteers over 24 years old in their experiment, they achieved the accuracy of both methods respectively 79.1% and 92.7%.

Jay Prakash Gupta et al., 2014, proposed the technique of human activity recognition is based on the foreground extraction, human tracking, feature extraction and recognition. Figure 3 shows the framework of the introduced human activity recognition system

using Gait to identify four basic human activities (i.e. walking, running, jogging and jumping). The proposed method has following main steps: Foreground Extraction, Human Tracking, Feature Extraction and Activity Recognition. In this framework, the video is given as an input to the system from the activity database and frames are extracted from that video. The parametric model of human is extracted from image sequences using motion/texture based human detection and tracking After that the results are displayed as the recognized activities like walking, running, jogging and jumping; and finally the performance of the method is tested experimentally using the datasets under indoor and outdoor environments.

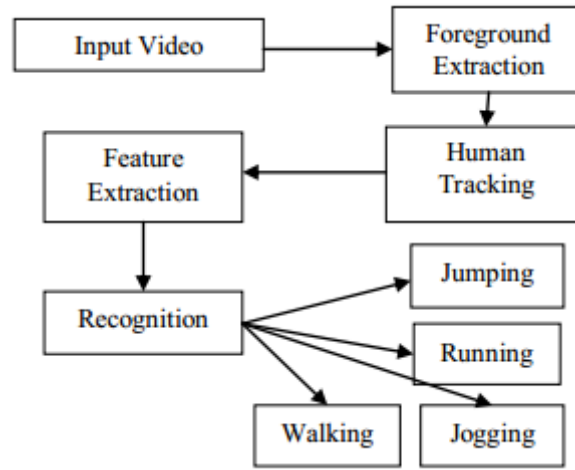


Fig. 3. Framework of Proposed System of Human Activity recognition

Prerna Arora, et al., 2015, Gait recognition is the term used for detection of Human based on the features. The Feature extraction and Feature Mapping is the main aspect to recognize the Gestures from the Database of features. Recognition of any individual is a task to identify people. Human recognition methods such as face, fingerprints, and iris generally require a cooperative subject, physical contact or close proximity. These methods are not able to recognize an individual at a distance therefore recognition using gait is relatively new biometric technique without these disadvantages. Human identification using Gait is method to identify an individual by the way he walk or manner of moving on foot. Gait recognition is a type of biometric recognition and related to the behavioral characteristics of biometric recognition. Gait offers ability of distance recognition or at low resolution. This project aims to recognize an individual using his gait features. However the majority of current approaches are model free

which is simple and fast but we will use model based approach for feature extraction and for matching of parameters with database sequences. After matching of Features, the Images have been identified and show the dataset from it matched. The Results are accurate and shows efficiency. In this firstly binary silhouette of a walking person is detected from each frame of an image. Then secondly, the features from each frame are extracted using the image processing operation. In the end SVM, K-MEANS and LDA are used for training and testing purpose. Every experiment and test is done on CASIA database. The results in this paper are better and improved from previous results by using SVM, K MEANS.

Biometrics is classified into two categories:

**Physiological:**

Fingerprint recognition

Iris recognition

Face recognition

DNA recognition

**Behavioral:**

Typing rhythm

Gait recognition

Voice recognition

### **3.3 Proposed Gait recognition method**

**A) The background subtraction:** in this step firstly the moving objects in the video are identified. Then some of the background extraction techniques are applied on it. The background subtraction generates binary image containing black and white (moving pixels) also known as binary silhouettes. There is no. of challenges in developing a good background subtraction algorithm.

- 1- It must be robust against changes in illumination.
- 2- It should avoid detecting non-stationary background objects such as moving leaves, rain, snow etc.
- 3- Its internal background model should react quickly to changes in background such as starting and stopping of vehicles. They are using CASIA database.

**B) Pre-processing:** silhouette segmentation is the first step towards gait recognition. Pre processing is done to reduce the presence of noise , then some filters are applied which in turn blur the frame of image, which helps in shadow removal, after pre-processing

motion detection is done. Background subtraction technique uses the difference of current image and background to detect the motion.

**CONCLUSION:** The Gait Recognition System has been developed based on Features. For Detection human gestures, the dataset has been taken of multiple scenarios. The Training has been provided to the algorithm and Features Mapping concept has been used. Initially the features have been extracted from the training dataset and stored in the templates. The new images have been provided for testing and validate the work, and then the new features has been extracted and mapped with the stored features. With mounting demands for visual surveillance systems, human identification at a distance has recently emerged as an area of significant interest. Gait is being considered as an impending behavioral feature and many allied studies have illustrated that it can be used as a valuable biometric feature for human recognition. The development of computer vision techniques has also assured that vision based automatic gait analysis can be gradually achieved. The proposed system has been tested on the gait databases and, the extensive experimental results on outdoor image sequences demonstrated that the proposed system possesses a pleasing recognition performance.

### **3.3.1 Structure-free Methods**

Baofeng Guo, et al., 2009, Feature subset selection is an important preprocessing step for pattern recognition, to discard irrelevant and redundant information, as well as to identify the most important attributes. In this paper, we investigate a computationally efficient solution to select the most important features for gait recognition. The specific technique applied is based on mutual information (MI), which evaluates the statistical dependence between two random variables and has an established relation with the Bayes classification error. The experimental results confirm the effectiveness of the method, removing about 50% of the model-based features and 95% of the symmetry map's pixels without significant loss in recognition capability, which outperforms correlation and analysis-of-variance-based methods.

### **3.3.2 Whole-body Movement Analysis**

Er. Shikha Chawla, et al., 2013, Gait recognition is one kind of biometric technology that can be used to monitor people without their cooperation. Controlled environments such as banks, military installations and even airports need to be able to quickly detect threats and provide differing levels of access to different user groups. Gait shows a particular way or manner of moving on foot and gait recognition is the process of identifying an individual by the manner in which they walk. Gait is less unobtrusive biom ethic, which offers the possibility to identify people at a distance, without any interaction or co-operation from the subject; this is the property which makes it so attractive. This paper proposed new method for gait recognition. In this thesis, first step is extraction of foreground objects i.e. human and other moving objects from input video sequences or binary silhouette of a walking person is detected from each frame and human detection and tracking will be performed. After getting binary silhouettes of human beings model based approach is used to extract the gait features of a person.

### **3.3.3 Holistic Approach**

Er.Ranjeet Kaur Sandhu, et al., 2013, Gait recognition is one kind of biometric technology that can be used to monitor people without their cooperation. Controlled environments such as banks, military installations and even airports need to be able to quickly detect threats and provide differing levels of access to different user groups. Gait shows a particular way or manner of moving on foot and gait recognition is the process of identifying an individual by the manner in which they walk. Gait is less unobtrusive biometric, which offers the possibility to identify people at a distance, without any interaction or co-operation from the subject; this is the property which makes it so attractive. This paper proposed new method for gait recognition. In this thesis they presented the review of gait recognition system, different approaches and classification categories of Gait recognition like Model Free Approach, GMM for background subtraction, Silhouette Segmentation. We combined used SVM (Support vector Machine) with NN (Neural Network) then compare it with BPNN.

### **3.3.4 Human Tracking**

Jay Prakash Gupta et al., 2014, in this paper they proposed human activity recognition based on the computer vision is the process of labeling image sequences with action labels. Accurate systems for this problem are applied in areas such as visual surveillance, human computer interaction and video retrieval. The challenges are due to variations in motion, recording settings and gait differences. Here we propose an approach to recognize the human activities through gait. Activity recognition through Gait is the process of identifying an activity by the manner in which they walk. The identification of human activities in a video, such as a person is walking, running, jumping, jogging etc are important activities in video surveillance. We contribute the use of Model based approach for activity recognition with the help of movement of legs only. Experimental results suggest that our method is able to recognize the human activities with a good accuracy rate and robust to shadows present in the videos.

Dr. Amit Dixit et al., 2011, this class of networks consists of multiple layers of computational units; usually interconnected in a feed forward way. Therefore each neuron in one layer has directed connections to the neurons of the subsequent layer. Many applications the units of these networks apply a sigmoid function as an activation function. Then universal approximation theorem for neural networks states that every continuous function that maps intervals of real numbers to some output interval of real numbers can be approximated arbitrarily closely by a multi-layer perception with just one hidden layer. The result holds only for restricted classes of activation functions, e.g. for the sinusoidal functions. Multi-layer networks use a variety of learning techniques; the most popular being back propagation.

Gait analysis is the systematic study of human locomotion; augmented by instrumentation for measuring body movements; body mechanics and the activity of the muscles. Gait based recognition is more suitable in video surveillance applications because of following advantages:

1. Recognition using gait do not need any user cooperation.
2. The gait of an individual can be captured at a distance.

3. Gait recognition does not require images of very high quality and provide good results in low resolution. Therefore Gait recognition aiming to identify the individuals by the way he walk or move.

### **3.4 Human being detection**

Deepti Singh et al., 2014 Human Identification Using Gait Recognition has been proposed previously but there have been always need for better Gait Recognition Technique. The existing Human Identification Using Gait Recognition doesn't consider some important parameters like distance between right hand and right leg and thus it is poor in quality. The existing Human Identification Using Gait Recognition algorithm is less accurate. Therefore, propose an enhanced Human Identification Using Gait Recognition algorithm which is based on PAL and PAL entropy and SVM (SUPPORT VECTOR MACHINE), LDA and k-Means. Our enhanced Human Identification Using Gait Recognition algorithm is more accurate. Our enhanced Human Identification Using Gait Recognition algorithm is fast and thus saves time.

#### **3.4.1 Tracking-by-detection**

Navneet Kaur et al., 2014 the appearance-based approaches suffer from changes in the appearance owing to the change of the viewing or walking directions. Model based approaches extract the motion of the human body by means of fitting their models to the input images. Model based methods are view and scale invariant. System will identify unauthorized individual and compare his gait with stored sequences and recognize him. Background subtraction is the common approach of gait recognition. Background subtraction method is used to subtract moving objects and to obtain binary silhouette.

#### **3.4.2 People tracking system**

In an image sequence, people detection can be implemented frame by frame. The output of the detection system such as position, shape and color of the object can be used to construct a tracking model. Because there exist a large correlation between consecutive video frames, the location and appearance of people do not undergo a dramatic change



within a certain period. Therefore the state of tracking method is focused on how to use these kinds of properties to solve feature matching problem in video sequences.

Based on the acquired video data tracking systems can be divided into monocular and multi-view approaches. Monocular approaches use the video image from a single view to perform tracking. Motion model is used for each person to estimate its position in subsequent frames; additionally a dynamic appearance model is constructed by combining gray-scale textural appearance and shape information of people. It is capable of tracking multiple persons simultaneously and people can still be tracked through interaction such as occlusion. The assumption of this method is that the motion of people is predictable such as walking and running. It is primarily designed for outdoor surveillance. As for the random motion of people in an indoor scene their motion model does not fit well. The distribution of color on human body is utilized to build 2D blob model of people. The body parts can be reliably labeled and located by 2D contour shape analysis. The color tracker can still track the hands in front of the body, although in this situation there is no contour evidence of hands. Tracking performance is significantly increased by taking color into account. The limitation of this method is that it describes the blob model as a single human which will cause problem for multiple user gesture recognition.

Multiple-view approaches utilize multiple cameras to record the video data which is effective to calculate the precise 3D position of people in complex environment. In the indoor video sequences are shot with four synchronized cameras and outdoor with three cameras. With these data, they build up a stochastic model to compute the optimal individual trajectories and then use optimization schemes to process the trajectories one after another. It is shown that this algorithm can follow up to six individuals with significant occlusion, but in the outdoor surveillance they do not mention the influence of other moving objects, for instance, a car that is parked in the scene, this sudden change in the background will degrade the performance dramatically.

Cai and Aggarwal, 1998, used a Bayesian tracking scheme to locate the most likely match of the object in next frame. The automatic camera switching selects the camera with the best view during tracking period, although they have developed techniques of multiple view transition tracking to deal with occlusion at a certain level by using distributed monocular cameras, it still remains a major obstacle in their tracking system.

Many current methods for detecting and tracking people rely on color contrast or movement to segment the image. Using color requires the target and the background to be significantly different and motion segmentation requires the target to be in constant motion relative to the background often requiring stationary cameras.

Sidenbladh and Kraigi, 1999, locate faces within the image. A closed-loop controller is then used to maintain tracking of the person by cameras. A more statistical approach comes from Tarokh and Ferarri, 2003, who use clothing color to segment the image. Statistical analysis is then used to determine which color blobs belong to a person and which do not. A more sophisticated approach was discussed by Kwon et al., 2005, who used color histograms to locate a person in a pair of stereo images followed by triangulation between the images to find the distance between the camera and the tracked person. Schlegel et al., 1998, used color histograms. They combined the histogram data with an edge-based detector to improve robustness of the algorithm at the expense of computation time.

Piaggio et al., 1998, approached the problem by threshold the velocity data and assuming that the person moves differently from the background.

Chivil`o, 2004, viewed velocity changes as a disturbance to be minimized by regulation (i.e. moving the camera or the robot to minimize the relative motion of the person, thereby providing tracking).

Munkelt et al., 1998, attempted to detect people in a video by calculating correspondence between features in the image and a 3D model of the internal structure

of the target. They presented a complex articulated model of a person and showed how it could be matched to a person in an image using color and 3D information. 3D scene information is provided by either a stereo camera setup or through matching basic shapes in the image with basic shapes in the model and determining depth and relative position information from this.

Inamura et al., 1998, presented a method of person detection and tracking that combines motion detection, edge detection and color detection with a voice recognition system. Krinidis et al., 2005, implemented a database of audio and visual data as a standard for testing of tracking systems. Their database covers a range of situations from a single person in a simple scene to multiple peoples moving in different directions. Each of these is repeated for various lighting conditions.

### **3.4.3 Feature selection for people detection**

Perez et al., 2004, presented the data fusion method to include color, stereo sound and motion cues to a human model. It is beneficial in a visual tracking system for the richness and complexity. When a predefined class of objects needs to be tracked other information such as shape cues can be involved to extend the model for various scenarios.

Sruti Das Choudhury, Tardi Tjahjadi, 2013, this paper presents a three-phase gait recognition method that analyses the spatio-temporal shape and dynamic motion (STS-DM) characteristics of a human subject's silhouettes to identify the subject in the presence of most of the challenging factors that affect existing gait recognition systems. In phase 1, phase-weighted magnitude spectra of the Fourier descriptor of the silhouette contours at ten phases of a gait period are used to analyze the spatio-temporal changes of the subject's shape. A component-based Fourier descriptor based on anatomical studies of human body is used to achieve robustness against shape variations caused by all common types of small carrying conditions with folded hands, at the subject's back and in upright position. In phase 2, a full-body shape and motion analysis is performed by fitting ellipses to contour segments of ten phases of a gait period and using a histogram

matching with Bhattacharyya distance of parameters of the ellipses as dissimilarity scores. In phase 3, dynamic time warping is used to analyze the angular rotation pattern of the subject's leading knee with a consideration of arm-swing over a gait period to achieve identification that is invariant to walking speed, limited clothing variations, hair style changes and shadows under feet. The match scores generated in the three phases are fused using weight-based score-level fusion for robust identification in the presence of missing and distorted frames, and occlusion in the scene. Experimental analyses on various publicly available data sets show that STS-DM outperforms several state-of-the-art gait recognition methods.

#### **3.4.4 Neural network in gait recognition**

Prabhsharan Kaur et al., 2015, used Gait recognition is the main field of biometric authentication system. Gait is a particular way of moving on foot. In this various approaches has been utilized for the process of gait recognition. On the basis of energy and enthalpy level available in different images these approaches utilized for feature extraction from silhouette Image's. Energy and enthalpy does not provide accurate information about gait. Gait recognition has main issue of the proper feature extraction due to variance in a gait cycle. To remove this issue in the field of gait recognition process the approach has to utilize which extract optimal feature for gait recognition process.

Adam Świtoński et al., 2015, We have proposed and evaluated features extracted from the gait paths based on the data from a motion capture for human identification task. We have used the following paths: skeleton root element, feet, hands and head. We have collected motion gait database containing 353 different motions of 25 actors. We have proposed four approaches to extract features from motion clips: statistical, histogram, Fourier transform and timeline We have prepared motion filters to reduce the impact of the capturing location and actor's height on the gait path and the method of steps detection. We have applied supervised machine learning techniques to classify gaits described by the proposed feature sets. We have prepared scenarios of the features selections for every approach

and iterated classification experiments. On the basis of obtained classifications results we have discovered most remarkable features for the identification task. We have achieved almost 97% identification accuracy for reliable normalized paths.

### **3.5 SUMMARY**

The literature has shown the usage of gait analysis in clinical analysis. The human movement analysis with fixed labeling of body parts and free body movement labeling are used. The angular movements of the different parts of the body considering leg and hand are very much used for gait analysis. Mostly close laboratory experiments have been considered in many PhD thesis. Standard database like Crouch dataset have been used. Many researchers have used hidden Markov model, Gabor, discrete wavelets, LDA, PCA for extracting representative features. All these analysis have been carried out only for controlled movement of the human walk in laboratory. Each feature extraction methods claim superior over other methods in different literature. Most of the work reported based on the algorithms like Gabor, wavelets, LDA, PCA which have been used in many another applications other than gait analysis. With regard to neural network and machine learning training, back-propagation, KNN, K-Means algorithms have been used by many researchers. As such counter propagation algorithm has been used by one researcher. However, the extraction of features has not been mentioned. Hence, in this thesis, CROUCH database is considered for implementation. This chapter has presented the different methods used for gait recognition. In each method, the quality of gait recognition is achieved to a limited extent. This has given scope for implementing Neural Networks (BP) for gait recognition.

## CHAPTER 4

### DESIGN AND IMPLEMENTATION

#### 4.1 Proposed Work

Figure 1 states the methodology for proposed system. It classifies the human locomotion as normal or pathological GAIT. The figure consists of training phase and testing phase. In the initial phase while doing training the dataset is fed with GAIT dataset (crouch2, crouch3, crouch4, normal) then the classification is done using neural network (back propagation), KNN and K-Means techniques. Then these techniques are compared. In testing phase, the testing input is given to classifier and classification system tells that human locomotion is normal or abnormal.

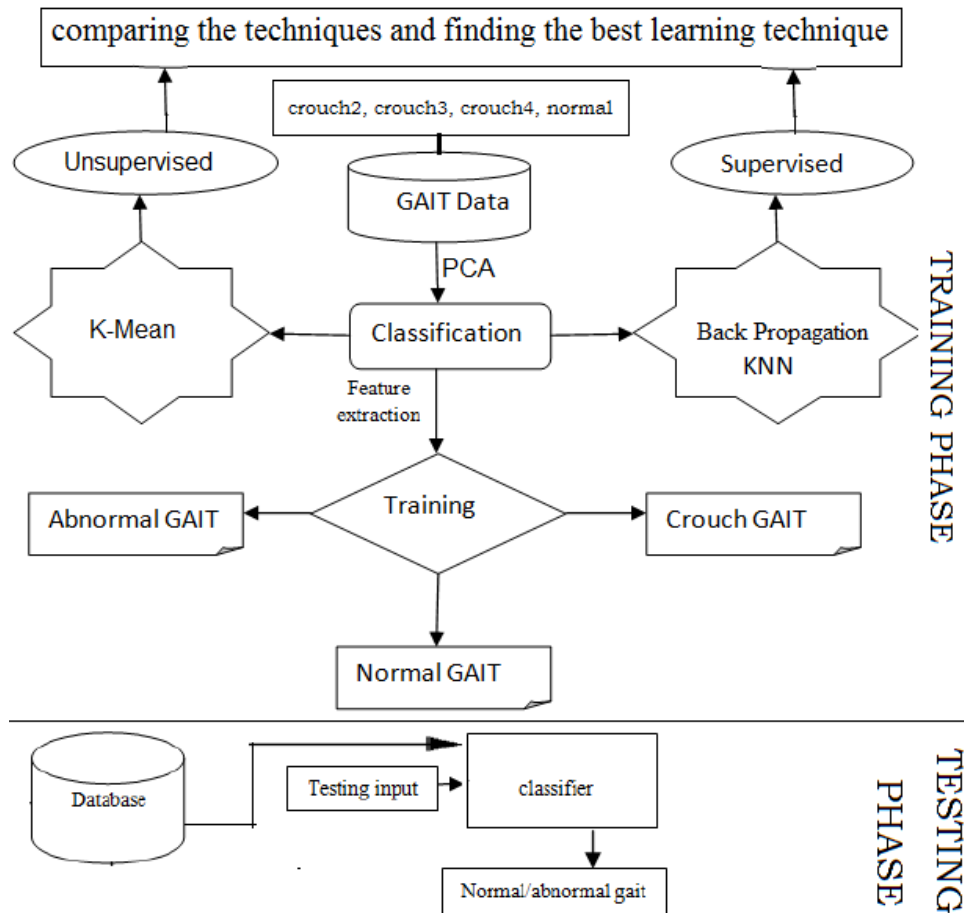


Fig. 1.Proposed GAIT classification system

## **4.2 Methodology**

Firstly we collect the GAIT data then features are extracted and we classify the GAIT dataset. Training and testing of system is done. After the classification is done result is shown as normal or pathological GAIT. Then 10 fold cross-validation test also conducted to validate the statistical significance of the results.

## **4.3 Data Collection**

### **4.3.1 HMDC**

To capture the human data we use HMCD. From this device different joint angles (knee, hip, and ankle) are captured to get the human GAIT data.

We capture the real human data using our developed Human Motion Capture Device (HMCD) suit. Fig.2 is the design of HMCD wearing by left hand object. We captured the different joint angles change (knee, hip, and ankle). The control reverse torques can be computed for the joints in a bipedal humanoid using the below equations:

$$\tau = M(\theta) \theta'' + C(\theta, \theta') + G(\theta)$$

Where:  $M(\theta)$  -inertial torque,  $C(\theta)$  - centripetal and coriolis forces,  $G(\theta)$  - gravitational force.

HMCD have following parts six potentiometers (100 K $\Omega$ ) for six joint Left and right Hip, Knee, and Ankle. When a force is applied, each joint move with certain angle, aluminum link with potentiometer attached in each joint will also move with certain degree (0° to 300°). These force and angle is measure in each reading by using FSR (force sensor) and potentiometer respectively. It gives value in digital counts (within range 0 to 999) from which we can further convert it into angular values using the given formula [6], given by equation (7) and (8).

$$\text{Angle (degree)} = [(\Theta - \Theta_0) \times 300/1000]$$

$$\text{Force (Newton)} = f \times 9.8/1000$$

Where,  $f$  = force in digital counts.

Force is applied from only one direction, due to limited area of force sensor, the range of FSR digital count is 1N to 100N and the sensing area of FSR 3105 is 14.5 cm<sup>2</sup>. We used a wooden hammer like structure to give push forces.



Fig. 2. Subject with HMCD Suit (a). Frontal (b). Back view

#### **4.3.2 Data Set**

The data set used for proposed system contains four data set namely “crouch2”, “crouch3”, “crouch4”, and “normal”. Each data set has 51 gaits and 16 features. Therefore to the classification system we pass  $51 \times 4 = 204$  gaits and train the classifier using machine learning techniques. Figure 3 shows the human GAIT of four different datasets. In figure 3 ‘a’, ‘b’, ‘c’ shows some abnormalities in human locomotion where as ‘d’ shows normal GAIT.



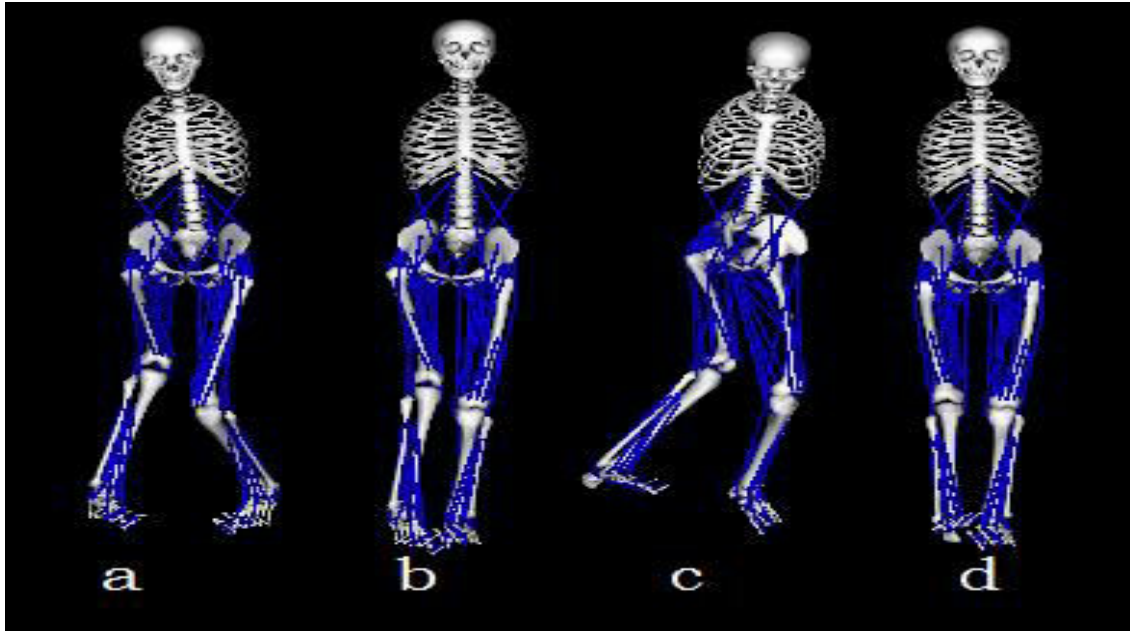


Fig. 3. View of the human GAIT of four different data sets

The data set when simulated in openSim, the human dataset motion has values and speed that can be seen in Figure 4, 5, 6, 7 which are captured while locomotion.

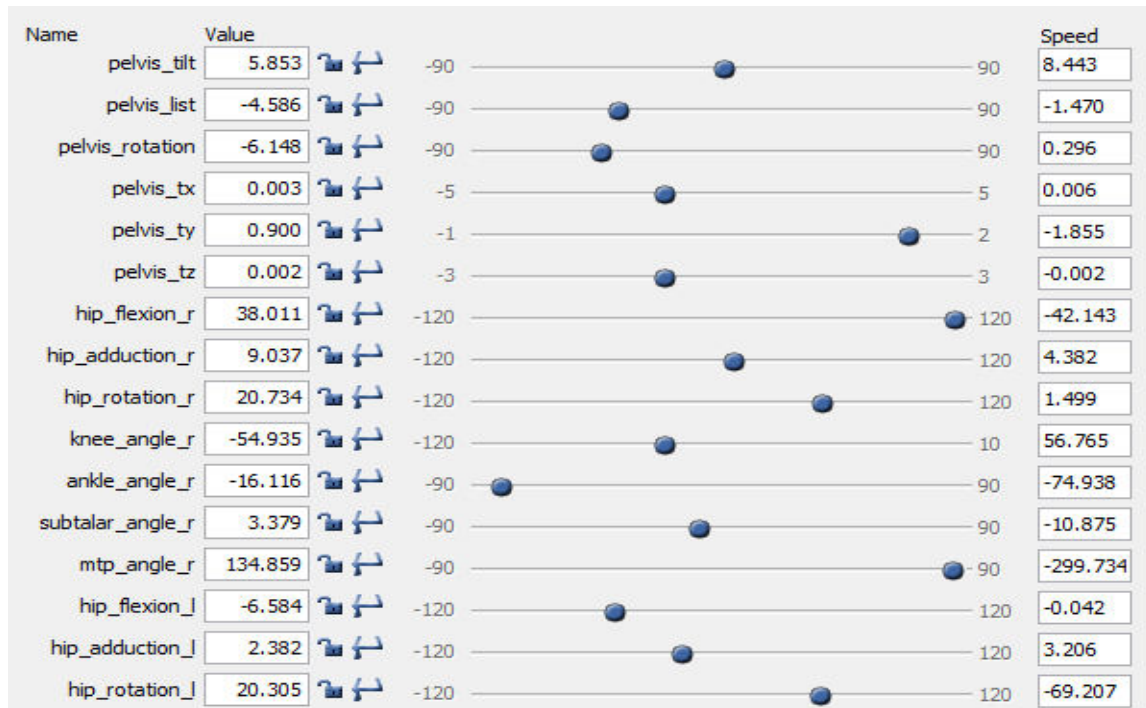


Fig. 4. Coordinates of crouch2 dataset

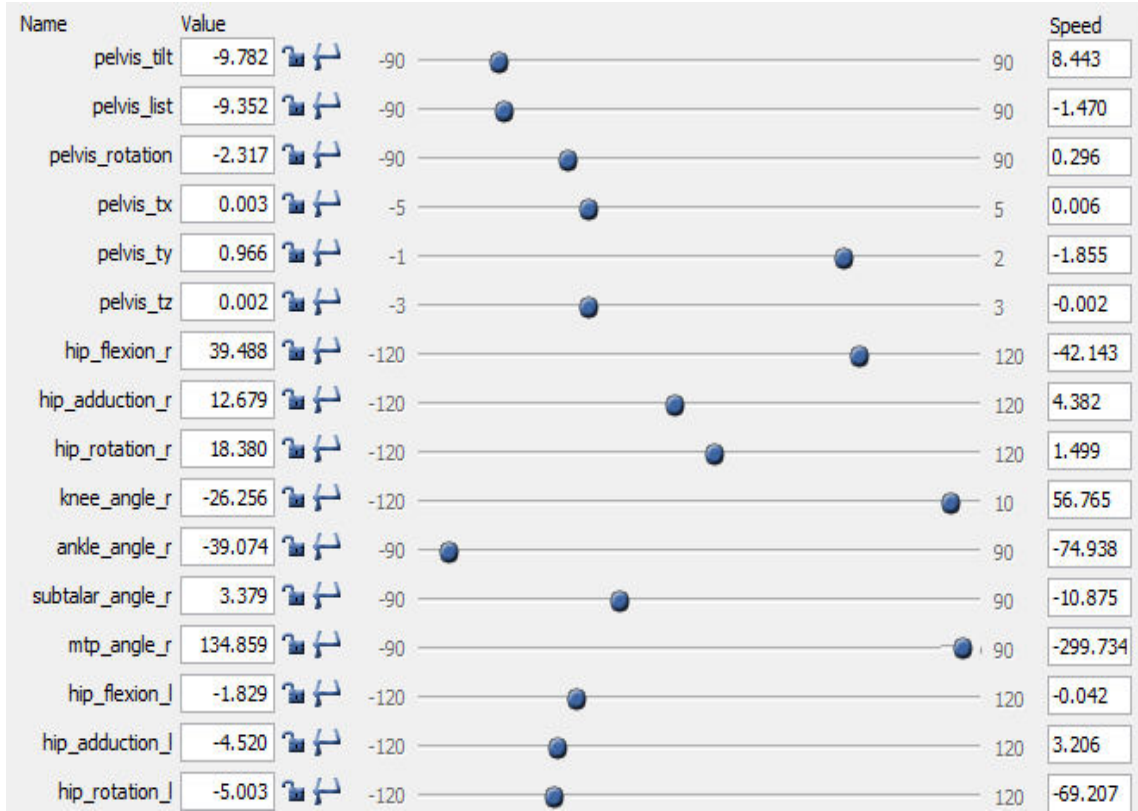


Fig. 5. Coordinates of crouch3 dataset

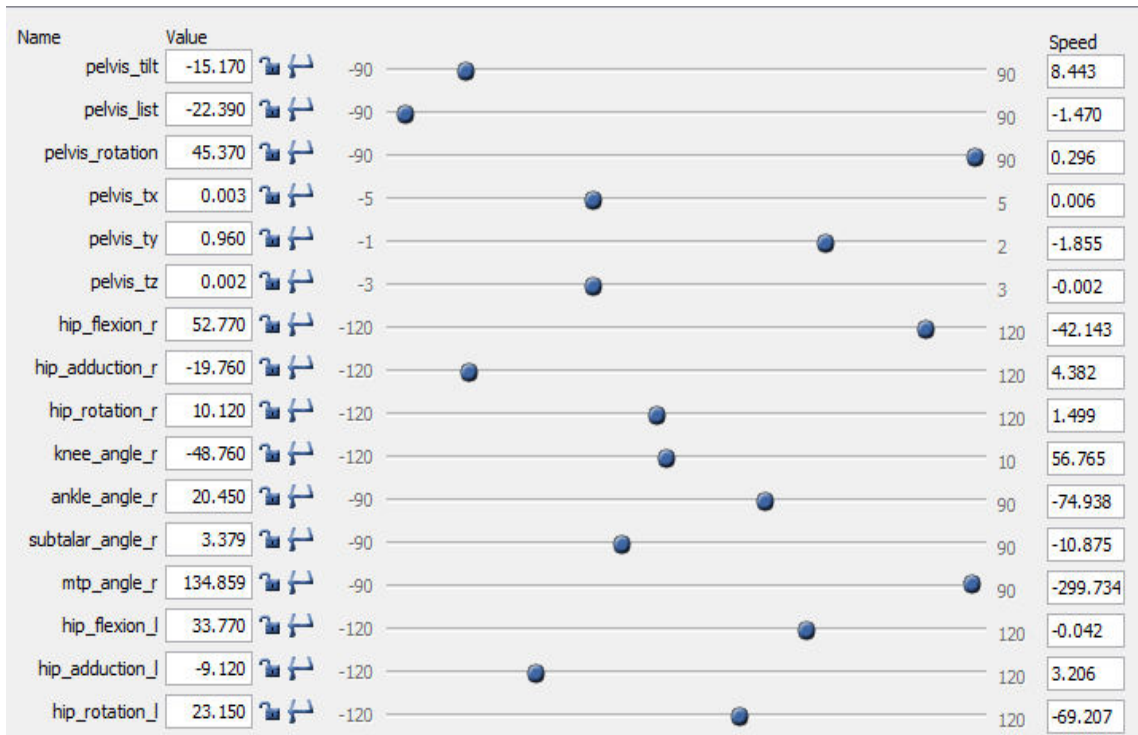


Fig. 6. Coordinates of crouch4 dataset

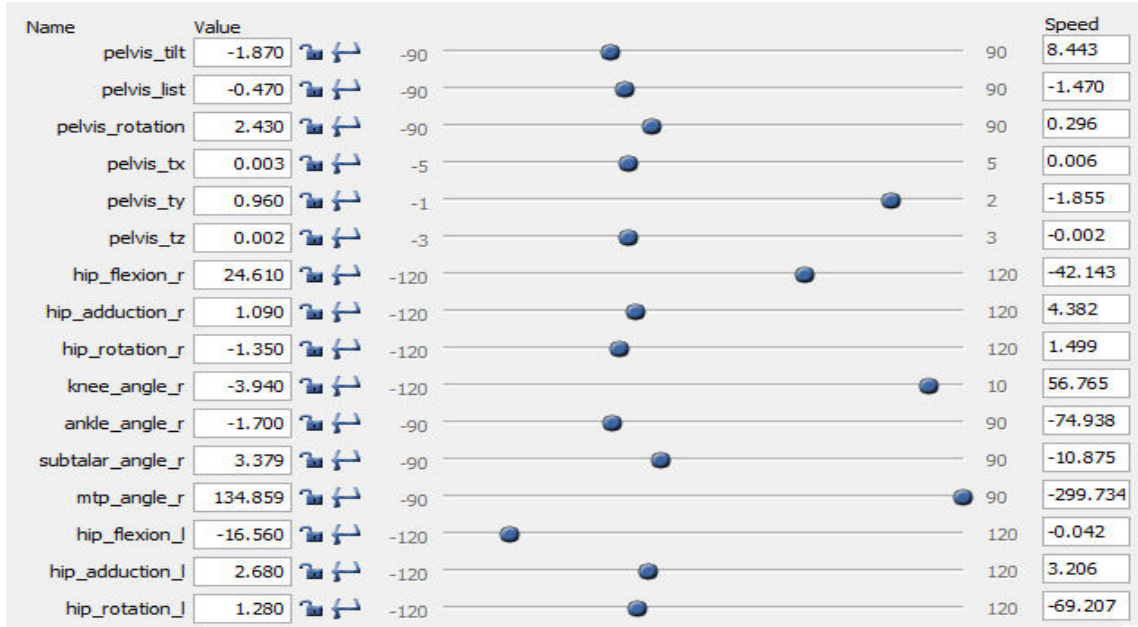


Fig. 7. Coordinates of normal dataset

When the figure.3 is simulated, then the coordinates of the human GAIT data set is captured and shown in figure.4, 5, 6, 7 for crouch2, crouch3, crouch4, normal datasets respectively.

Figure 8 shows plot of knee angle curve of data set crouch2 (red), crouch3 (blue), and crouch4 (green), normal (purple) respectively.

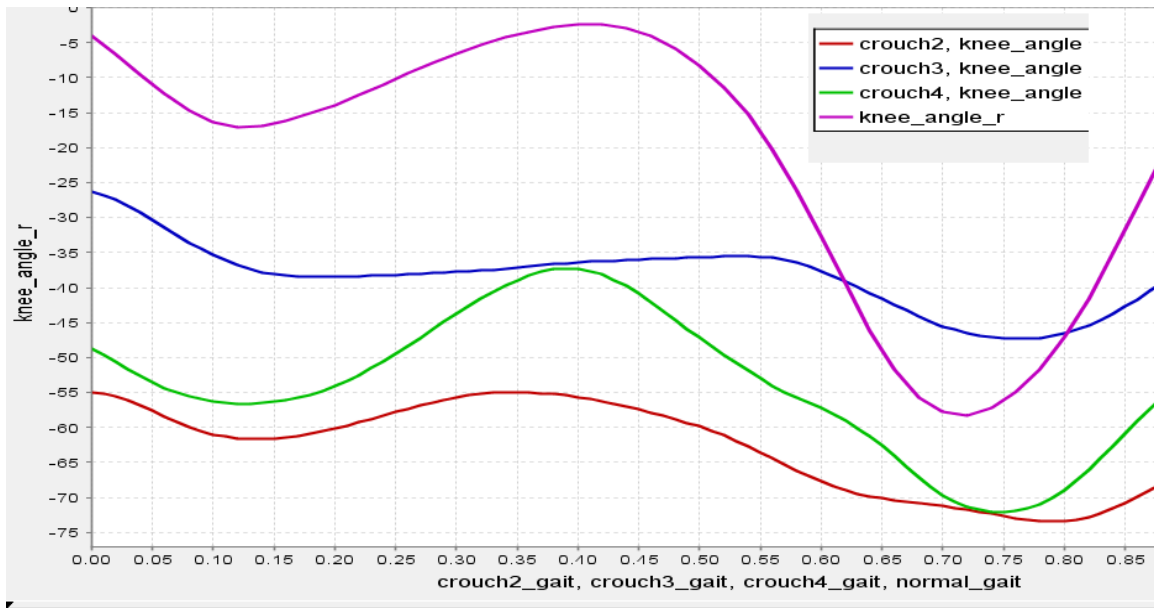


Fig. 8. Plot of crouch2, crouch3, crouch4, and normal datasets respectively.

### **4.3.3 Feature Selection**

Here 16 features are extracted. The features are collected from different joints while locomotion [1]. In all the four datasets 16 features are extracted which is listed in table 1.

TABLE I. list of 16 features of gait dataset

| Feature Category | Feature Name    |
|------------------|-----------------|
| F1               | Pelvis_tilt     |
| F2               | Pelvis_list     |
| F3               | Pelvis_rotation |
| F4               | Pelvis_tx       |
| F5               | Pelvis_ty       |
| F6               | Pelvis_tz       |
| F7               | hip_flexion_r   |
| F8               | hip_adduction_r |
| F9               | hip_rotation_r  |
| F10              | knee_angle_r    |
| F11              | ankle_angle_r   |
| F12              | ankle_angle_l   |
| F13              | knee_angle_l    |
| F14              | hip_flexion_l   |
| F15              | hip_adduction_l |
| F16              | hip_rotation_l  |

This chapter describes the extraction of individual features of the acquired gait data. During the development, ample priority has been given to computationally efficient algorithms.

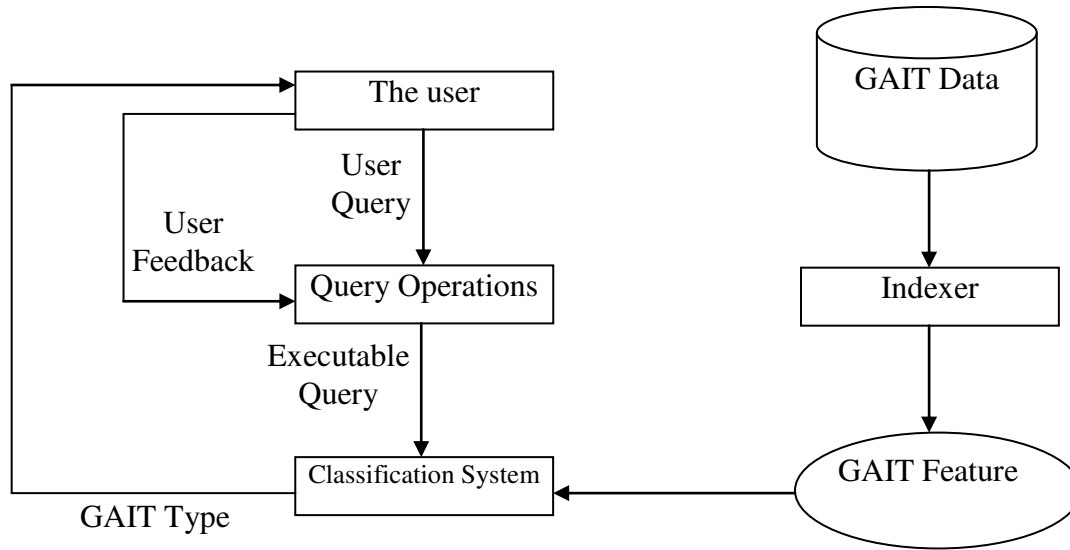


Fig. 9. Example of a scatter plot

Numerous different ways of extracting discriminative features from gait sequences have been proposed. Despite the broad variability of the methods they can be divided into two distinct groups: (1) systems that need to locate one step or a complete gait cycle and (2) systems that do not need to locate the gait cycle within the acquired data.

#### What Features should be Used?

In order to achieve a high recognition rate, discriminative features must be extracted from the available data in figure 9. From a naive point of view, it would seem obvious to try to recognise people through their stride length, walking cadence, body weight, body height, and so forth. However tempting the aforementioned features might be, they are not ideal for a biometric system. In fact, they are highly insecure due to their static nature that allows an impostor to easily mimic them; e.g. an impostor can easily adjust his body weight, stride length, and cadence to match that of a legitimate user and try to gain access to a restricted area.

Conversely, the dynamic properties of human gait are far more difficult to imitate, since they depend on physiological properties of the user's body, such as bone structure. The use of dynamic properties of human gait is therefore considered in this thesis.

#### **4.3.4 Data Correction and Smoothing**

The ideal curve for hip, knee and ankle are disturbed due to push applied from behind, so we smoothed the data using cubic spine. The reading initial reading which we captured has certain value initially due to noise, so we corrected data using zero correction i.e. subtracted initial value from the all measured value.

#### **4.4 Classification**

Classification is a collection of different methods for categorizing data. The problem of classification can be illustrated in a scatter plot as seen in figure 10. To correctly classify the data one wishes to find features that separate each group of data into clusters. Before classification it is common practice to normalize the data. Doing so will make the distances in different directions to carry the same weight, i.e. assuming that all features have the same importance.

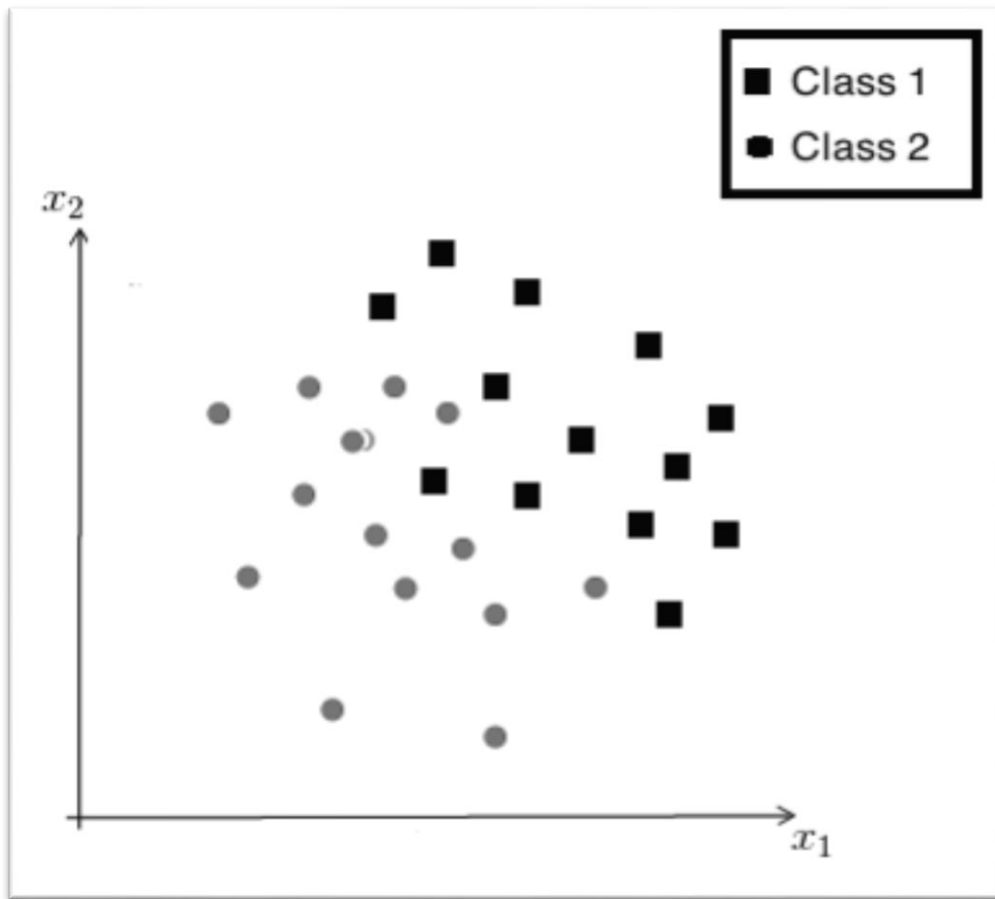


Fig. 10. Example of a scatter plot showing two features  $x_1$  and  $x_2$  plotted against each other. Circles belong to one class and squares to another class

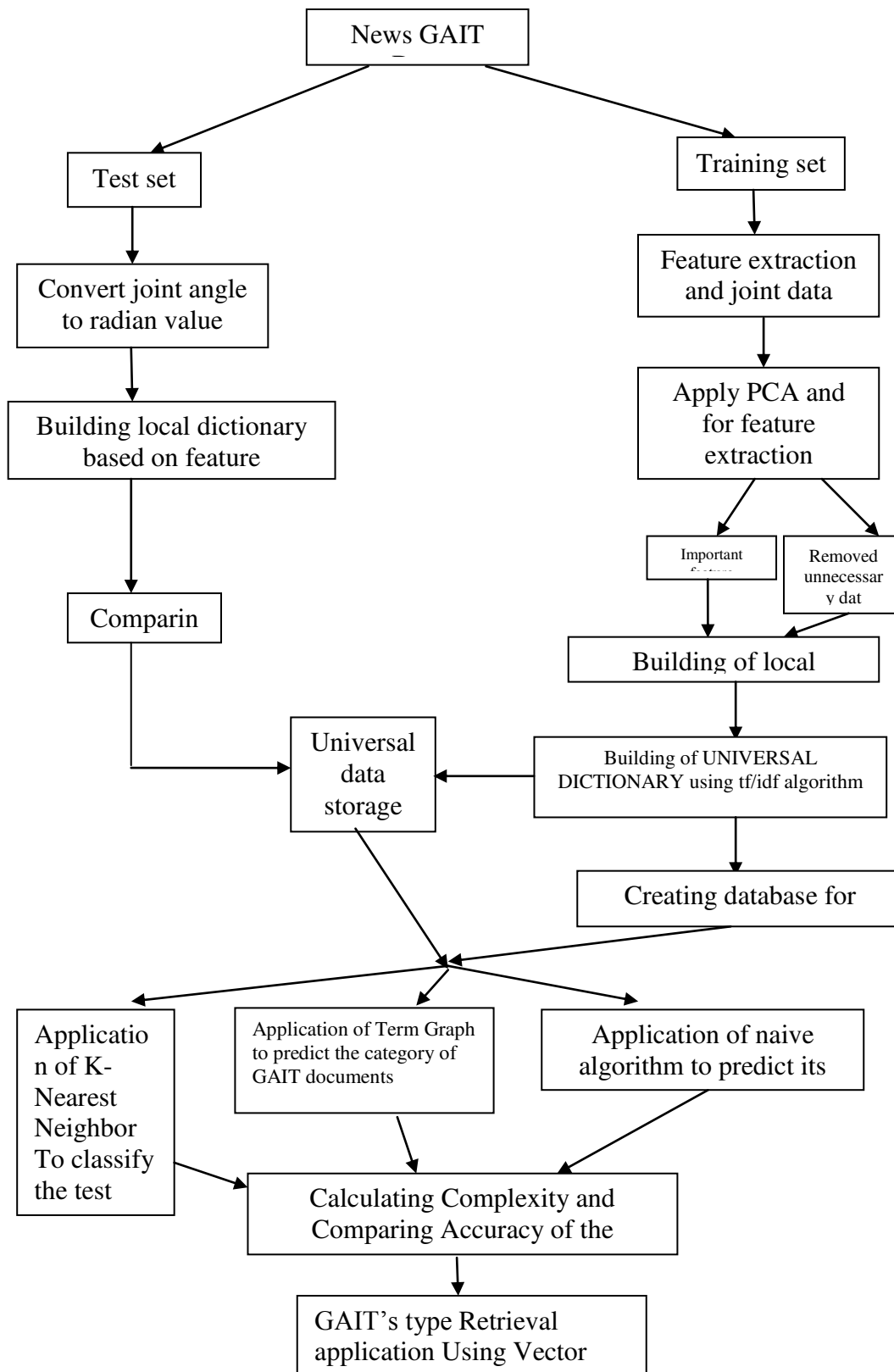


Fig. 11. Flow chart of the GAIT Data Classification

#### **4.4.1 Classification Techniques**

Classification of GAIT data is done through two following techniques:

- i. KNN
- ii. BP
- iii. K-Means

Several classifiers have been used for classification, e.g. K-Nearest Neighbor, (KNN), Neural networks (Back Propagation), and K-Means.

According to their results Back Propagation outperformed the other methods which are why it is the chosen method for this project

##### **4.4.1.1 Neural Network (Back Propagation)**

Back propagation, an abbreviation for "backward propagation of errors", is a common method of training artificial neural networks used in conjunction with an optimization method such as gradient descent. The method calculates the gradient of a loss function with respect to all the weights in the network. The gradient is fed to the optimization method which in turn uses it to update the weights, in an attempt to minimize the loss function.

Back propagation requires a known, desired output for each input value in order to calculate the loss function gradient. It is therefore usually considered to be a supervised learning method, although it is also used in some unsupervised networks such as auto encoders. It is a generalization of the delta rule to multi-layered feed forward networks, made possible by using the chain rule to iteratively compute gradients for each layer. Back propagation requires that the activation function used by the artificial neurons (or "nodes") be differentiable.

The back propagation learning algorithm can be divided into two phases: propagation and weight update.

##### **I. PHASE 1: PROPAGATION**

Each propagation involves the following steps:



1. Forward propagation of a training pattern's input through the neural network in order to generate the propagation's output activations.
2. Backward propagation of the propagation's output activations through the neural network prout uses the training pattern target in order to generate the deltas (the difference between the targeted and actual output values) of all output and hidden neurons.

## **II. Phase 2: Weight update**

For each weight-synapse follow the following steps:

1. Multiply its output delta and input activation to get the gradient of the weight.
2. Subtract a ratio (percentage) of the gradient from the weight.

This ratio (percentage) influences the speed and quality of learning; it is called the *learning rate*. The greater the ratio, the faster the neuron trains; the lower the ratio, the more accurate the training is. The sign of the gradient of a weight indicates where the error is increasing; this is why the weight must be updated in the opposite direction.

Repeat phase 1 and 2 until the performance of the network is satisfactory.

### **Algorithm**

The following is a stochastic gradient descent algorithm for training a three-layer network (only one hidden layer):

1. initialize network weights (often small random values)
2. **do**
3.     **forEach** training example named ex
4.         prediction = neural-net-output(network, ex) // forward pass
5.         actual = teacher-output(ex)
6.         compute error (prediction - actual) at the output units
7.         compute delta  $w_h$  for all weights from hidden layer to output layer // backward pass
8.         compute delta  $w_i$  for all weights from input layer to hidden layer // backward pass continued
9.         update network weights // input layer not modified by error estimate

10. **until** all examples classified correctly or another stopping criterion satisfied
11. **return** the network

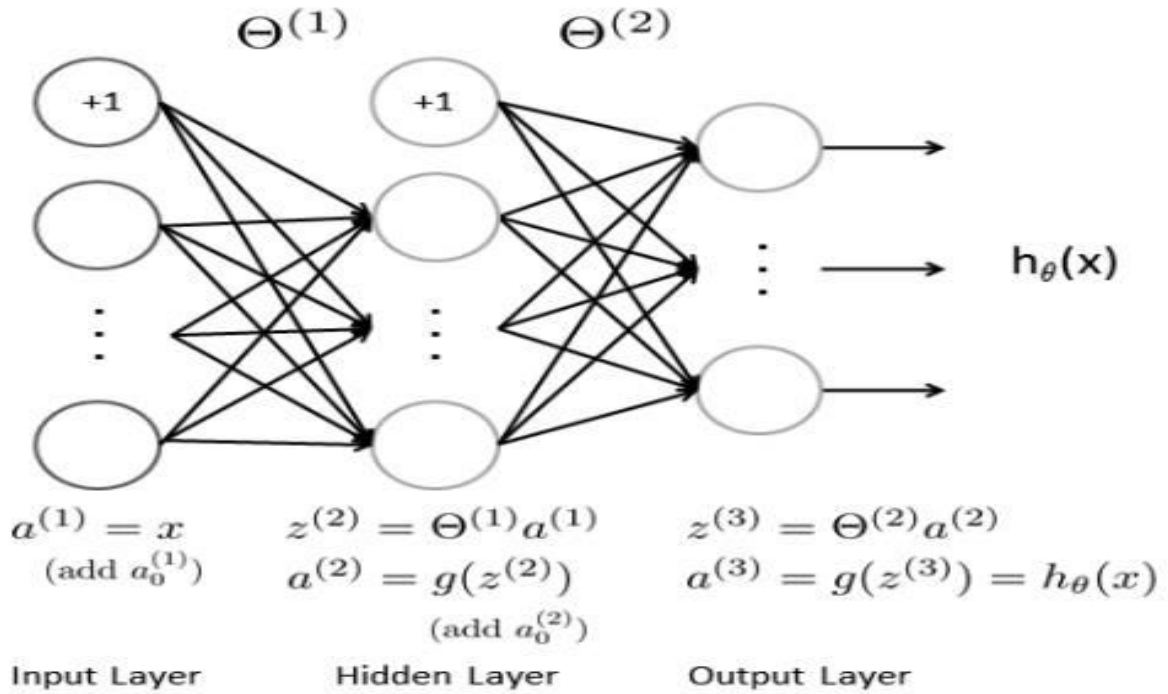


Fig. 12. Neural network model

In this technique we train the system with 16 input neurons in the input layer and we get 4 neurons in the output layer. For adjusting the weights of the network, training is done by back-propagation algorithm. The Accurate classification (training and testing) of back propagation comes out to be 69.825%

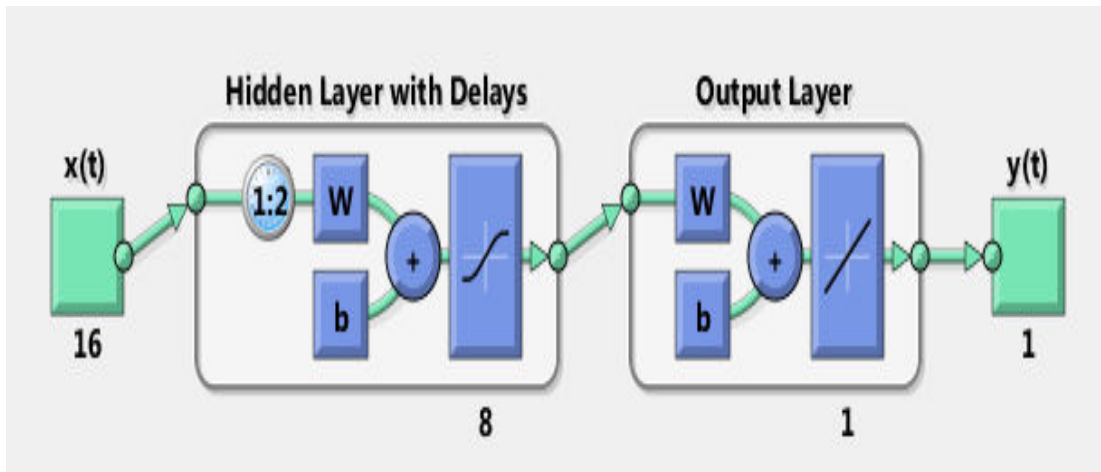


Fig. 13. Model of Back Propagation algorithm

Now the back propagation algorithm is used to train and test the system. Training of system shows 100% accuracy and testing of system shows 43% accuracy in figure 14.

Figure 14 shows accuracy percentage on y axis and number of training and testing iterations are performed on x axis.

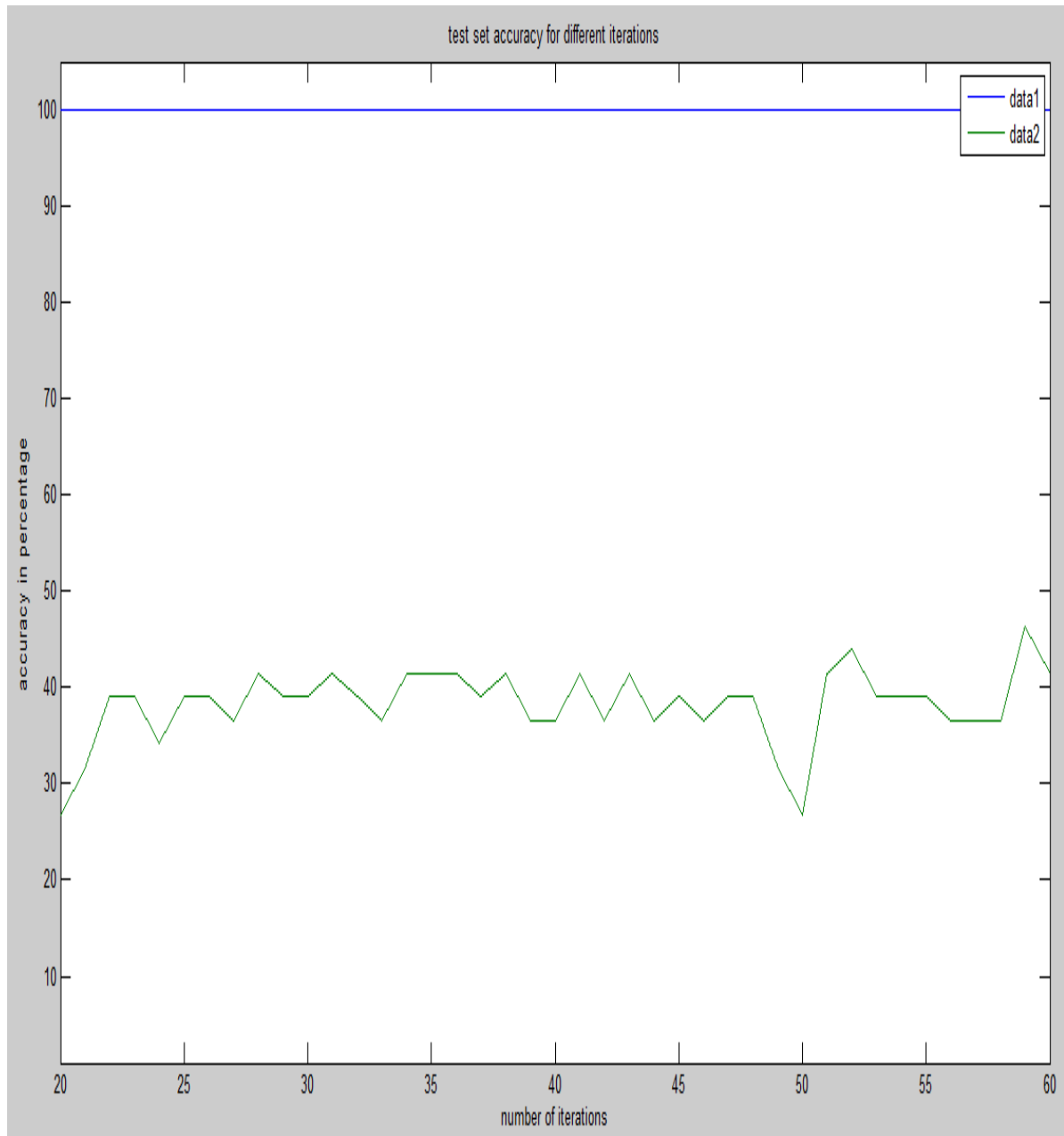


Fig. 14. Training(blue line-100%) and testing(green line-43%) accuracy of back propagation

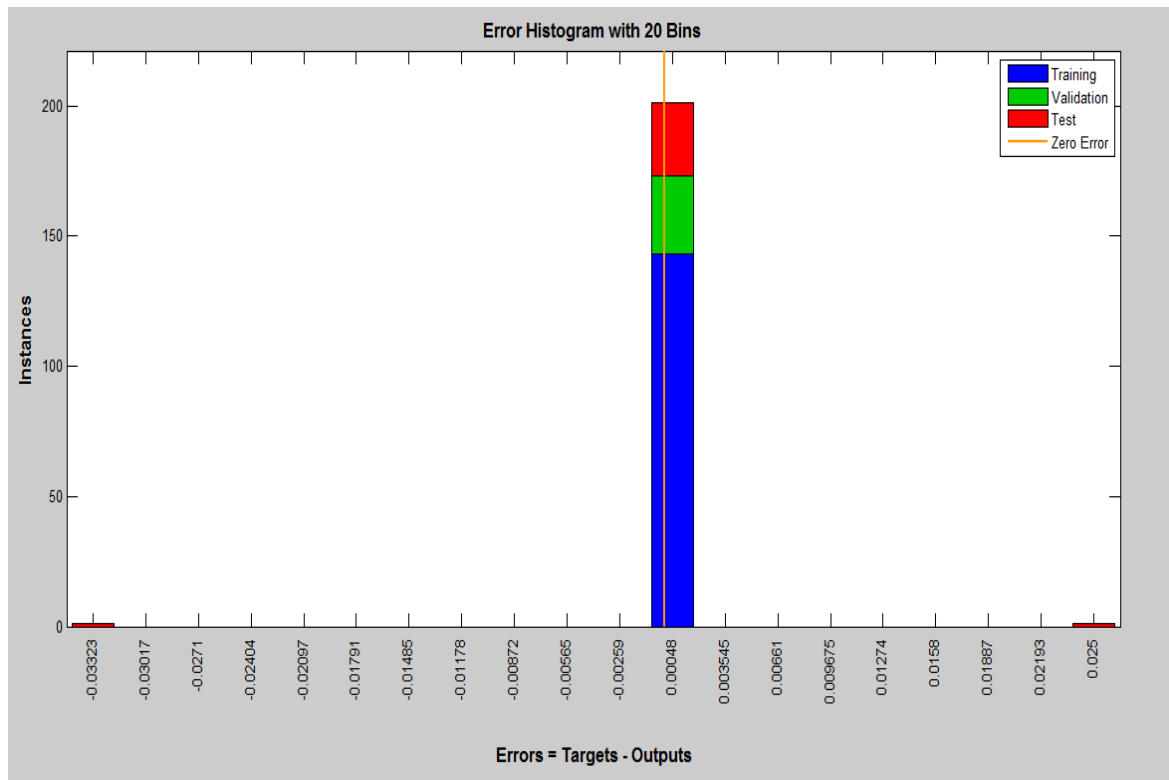


Fig. 15. Error histogram

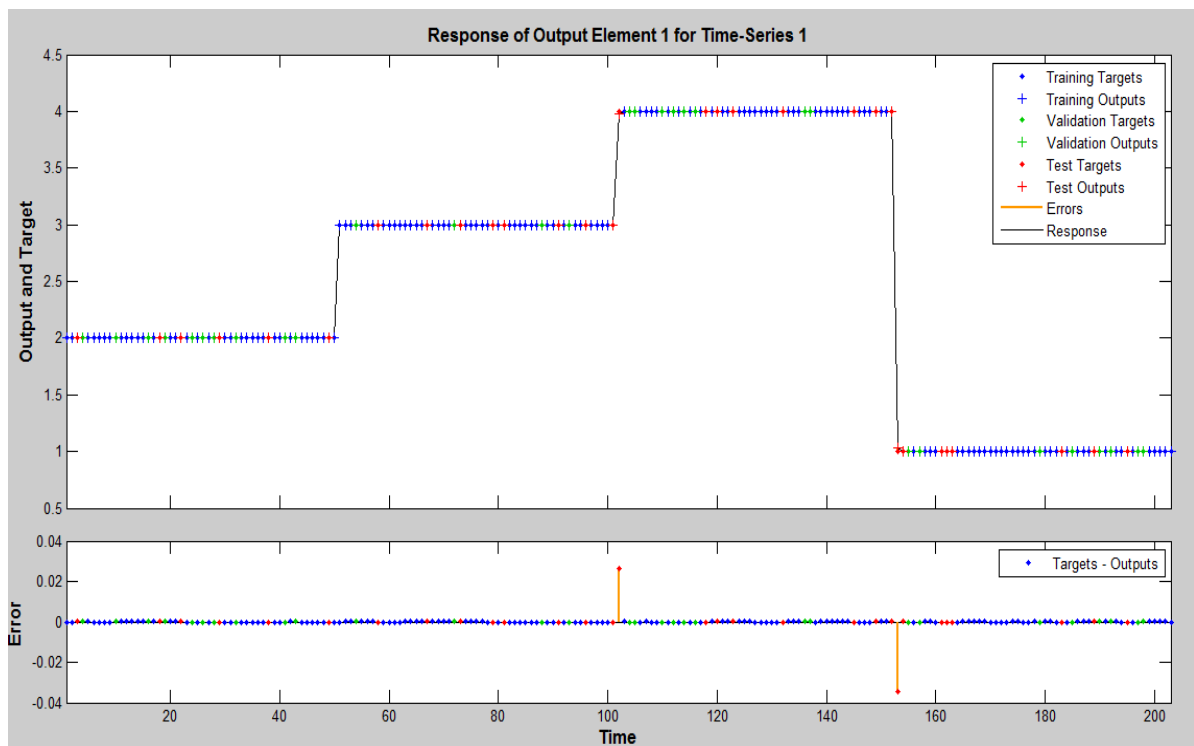


Fig. 16. Plot response

Figure 15 shows error histogram in which on y-axis there are instances and on x axis it shows error as output. Figure 16 shows the plot response where on y-axis there are outputs and targets and on x axis is plot against time. Figure 17 shows error autocorrelation diagram. Figure 18 shows performance of the back propagation system and the best validation performance is  $6.1068e-011$  and on y-axis mean square error count is taken. . Figure 19. Gives the training state gradient value= $5.4746e-006$ ,  $\mu=1e-008$  and validation checks=0. Figure 20. Shows at training  $r=1$ ,  $\text{output}=1*\text{target}+8.4e-008$ ; validation  $r=1$ ;  $\text{output}=1*\text{target}+-8.8e-007$ ; test  $r=0.99998$ ,  $\text{output}=1*\text{target}+0.0062$ ; target  $r=1$ ;  $\text{output}=1*\text{target}+0.00094$ . Figure 21 shows Auto Correlation between input 1 and error 1. Figure 22 shows Cross Correlation between input 1 and error 1.

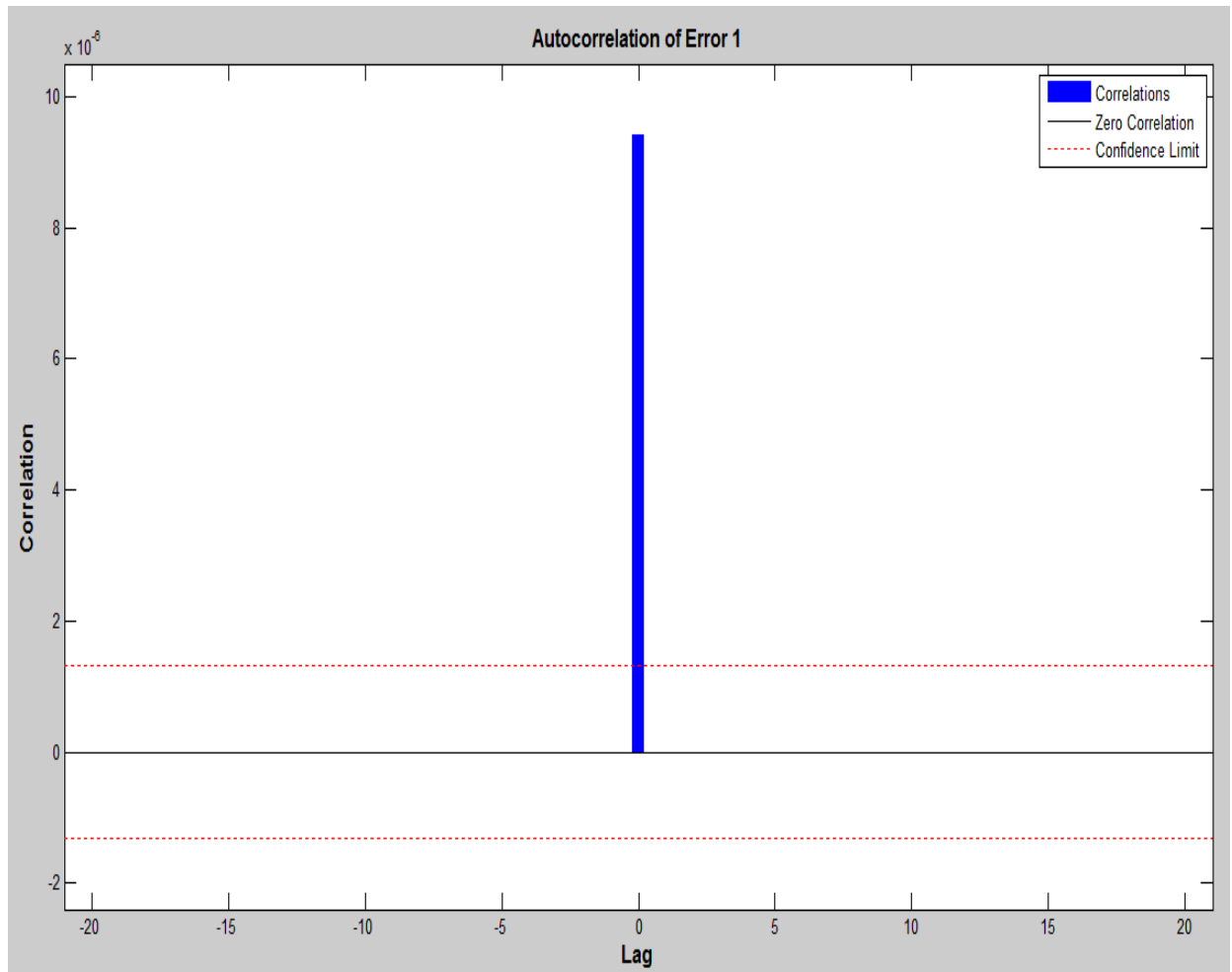


Fig. 17. Error autocorrelation

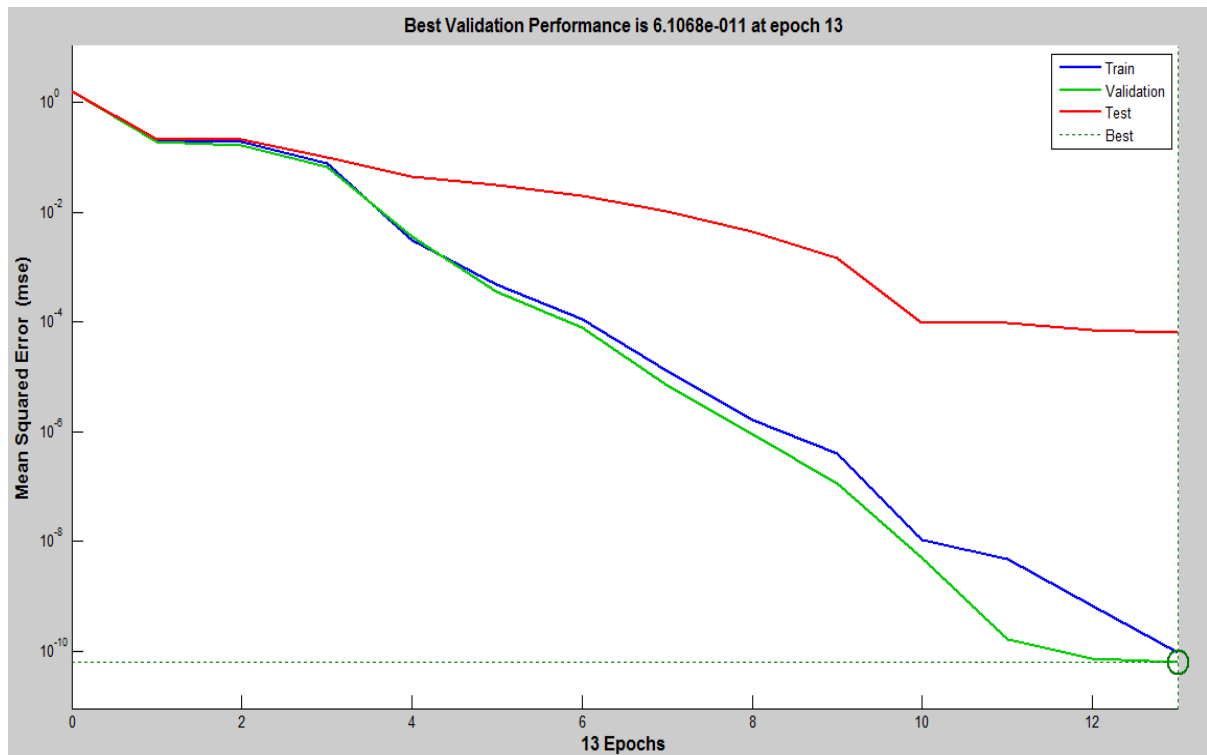


Fig. 18. Model of Back Propagation algorithm

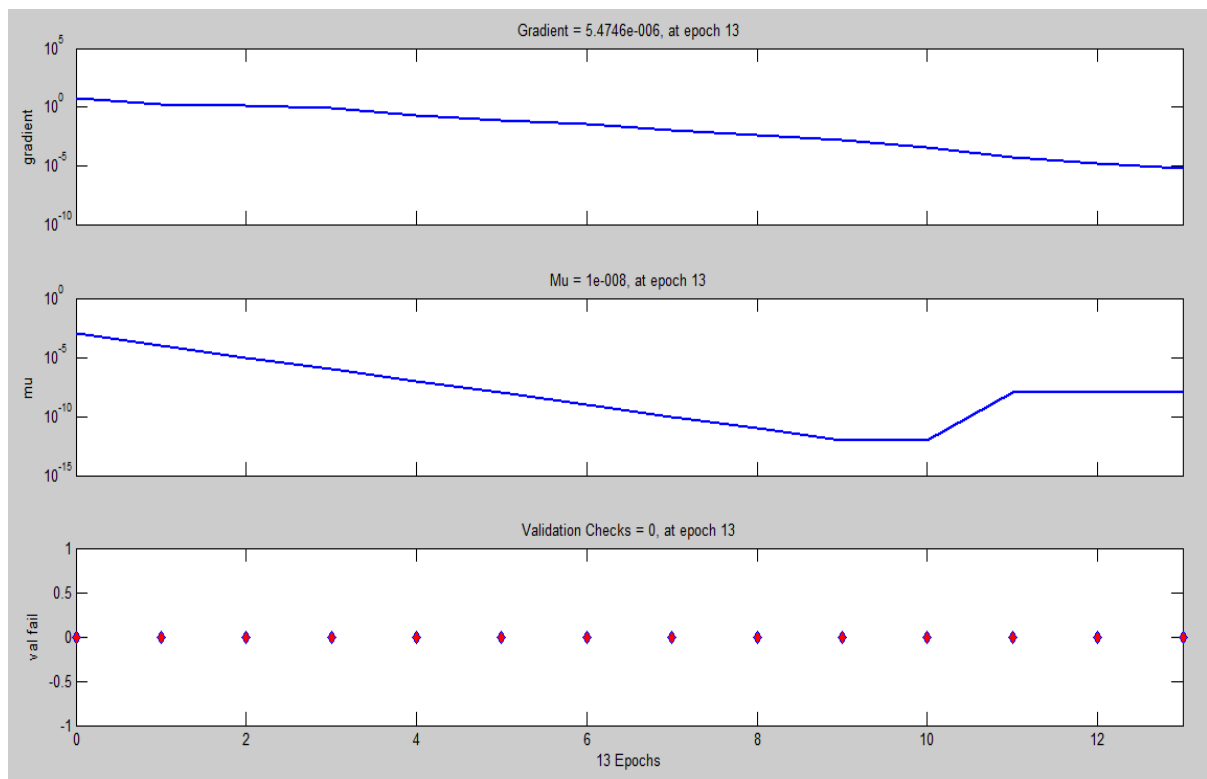


Fig. 19. Training state

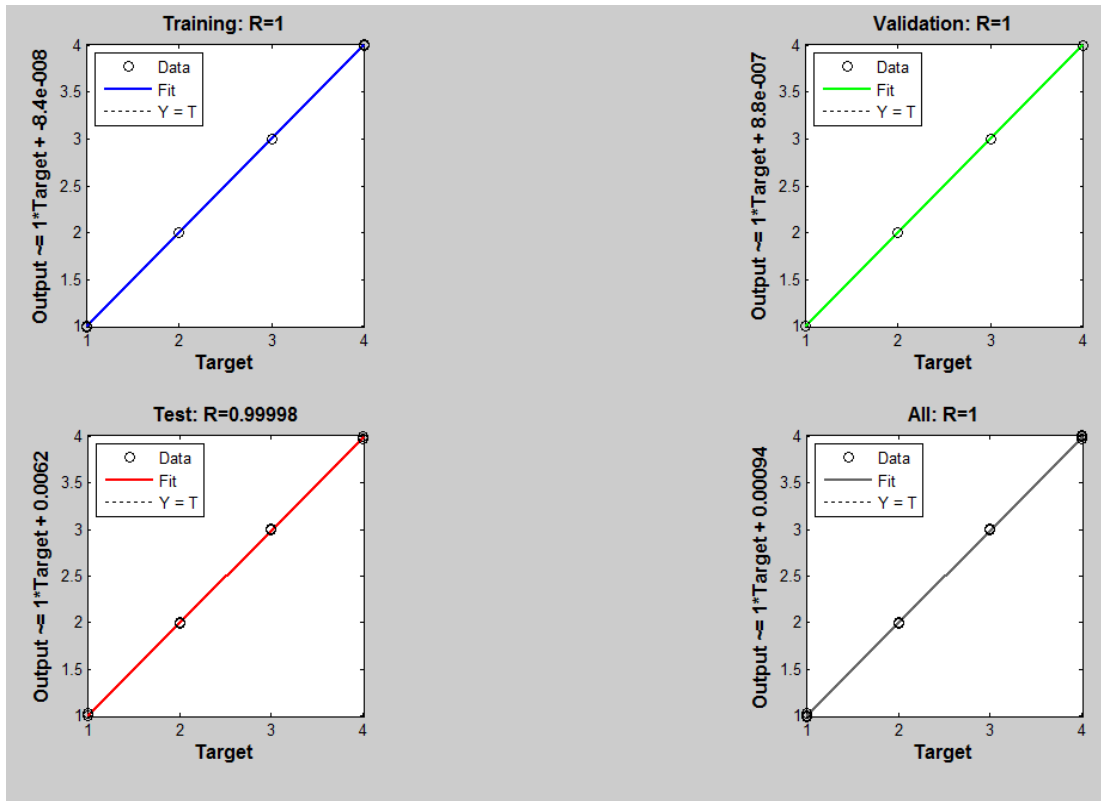


Fig. 20. Regression

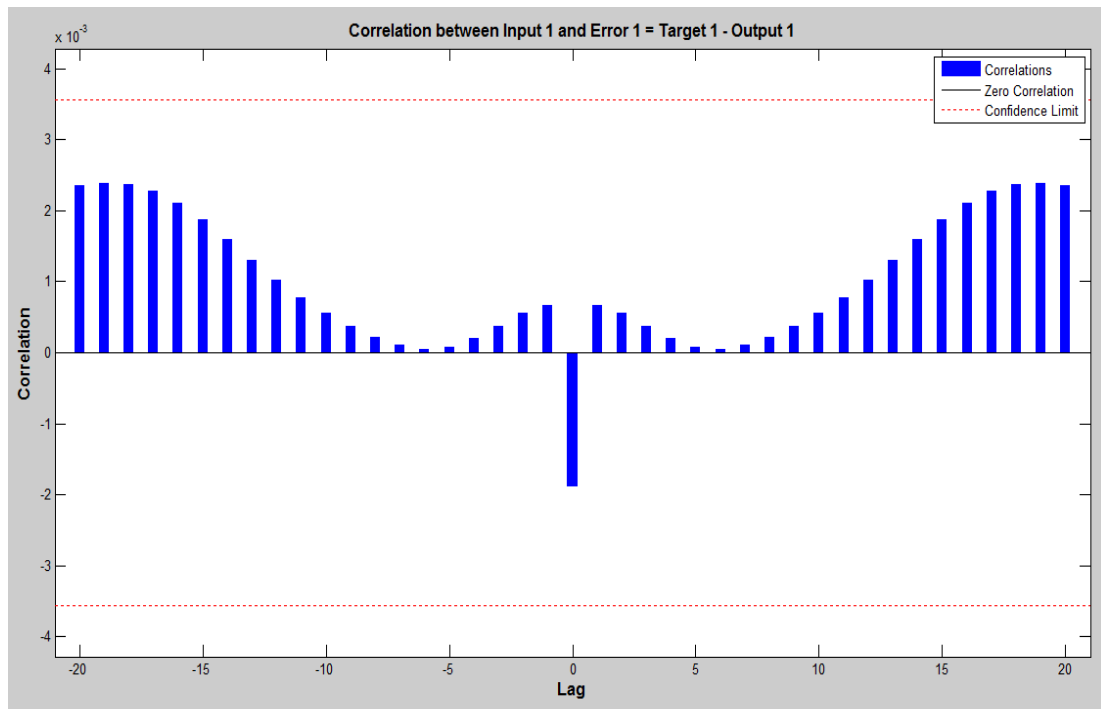


Fig. 21. Auto Correlation between input 1 and error 1

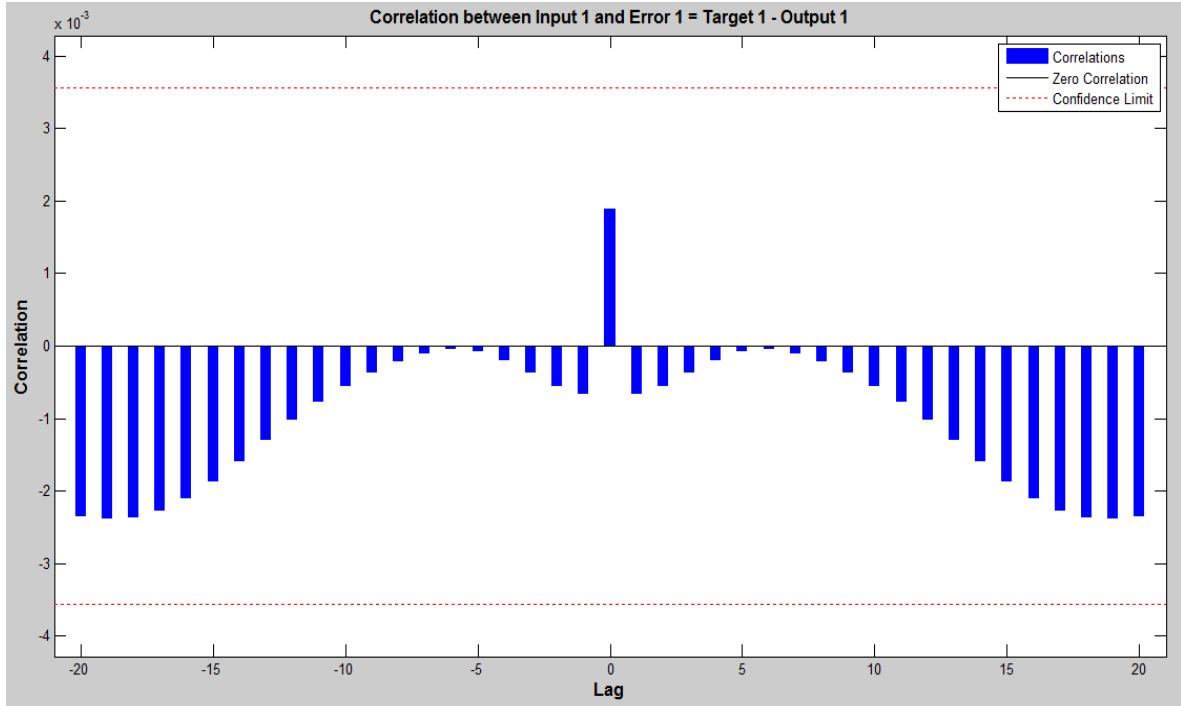


Fig. 22. Cross Correlation between input 1 and error 1

#### 4.4.1.2 KNN

*K-Nearest* Neighbours algorithm (or *k*-NN for short) is a non-parametric method used for classification. In this, the input consists of the *k* closest training examples in the feature space. The output depends on whether *k*-NN is used for classification:

- In *k*-NN classification, the output is a class membership. An object is classified by a majority vote of its neighbours, with the object being assigned to the class most common among its *k* nearest neighbours (*k* is a positive integer, typically small). If *k* = 1, then the object is simply assigned to the class of that single nearest neighbour.

The initial application of *k*-Nearest Neighbors (*k*-NN) to GAIT categorization was reported by Mas and colleagues. The basic idea is to determine the category of a given query based not only on the GAIT data that is nearest to it in the GAIT space, but on the categories of the *k* GAIT that are nearest to it. Having this in mind, the Vector method can be viewed as an instance on the *k*-NN method, where *k*=1. This work uses a vector-



based, distance-weighted matching function, as did Yang, by calculating gait's similarity like the Vector method.

It classifies an unknown sample on the vote of  $k$  nearest neighbor rather than only single neighbor. The centroids of the cluster are recomputed as soon as a new sample joins the cluster.

In KNN classification the number of  $k$  values such as 3, 5, 7 and 9 were used in the training phase. The training and the testing was performed and the accurate classification of KNN (training and testing) comes out to be 66.25%.

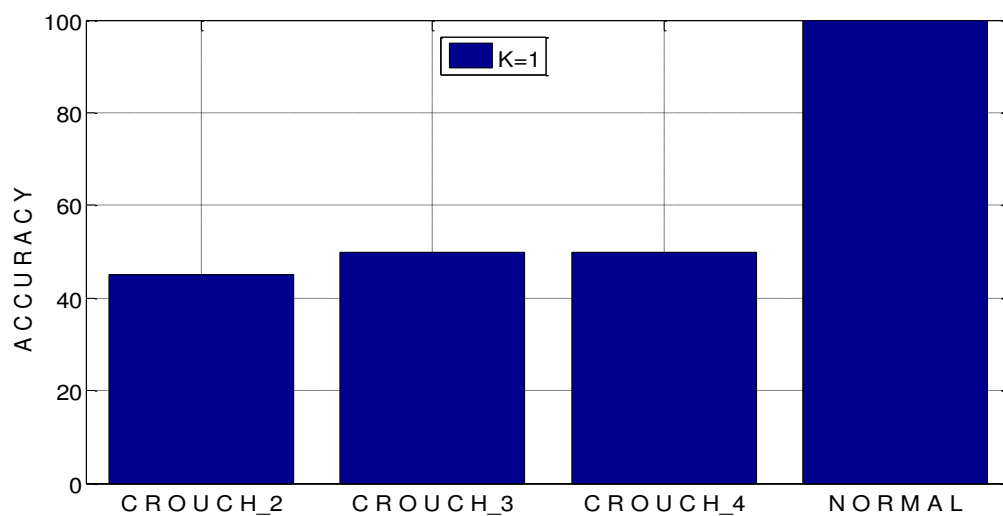


Fig. 23. Accuracy for KNN when  $k=1$

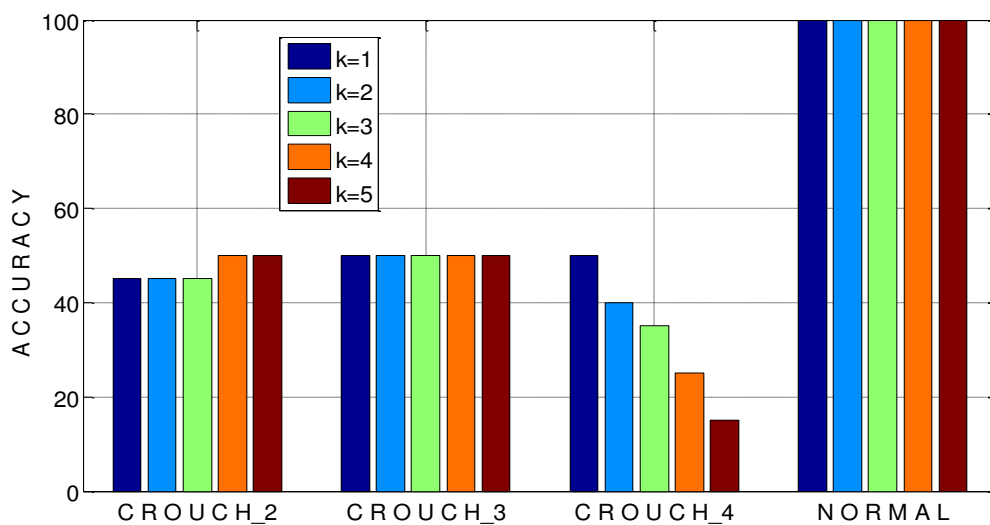


Fig. 24. Accuracy of classification of KNN when  $k=1,2,3,4,5$

TABLE II. Accuracy for KNN

| Category/Method | KNN(k=1,2,3,4,5)    |
|-----------------|---------------------|
| Normal Gait     | 100,100,100,100,100 |
| Crouch 2 Gait   | 50,50,50            |
| Crouch 3 Gait   | 50,50,50            |
| Crouch 4 Gait   | 45,40               |

#### 4.4.1.3 K-Means

- The K-Means algorithm is a method to cluster objects based on their attributes into k partitions.
- It assumes that the k clusters exhibit Gaussian distributions.
- It assumes that the object attributes form a vector space.
- The objective it tries to achieve is to minimize total intra-cluster variance.

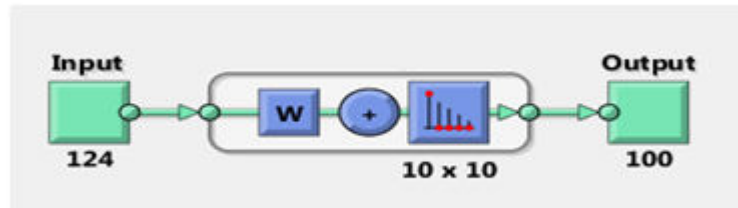


Fig. 25. network architecture

#### Algorithm Fitness Function

The K-Means algorithm attempts to minimize the squared error for all elements in all clusters.

The error equation is:

$$E = \sum_{i=1}^k \sum_{p \in C_i} |p - m_i|^2$$

Where E is the sum of the square error for all elements in the data set; p is a given element; and  $m_i$  is the mean of cluster  $C_i$

### **K-Means algorithm.**

The k-Means algorithm is generally used for compact clustering. It is very much sensitive for outlier and noise. It is applicable only for numerical data.

### **Pseudo Code:**

We have input as Feature vector  $X=\{x_1, x_2, x_3, x_n\}$  where  $X \in R^d$

Number of cluster to be denoted-k.

Convergence threshold z

### **Algorithm steps:-**

BEGIN

1-Select the random vector means

At  $t=0$ ,

Mean would be  $[m_1, m_2, m_3, \dots, m_k]^t$  where k is number of cluster head we decided.

2-Then classify the all raw data according to distance from mean at  $t=0$ .

REPEAT /\* **Untill Convergence criteria not satisfied** i.e. threshold value reached\*/

3-Re compute the mean vector for  $t=t+1$  and classified accordingly.

Update  $t=t+1$ ;

Convergence criteria

Compute Calculate the Euclidean distance  $\|m(t)-m(t-1)\| < z$

Use  $m(t)$  as solution

Else

Go back to step 2.

END LOOP

UNTILL the data not classified/\*classification gait={normal,croach1,croach2,croach3,croach4 }\*/

END

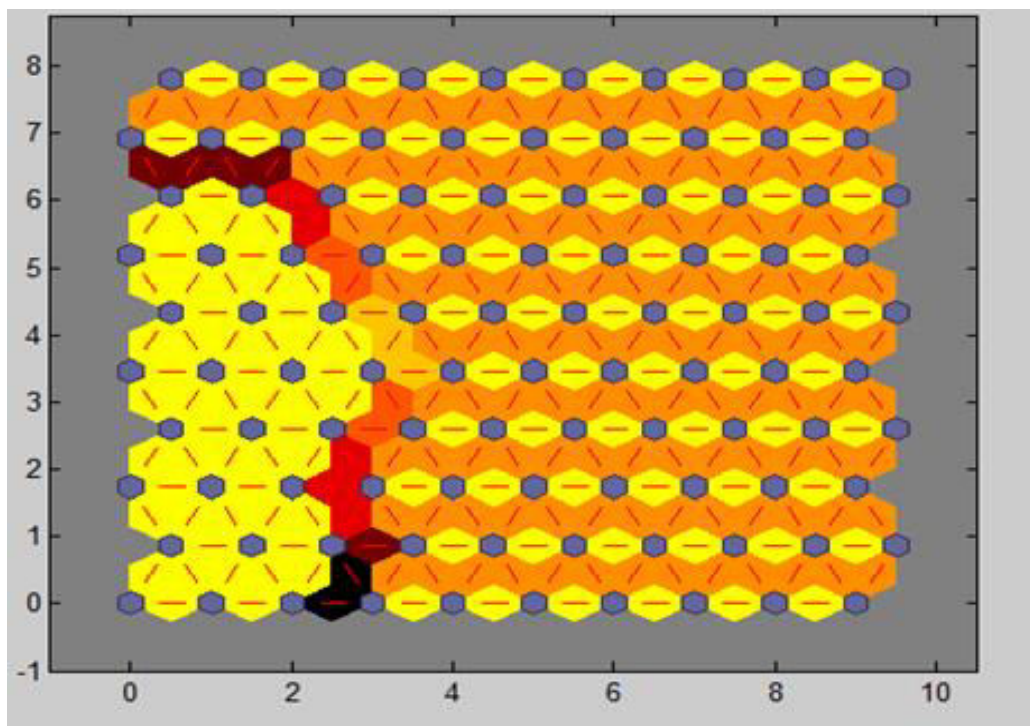


Fig. 26. neighboring distances

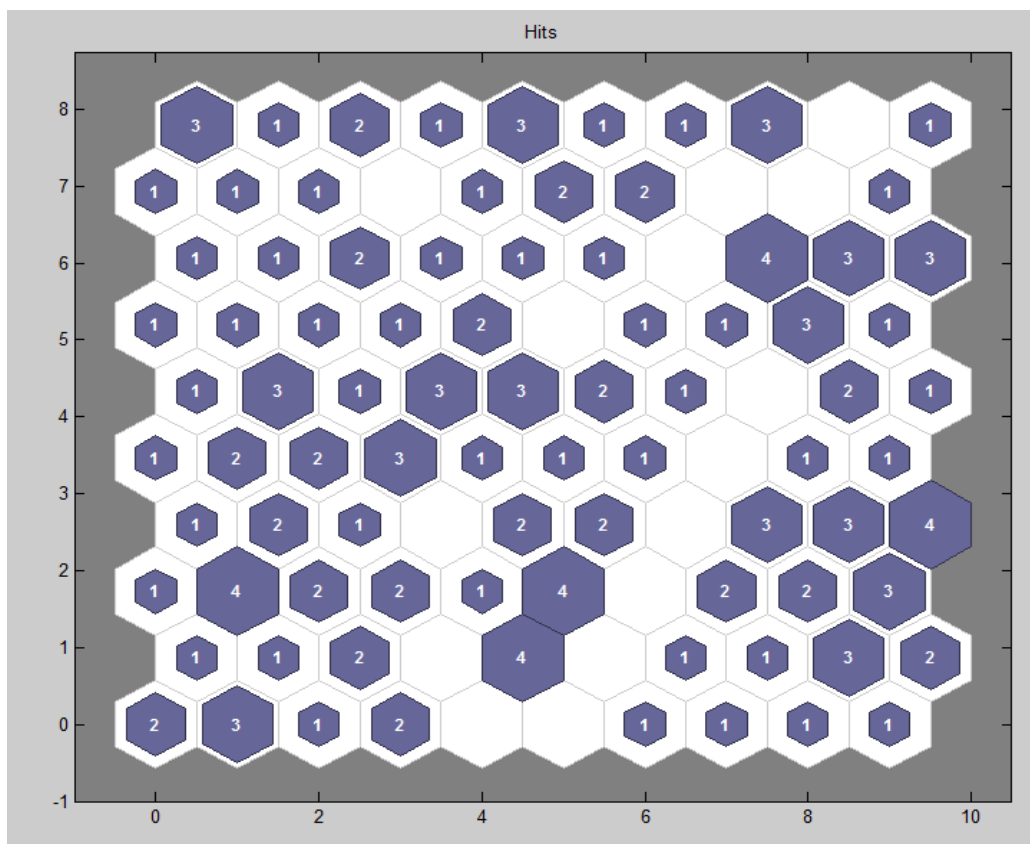


Fig. 27. Sample Hits

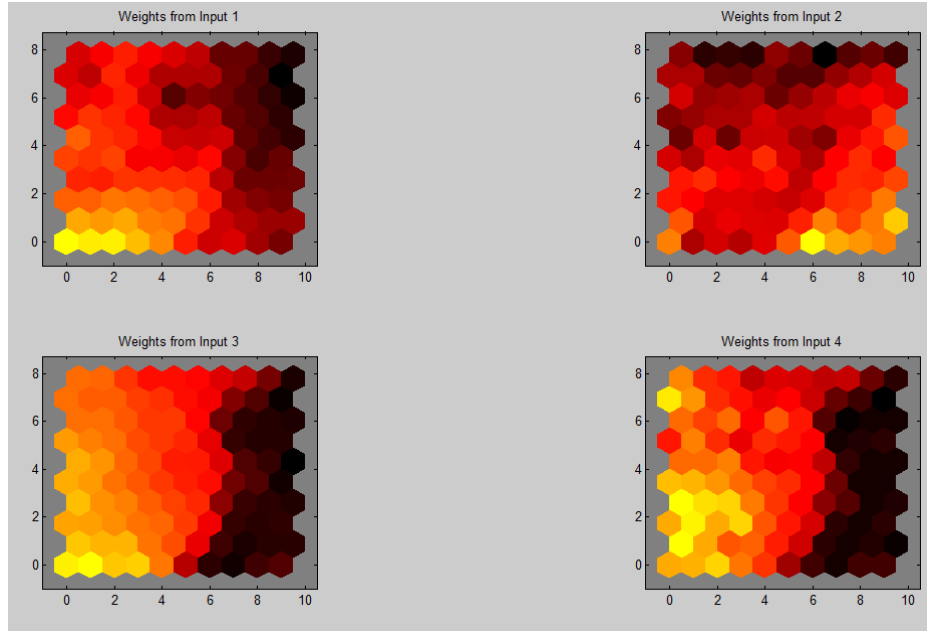


Fig. 28. weight Planes

#### **4.4.2 Validation**

|                             |          |
|-----------------------------|----------|
| Correlation coefficient     | 0.9955   |
| Mean absolute error         | 8.2394   |
| Root mean squared error     | 117.3938 |
| Relative absolute error     | 0.741 %  |
| Root relative squared error | 9.4381 % |
| Total Number of Instances   | 203      |

#### **4.5 Principal Component Analysis (PCA)**

Principal component analysis (PCA) is a mathematical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components. The number of principal components is less than or equal to the number of original variables. This transformation is defined in such a way that the first principal component has the largest possible variance (that is, accounts for as much of the variability in the data as possible), and each succeeding component in turn has the highest variance possible under the

constraint that it be orthogonal to (i.e., uncorrelated with) the preceding components. Principal components are guaranteed to be independent only if the data set is jointly normally distributed. PCA is sensitive to the relative scaling of the original variables.

PCA was invented in 1901 by Karl Pearson. It is mostly used as a tool in exploratory data analysis and for making predictive models. PCA can be done by eigenvalue decomposition of a data covariance (or correlation) matrix or singular value decomposition of a data matrix, usually after mean centering (and normalizing or using Z-scores) the data matrix for each attribute. The results of a PCA are usually discussed in terms of component scores, sometimes called factor scores (the transformed variable values corresponding to a particular data point), and loadings (the weight by which each standardized original variable should be multiplied to get the component score).

PCA is the simplest of the true eigenvector-based multivariate analyses. Often, its operation can be thought of as revealing the internal structure of the data in a way that best explains the variance in the data. If a multivariate dataset is visualised as a set of coordinates in a highdimensional data space (1 axis per variable), PCA can supply the user with a lower-dimensional picture, a "shadow" of this object when viewed from its (in some sense; see below) most informative viewpoint. This is done by using only the first few principal components so that the dimensionality of the transformed data is reduced.

PCA is mathematically defined as an orthogonal linear transformation that transforms the data to a new coordinate system such that the greatest variance by any projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.

Define a data matrix,  $X^T$ , with zero empirical mean (the empirical (sample) mean of the distribution has been subtracted from the data set), where each of the  $n$  rows represents a different repetition of the experiment, and each of the  $m$  columns gives a particular kind of datum (say, the results from a particular probe). The singular value decomposition of  $X$  is  $X = W\Sigma V^T$ ,

where the  $m \times m$  matrix  $\mathbf{W}$  is the matrix of eigenvectors of the covariance matrix  $\mathbf{X}\mathbf{X}^T$ , the matrix  $\Sigma$  is an  $m \times n$  rectangular diagonal matrix with nonnegative real numbers on the diagonal, and the  $n \times n$  matrix  $\mathbf{V}$  is the matrix of eigenvectors of  $\mathbf{X}^T\mathbf{X}$ . The PCA transformation that preserves dimensionality (that is, gives the same number of principal components as original variables)  $\mathbf{Y}$  is then given by:

$$\begin{aligned}\mathbf{Y}^T &= \mathbf{X}^T\mathbf{W} \\ &= (\mathbf{W}\Sigma\mathbf{V}^T)^T\mathbf{W} = \mathbf{V}\Sigma^T\mathbf{W}^T\mathbf{W} = \mathbf{V}\Sigma^T\end{aligned}$$

$\mathbf{V}$  is not uniquely defined in the usual case when  $m < n - 1$ , but  $\mathbf{Y}$  will usually still be uniquely defined. Since  $\mathbf{W}$  (by definition of the SVD of a real matrix) is an orthogonal matrix, each row of  $\mathbf{Y}^T$  is simply a linear transformation of the corresponding row of  $\mathbf{X}^T$ . The first column of  $\mathbf{Y}^T$  is made up of the “scores” of the cases with respect to the “principal” component, the next column has the scores with respect to the “second principal” component, and so on.

Given a set of points in Euclidean space, the first principal component corresponds to a line that passes through the multidimensional mean and minimizes the sum of squares of the distances of the points from the line. The second principal component corresponds to the same concept after all correlation with the first principal component has been subtracted from the points. The singular values (in  $\Sigma$ ) are the square roots of the eigenvalues of the matrix  $\mathbf{X}\mathbf{X}^T$ . Each eigenvalue is proportional to the portion of the “variance” (more correctly of the sum of the squared distances of the points from their multidimensional mean) that is correlated with each eigenvector. The sum of all the eigenvalues is equal to the sum of the squared distances of the points from their multidimensional mean. PCA essentially rotates the set of points around their mean in order to align with the principal components. This moves as much of the variance as possible (using an orthogonal transformation) into the first few dimensions. The values in the remaining dimensions, therefore, tend to be small and may be dropped with minimal loss of information. PCA is often used in this manner for dimensionality reduction. PCA has the distinction of being the optimal orthogonal transformation for keeping the subspace that has largest “variance”.

## CHAPTER 5

### RESULT AND DISCUSSION

This thesis shows three main approaches in gait classification i.e. back propagation, K-Means and KNN. After comparing the testing and training results we get better results using back propagation learning technique. The whole work of thesis is to describe the classification technique of different type of GAIT into following four categories: Normal, crouch2, crouch3, crouch4 using back propagation, K-Means and KNN. When the training of data is done then the output of will be either four options i.e. Normal, crouch2, crouch3, crouch4. If the testing data set is tested on the proposed system then the output must be normal GAIT if the classifier classifies it as normal data or the output will be abnormal GAIT data if the classifier classifies it as crouch2, crouch3, crouch4 in any of these three categories.

#### **5.1 Performance**

After comparing the overall accuracy of back propagation, K-Means and KNN, the performance of back propagation is best for 16 features of human GAIT.

- *Back propagation*: The Accurate classification (training and testing) of neural network(back propagation) comes out to be 69.825%
- *KNN*: The accurate classification of KNN (training and testing) comes out to be 61.25%.
- *K-Means*: The accurate clustering of K-Means (training and testing) comes out to be 50%.

The result shown in table 1 describes that back propagation performs better than KNN if we increase the number of neighbors in KNN the classification accuracy decreases.

TABLE I. COMPARISON

| Accuracy | Category/Method | K-Means | NN(BP) | KNN(k=1,2,3,4,5) |
|----------|-----------------|---------|--------|------------------|
|----------|-----------------|---------|--------|------------------|



|                   |               |       |      |                     |
|-------------------|---------------|-------|------|---------------------|
| Training Accuracy |               | 100%  | 100% | 60%                 |
| Testing Accuracy  | Normal Gait   | 54.68 | 56   | 100,100,100,100,100 |
| Testing Accuracy  | Crouch 2 Gait | 51.43 | 43   | 50,50,50            |
| Testing Accuracy  | Crouch 3 Gait | 33.19 | 43   | 50,50,50            |
| Testing Accuracy  | Crouch 4 Gait | 51.80 | 53   | 45,40               |

TABLE II. ACCURACY COMPARISON FOR BACK PROPAGATION, K-MEANS AND KNN

|                       | Table Column Head           |            |                |
|-----------------------|-----------------------------|------------|----------------|
|                       | <i>Back<br/>Propagation</i> | <i>KNN</i> | <i>K-Means</i> |
| Accuracy<br>(Overall) | 69.825%                     | 66 %       | 50%            |

## 5.2 Classifier Output

Scheme: weka.classifiers.rules.

Relation: all

Instances: 203

Test mode: split 66.0% train, remainder test

=== Classifier model (full training set) ===

Correlation coefficient 1  
Mean absolute error 0  
Root mean squared error 0  
Relative absolute error 0 %  
Root relative squared error 0 %  
Total Number of Instances 69

## 5.3 Cluster Output

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-

Relation: all

Instances: 203

Test mode: evaluate on training data

Number of iterations: 4

Within cluster sum of squared errors: 199.3088048905809

Final cluster centroids:

Clustered Instances

0 101 (50%)

1 102 (50%)

## **5.4 Conclusion**

In this chapter, the evaluation results of classification and clustering is shown. In order to demonstrate the robustness and performance of these classifiers a thorough analysis in terms of accuracy, computation, and memory was conducted. Also, we introduced several databases and described the benchmark setup and the rationale behind choosing different parameters of the classifiers. The impact of these parameters on the final recognition rates were illustrated by providing several graphs.

Accuracy analysis clearly demonstrates the robustness and the generalization capability of our classifiers in unconstrained environments with difficult illumination conditions. Also, the complete computational and memory analysis that were presented in this chapter could acknowledge the very low computational and memory requirements of our approach.

## CHAPTER 5

### CONCLUSIONS AND FUTURE SCOPE

In this chapter, we conclude this thesis with a summary of the material presented in the previous chapters, and discuss a number of limitations and potential solutions as our future work to extend the capabilities of our age and gender classification system.

#### **6.1 Conclusion**

This thesis has presented a complete framework for real-time and accurate normal and pathological human GAIT classification on embedded systems in unconstrained environments. However this approach added a large amount of redundancy to the feature vector, and increased its dimensionality, significantly. To counter this problem, we introduced an enhanced segmental dimensionality reduction technique utilizing PCA to extract the most descriptive and discriminative features. This technique not only improved the accuracy by reducing the noise and redundancy, but also enabled the implementation of our classifier on the resource-limited embedded systems.

In spite of the memory-efficient and real-time performance of our method, the recognition rates are closely comparable with other state-of-the-art but resource-demanding approaches. For instance, we have improved the classification rate up to 3.825% compared to the other classification approaches.

Thanks to this strategy we could achieve 100% training accuracy of classification and 43% testing accuracy. For 16 input features the overall accuracy we get from back propagation algorithm is 69.825%

Therefore the proposed system can be used to identify the human gait and to know which locomotion provides normal GAIT and abnormal GAIT.

## **6.2 Limitations and Future Scope**

Essentially, for a given number of training samples, there is a limit for the maximum number of features, or the so-called dimensionality of the feature vector. With the growth of dimensionality (i.e., volume of the space), the number of required training samples must grow exponentially, otherwise, the data in high-dimensional space will become sparse.

This scarcity of data can lead to over fitting since there are not enough training samples. Also, utilizing PCA before LDA to reduce the dimensionality cannot solve this ill-conditioned problem, because the available samples are too few and PCA is a “lossy” compression method that overly discards useful texture information and deteriorates the recognition rate, significantly.

The gender of a subject can also affect the age classification. The future scope is the proposed system can be used in surveillance systems or to detect any upcoming disease related to GAIT human locomotion; reorganization system and testing accuracy of the system can be improved.

There are many new directions in which to extend the proposed approaches presented in this thesis, including both the algorithms as well as the current and novel applications. The first direction is large-scale discriminative learning (T. INDUMATHI *et al.* 2014). Kernel methods, especially GP, provide a flexible way to train probabilistic discriminative models (e.g., given the predictive predictions and Bayesian framework to learn model hyper parameters). However, the size and computational time of these non-parametric models scales to the number of training examples.

The second direction, somewhat related to large-scale learning as well, is adaptive and online learning, where the discriminative model can be adapted incrementally and sequentially during operation to improve its performance in the environment which is changing over time. Besides the adaptation capability in a dynamic environment, online

sequential learning (i.e., the setting where training examples typically become available over time) can help in large-scale problems, when compared with batch learning (i.e., the setting where the whole dataset is available a priori), due to less computation and memory resources needed to update the hypothesis (Li Rong Zhang *et al.* 2005). These ideas have been used to extend standard kernel machines in many different approaches (Csato & Opper 2002, Sato & Oba 2002, Quinonero-Candela & Winther 2003, Cauwenberghs & Poggio 2000, Bordes & Bottou 2005, Bordes *et al.* 2005, Crammer *et al.* 2004, Crammer & Singer 2003, Kivinen *et al.* 2004, Weston *et al.* 2005).

The third direction is related to available class labels. To train an effective discriminative model, labeled data for supervised learning is needed. However, in many applications, labeling is difficult and time-consuming, whereas it is easy to collect unlabeled datasets. In recent years, the problem of semi-supervised learning, where a small number of labeled examples and many unlabeled examples are available, has been very actively addressed in the machine learning community (Davrondzhon Gafurov), with some promising results even in discriminative kernel-based learning (Joachims 1999, Lawrence & Jordan 2005, Sindhwani *et al.* 2007, Rogers & Girolami 2007). Combined with complex structured input-output learning methods, semi-supervision can improve the practicality of applications in many fields, including the ones presented in this thesis. In addition, other incomplete data scenarios, such as temporally missing sensor sources (i.e., input features), should be handled in multi-modal learning (Kapoor *et al.* 2005). For more general and practical use, particle-filter- and machine-learning-based object tracking could be extended with semi-supervision and incomplete data modeling, including general data association problems such as an unknown number of targets.

## REFERENCES

- [1] T INDUMATHI, M.PUSHPARANI; MULTIMODAL HUMAN IDENTIFICATION SYSTEM BASED FUSION USING GAIT BIOMETRICS; International Journal of Soft Computing and Artificial Intelligence, ISSN: 2321-404X, Volume-2, Issue-1, May-2014.
- [2] Rong Zhang, Christian Vogler, Dimitris Metaxas; Human Gait Recognition; IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 27, NO. 11, NOVEMBER 2005.
- [3] Davrondzhon Gafurov; A Survey of Biometric Gait Recognition: Approaches, Security and Challenges, IEEE.
- [4] Ming-Gang Wen; Human Gait Classification Using Motion Information; Journal of Information Technology and Applications Vol. 1 No. 1, May, 2006, pp. 1- 8
- [5] Adam ŚWITOŃSKI, Romualda MUCHA, Dariusz DANOWSKI, Monika MUCHA, Grzegorz CIEŚLAR, Konrad WOJCIECHOWSKI, Aleksander SIEROŃ, "Human identification based on a kinematical data of a gait" in PRZEGLĄD ELEKTROTECHNICZNY (Electrical Review), ISSN 0033-2097, R. 87 NR 12b/2011
- [6] Adam ŚWITOŃSKI, Romualda MUCHA, Dariusz DANOWSKI, Monika MUCHA, Andrzej POLAŃSKI, Grzegorz CIEŚLAR, Konrad WOJCIECHOWSKI, Aleksander SIEROŃ, "Diagnosis of the motion pathologies based on a reduced kinematical data of a gait," in PRZEGLĄD ELEKTROTECHNICZNY (Electrical Review), ISSN 0033-2097, R. 87 NR 12b/2011.
- [7] A. Hayder, J. Dargham, A. Chekima, and G. M. Ervin, "Person Identification Using Gait," International Journal of Computer and Electrical Engineering, Vol. 3, No. 4, August 2011.
- [8] Hoang Minh Thang, Vo Quang Viet, Nguyen Dinh Thuc, Deokjai Choi, "Gait Identification Using Accelerometer on Mobile Phone," in Advances in Cryptology Eurocrypt. Berlin, Germany: Springer, vol. 3027, Lecture Notes in Computer Science, pp. 523–540.
- [9] Jay Prakash Gupta, Pushkar Dixit, Nishant Singh, Vijay Bhaskar Semwal, "Analysis of Gait Pattern to Recognize the Human Activities," International Journal of Interactive Multimedia and Artificial Intelligence July 2014.
- [10] Prerna Arora, Rajni, "Features Mapping Based Human Gait Recognition" International Journal on Recent and Innovation Trends in Computing and Communication ISSN: 2321-8169.
- [11] Baofeng Guo and Mark S. Nixon, "Gait Feature Subset Selection by Mutual Information"; IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS—PART A: SYSTEMS AND HUMANS, VOL. 39, NO. 1, JANUARY 2009.

- [12] Supreet Kaur and Er. Shikha Chawla, "An Approach for Enhancing the Human Identification Using Gait Recognition Techniques," International Journal of Application or Innovation in Engineering & Management (IJAIEM).
- [13] Supreet Kaur, and Er.Ranjeet Kaur Sandhu, "A Survey on Enhanced Human Identification Using Gait Recognition based on Neural Network And Support Vector Machine," International Journal of Application or Innovation in Engineering & Management (IJAIEM).
- [14] Jay Prakash Gupta, Pushkar Dixit, Nishant Singh, Vijay Bhaskar Aemwal," Analysis of Gait Pattern to Recognize the Human Activities" International Journal of Artificial Intelligence and Interactive Multimedia, Vol. 2, N° 7.
- [15] Karmjeet Kaur , Prabhsharan Kaur, "GAIT RECOGNITION USING V2DPCA (VARIABLE 2D-PCA) – A REVIEW," International Journal of Application or Innovation in Engineering & Management (IJAIEM) Volume 4, Issue 7, July 2015.
- [16] Deepti Singh , Dr. Amit Dixit, "Review on; Human Identification using GAIT Recognition Technique with PAL and PAL entropy, SVM and k-Means with LDA," Deepti Singh et al, / (IJCSIT) International Journal of Computer Science and Information Technologies, Vol. 5 (6) , 2014, 7136-7141.
- [17] Navneet Kaur, Samandeep Sing; "Review On: Gait Recognition for Human Identification using NN"; Navneet Kaur et al, / (IJCSIT) International Journal of Computer Science and Information Technologies, Vol. 5 (3) , 2014, 3991-3993.
- [18] Sruti Das Choudhury, Tardi Tjahjadi; "Gait Recognition Based on Shape and Motion Analysis of Silhouette Contours"; Computer Vision and Image Understanding, Volume 117 (Number 12). pp. 1770-1785.
- [19] Adam Świtoński, Andrzej Polański, Konrad Wojciechowski; "Human Identification Based on Gait Paths"; IEEE.
- [20] Michael Fitzgerald Nowlan; Human Identification via Gait Recognition Using Accelerometer Gyro Forces"; IEEE Conference on Computer Vision and Pattern Recognition, 2007. pgs 1-21.
- [21] Rong Zhang, Christian Vogler, Dimitris Metaxas; "Human Gait Recognition"; Conference on Computer Vision and Pattern Recognition.
- [22] Dr. Jan Peters; "Machine Learning for Active Gait Support with a Powered Ankle Prosthesis" ; Maschinelles Lernen zur gesteuerten Gangunterstützung durch eine aktive Fußgelenkprothese.
- [23] Arun Joshi, Mr. Shashi Bhushan, Ms. Jaspreet Kaur; "GAIT RECOGNITION OF HUMAN USING SVM AND BPNN CLASSIFIERS"; International Journal of Computer Science and Mobile Computing A Monthly Journal of Computer Science and Information Technology.
- [24] Rattiya Thanasoontornerk, Chatkaew Pongmala, Areerat Suputtiada: Tree Induction for Diagnosis on Movement Disorders Using Gait Data. In: 2013 5th International Conference on Knowledge and smart technology (KST)
- [25] Kiichi\_Katoh,, *Makoto Katoh.*: Using FS in Tuning a Biped Locomotion Surrogate Robot for Walking Training. In: 2011 4th International Conference on Biomedical Engineering and Informatics (BMEI)

- [26] Mitsuru Yoneyama, Yosuke Kurihara, Kajiyo Watanabe,: Accelerometry-Based Gait Analysis and Its Application to Parkinson's Disease Assessment—Part 2: IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING, VOL. 21, NO. 6, NOVEMBER 2013
- [27] M. Ogata<sup>1</sup>, T. Karato<sup>1</sup>, T. Tsuji<sup>1</sup>, H. Kobayashi<sup>1</sup> and K. Irie<sup>2</sup>.: Development of Active Walker by using Hart Walker. In: SICE-ICASE International Joint Conference 2006 Oct. 18-21, 2006 in Bexco, Busan, Korea
- [28] Hiroshi Kobayashi, Takeo Karato, and Toshiaki Tsuji.: Development of an Active Walker as a New Orthosis. Proceedings of the 2007 IEEE International Conference on Mechatronics and Automation August 5 - 8, 2007, Harbin, China
- [29] Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami.: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. In Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
- [30] Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
- [31] Anubha Parashar, Somya Goyal, Apoorva Parashar, Bharat Sahjlan: Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Means. ICTCS 2016 - CSI-Udaipur Chapter.
- [32] Semwal, Vijay Bhaskar, et al : Biped model based on human Gait pattern parameters for sagittal plane movement.Control, Automation, Robotics and Embedded Systems (CARE), 2013 International Conference on. IEEE, 2013.
- [33] Semwal, Vijay Bhaskar, et al.: Biologically-inspired push recovery capable bipedal locomotion modeling through hybrid automata. Robotics and Autonomous Systems 70 (2015): 181-190.
- [34] Semwal, Vijay Bhaskar, Manish Raj, and Gora Chand Nandi. :Biometric gait identification based on a multilayer perceptron. Robotics and Autonomous Systems 65 (2015): 65-75.
- [35] Hiroshi Kobayashi, Takeo Karato, and Toshiaki Tsuji.: Development of an Active Walker as a New Orthosis. Proceedings of the 2007 IEEE International Conference on Mechatronics and Automation August 5 - 8, 2007, Harbin, China.
- [36] Byungyun Lee, Sungjun Hong, Heesung Lee, Euntai Kim;Gait Recognition System using Decision-Level Fusion; 2010 5th IEEE Conference on Industrial Electronics and Applications.
- [37] Slobodan Ribaric, Ivan Fratric; A Biometric Identification System Based on Eigenpalm and Eigenfinger Features; IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 27, NO. 11, NOVEMBER 2005.
- [38] C. Nandini, C. N. RaviKumar; An approach to Gait Recognition, IEEE.



## Appendix –I

# Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Mean

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## ABSTRACT

Push recovery is prime ability that is essential to be incorporated in the process of developing a robust humanoid robot to support bipedalism. In real environment it is very essential for humanoid robot to maintain balance. In this paper we are generating a control system and push recovery controller for humanoid robot walking. We apply different kind of pushes to humanoid robot and the algorithm that can bring a change in the walking stage to sustain walking. The simulation is done in 3D environment using Webots. This paper describes techniques for feature selection to foreshow push recovery for hip, ankle and knee joint. We train the system by K-Mean algorithm and testing is done on crouch data and tested results are reported. Random push data of humanoid robot is collected and classified to see whether push lie in safer region and then tested on given proposed system.

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## Keywords:

Bipedal walking, push recovery, humanoid robot, K-Mean classification, Zero Moment Point, Linear Inverted Pendulum

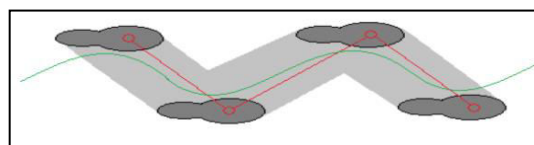
## 1. INTRODUCTION

Now a day's human robots are developed to perform many human activities. Bipedal robots are comparable to human walking. Though we can say human locomotion seems effortless but it is exceptionally complex [1].

When we talk about the push recovery features in humanoid robot we face difficulties as compared to animals and humans. Humans and animals are completely versed where as humanoid robots have certain other features [2]. It's contemplated to make a humanoid robot that can work smoothly and practically. There are many uncertain disturbances that emerge like push is one of the common problems that get created during locomotion. The goal of this paper is to search easy, lucid way for push recovery walking in humanoid robots. So we collect the push data by pushing humanoid robot from behind and we train the system by applying K-Mean algorithm and using crouch as data. Therefore we get a trained system which can analyse and work on any push recovery strategy. [18]- [23]

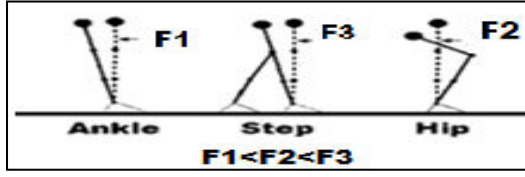
In this paper we apply four different types of pushes. The first difficulty that arises in humanoid robot is mainly due to ZMP (Zero Moment Point). A humanoid robot will be stable if and only if its ZMP is within the geometry of its foot.

In figure 2 the red line tells how the ZMP must lie within the sole of the robot through the complete motion. Green line shows that the trajectory of robot must emphasis on keeping the centre of gravity (COG) between the two feet of robot. Dark grey division show safe region of the foot for single support phase of humanoid motion. And light grey region shows safe region for double support phase of humanoid motion.



**Fig. 1 Proposed Trajectory of Humanoid's ZMP**

When an external push is applied to a humanoid robot we have 3 approaches, i.e. ankle, step and hip strategy shown in figure 2.

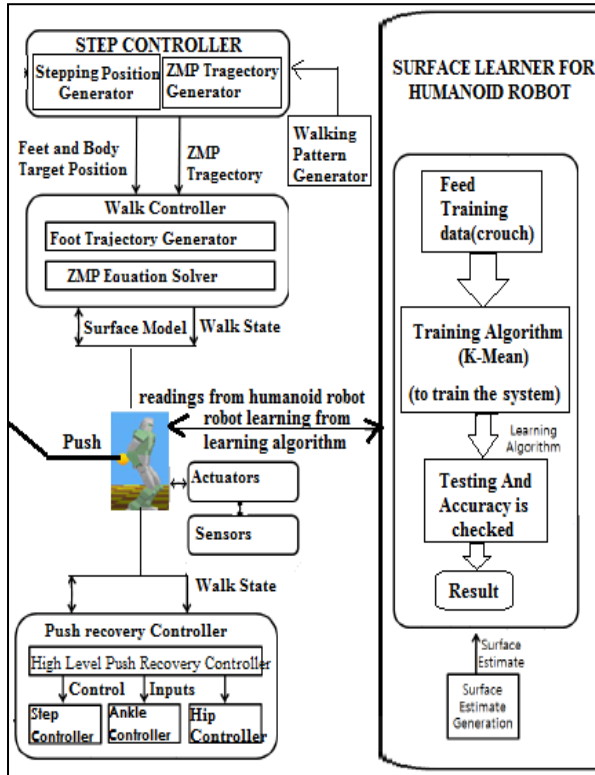


**Fig. 2 Three basic balancing strategies in the presence of external forces**

It is troublesome to obtain perfect motion like human gait when there is high Degree of Freedom and variable mechanical structure of a human body. Inverted Pendulum Model (IPM) is utilized for gathering human motion gait [8] [9] [10] [11] robotics and biomechanics [12] [13] [14] [15] [16].

The IPM presumes reaction force Counter-revolution obtained from the floor which is produce from the point and penetrates the COG (Centre of Gravity) of the IPM, resulting as absence of angular moments around COG. The linear acceleration of centre of mass will be more than that of the gravity acceleration when the humanoid is pushed very hardly on the Centre of Mass (COM). And as surface always has friction force therefore upper part of body will move faster than lower part of body. So in order to avoid this we eradicate the undesirable momentum induced from push therefore, humanoid robot may move steadily.

## 1.1 Proposed system



**Fig. 3 Proposed System for Push Recovery for Humanoid Robot and Classifying the Data Using K-Mean**

In the context of this paper first stage involves to analyse the push recovery pattern. Push recovery pattern can be analysed by collecting different data. To find this pattern we need machine learning Technique. In second stage we gather the data of humanoid robot by pushing it from a rigid ball from behind. In third stage we calculate leg joint angles using inverse kinematics [4] [5] [6]. In fourth stage we train the system using K-Mean classification and crouch data is tested on it. In fifth stage we find out the push recovery strategy of humanoid robot on given training system to get better results or push recovery. Then from the classifier we see whether the ZMP lie in safe region (from fig 1) or not. If it doesn't fall in safe region, we make the humanoid robot learn by modifying the algorithm and to sustain its walking.

## 1.2 Push recovery controller

The work of Push Recovery Controller (PRC) is to remove anxiety raised because of external disturbances or modelling error. On uneven surface chances of perturbations is very high so PRC is more required. PRC can be categorised into 3 categories (based on Fig.2)

### 1.2.. Ankle Controller

To keep COM within the base support the torque is applied on the ankle joints by the ankle controller. It is effectuated as,

$$\Delta\theta_{\text{ankle}} = x_{\text{ankle}}$$

here  $x_{\text{ankle}}$  denotes the input of ankle controller and  $\Delta\theta$  denotes the joint angle bias.

### 1.2.2 Step Controller

It controls the moves of the base to support in taking steps. With the relative target foot position a new step is inserted by overriding the step controller as  $x_{\text{capture}}$ .

### 1.2.3 Hip Controller

It puts the angular acceleration of torso and limbs in use to create backward ground reaction force to pull COM back towards base support. To implement it we use,

$$Z_{\text{hip}}(w) = \tau_{\text{maxi}} a(w) - 2\tau_{\text{maxi}} a(w - Z_{Y1}) + \tau_{\text{maxi}} a(w - Z_{Y2})$$

where  $\tau_{\text{maxi}}$  is maximum torque that joint can apply,  $a(w)$  is unit step function,  $Z_{Y1}$  is the time when torso stops accelerating and  $Z_{Y2}$  is the time at which torso stops.

## 1.3 Hip strategy stability criteria

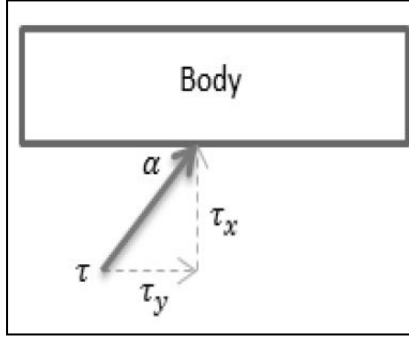
Hip strategy implements push recovery by intensifying the torque of the hip joint to overcome the forces of push. The torque of the hip is applied closer to COM of the robot, and therefore produces greater reaction force [18]. The hip strategy can create greater counter forces therefore it is used for recovering from larger pushes in comparison with ankle strategy.

In general Linear Inverted Pendulum (LIP) is applied to humanoid robot [7] but this strategy deficit rotational inertia,

which is very essential for hip strategy to balance push recovery. Therefore we exercise Angular Moment with Inverted Pendulum.

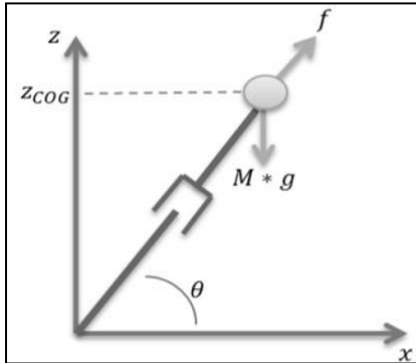
The classification of angular disturbance (in fig 4) can be done by control algorithm by comparing the sagittal and coronal torques [19]. The angular torque can be represented by following equation,

$$\tau = \tau_x \cos(\alpha) + \tau_y \sin(\alpha)$$



**Fig. 4 Angular Disturbance**

Figure 5 is used to find the control for humanoid. The motion of COG is controlled to make sure robot is stable during walking. The motion of COG of robot is affected by gravity and the contractility of the robot. Due to this the derivation of the robot's kinematic equations [17] comes into picture.



**Fig. 5 Linear Inverse Pendulum Model**

By analysing the humanoid robot's COG in figure 4, the equation can be obtained,  $Mx'' = f \sin(\theta)$ .

Based on the above discussion hip controller algorithm is designed.

---

Algorithm-1 stiffness controller working and

Target angle  $\phi_{hip}$

---

```

 $\phi_{hip} = \phi_{hip, initz}$ 
stiffness = stiffnessinitz
else
  if b= not true and push detected and ground contact then
     $\phi_{hip} = \phi_{hip, curr}$ 
    stiffness= stiffnessminimum
    b=true

```

```

end if
if  $\phi_{hip} \in [\phi_{hip, min}, \phi_{hip, max}]$  and stiffness <= stiffnessmaximum then
   $\phi_{hip} = \phi_{hip, curr}$ 
  stiffness = stiffness +  $\Delta$ stiffness
else
  A= false
   $\phi_{hip} = \phi_{hip, initz}$ 
  stiffness = stiffnessinit
end if
end if

```

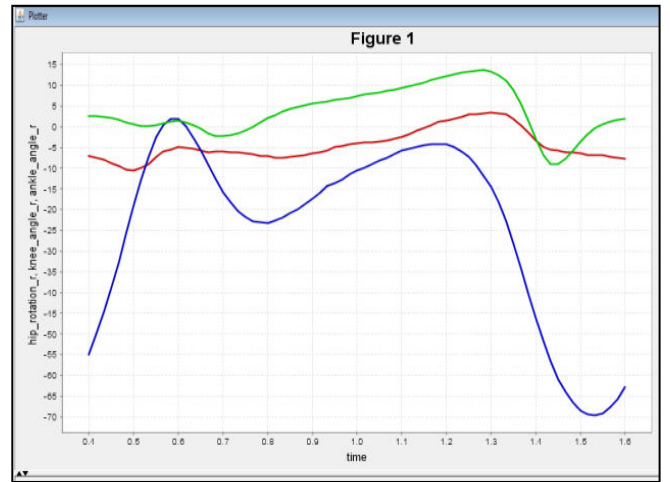
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The stiffness controller gets real position and the strength of activation of joint. It then calculates the strength of activation and the target joint position. To calculate the torque for the joint, the output from the stiffness controllers is passed to low-level joint setup.

## 2. METHODOLOGY

### 2.1 To gather data and extract feature

Data is captured using body sensor based Human Motion Capture Device (HMCD) and potentiometer. Then data is collected using accelerometer sensor. It gives the meticulous acceleration and user can analyse the moment.



**Fig. 6 hip ankle and knee joint**

Figure 6 shows the simulation of hip, ankle and knee joint with respect to time. Red curve is for hip, blue curve represent knee and green curve show ankle. Then we apply the push, i.e. four sorts of pushes are applied categorized as low, medium, medium high and high. Then the data is collected and Force sensing resistor (FSR 3105) is used to estimate push from back. To obtain joint angles of hip and knee we use inverse kinematics, and it employs to control the position of a humanoid's extremities.

### 2.2 Joint angle calculation

Inverse kinematics is used to gather the result of link manipulator. Now the human leg is represented as two link manipulator in Fig.7 [8]

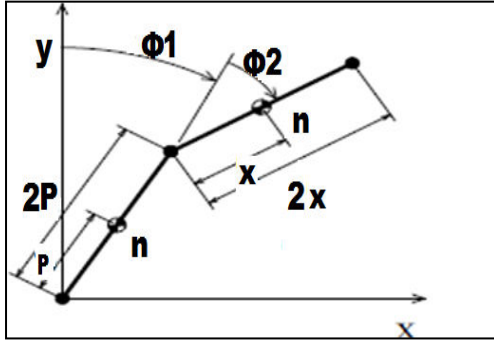


Fig.7 two link manipulator

The dynamics of 2-link manipulator is shown by,

$$N(\Phi(a)) \ddot{\Phi}(a) + X(\Phi(a), \dot{\Phi}(a)) + D\Phi(a) = \kappa(a-P)$$

Where N is inertia matrix, X is centripetal and coriolis torque,  $\kappa$  is input torques and P is a dead line time. Each element is calculated as

$$N(\Phi(a)) = \begin{bmatrix} j_1 + j_2 + 2\beta \cos \Phi_2(a) & j_2 + \beta \cos \Phi_2(a) \\ j_2 + \beta \cos \Phi_2(a) & j_2 \end{bmatrix}$$

$$X(\Phi(a), \dot{\Phi}(a)) = \begin{bmatrix} \Phi_2 \dot{\Phi}_1(a) \dot{\Phi}_2(a) + \Phi_2(a)^2 \sin \Phi_1(a) \\ \beta \Phi_1(a) 2 \sin \Phi_2(a) \end{bmatrix}$$

$$D = \begin{bmatrix} D_1 & 0 \\ 0 & d_2 \end{bmatrix}, \quad \kappa = (a-L) = \begin{bmatrix} k_1(a-P) \\ k_2(a-P) \end{bmatrix}$$

$$J_1 = I_1 + (m_1 + 4m_2) l_1^2, \quad J_2 = I_2 + (m_1 + 4m_2) l_2^2$$

$$B = 2m_2 l_1 l_2, \quad I_1 = 1/3 m_1 l_1^2, \quad I_2 = 1/3 m_2 l_2^2$$

Now the code is written for classification using Matlab. This is unsupervised classification technique. Appropriate area of training is selected for each class. Here .CSV (comma separated value) file is used to store training areas and the corresponding class.

### 2.3 Training

The system is trained using K-mean algorithm which is an unsupervised classification technique.

---

Algorithm 2: train the system using K-Mean algorithm

---

**START**

- [1] Choose any arbitrary vector mean when  $a=0$ ,
- [2] Mean will be  $[w_1, w_2, w_3 \dots w_z]^a$  where z is no. of cluster head.
- [3] After this classification of raw data is done according to distance from mean at  $a=0$ .
- [4] **DO** (Until threshold value is not reached that is the convergence criteria is not satisfied).
- [5] Again calculate mean vector for  $a=a+1$  and consequently do the classification.
- [6] Modify  $a=a+1$ ;

[7] Convergence criteria

[8] Calculate the Euclidean distance  
 $\|w(a) - w(a-1)\| < z$

[9] Use  $w(a)$  as solution  
**ELSE**

[10] Go to step 5

**END**

**UNTIL** the data not classified/\*classification gait= {normal, croach1, croach2, croach3, croach4}\*/

**END**

---

Now the K-Mean algorithm (Algorithm-2) is framed in order to classify the crouch data, we make a minor modification in Algorithm-2, in order to classify the trajectories (in fig-1) of humanoid robot so as to find out which trajectory is safe push and which trajectory is not and we make four clusters (from fig-1) and see where trajectory lies i.e. grey, dark grey, green line or outside this.

So there will be 3 centroids and we use K-Mean algorithm to minimize an *objective function*,

$$J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$$

$\|x_i^{(j)} - c_j\|^2$  is distance between data point  $x_i^{(j)}$ ,  $c_j$  is cluster centre which indicates cluster centres of the distance of n data points.

Suppose n is the sample feature vectors  $x_1, x_2, \dots, x_n$ , and falls under k compact clusters. Say  $z_i$  is in cluster i depicts mean of the vectors, where i represents 3 clusters i.e. in grey region, dark grey, green line or if not anyone then it lies outside this, If the clusters are properly distinguished, we can use a minimum-distance classifier to distinguish them, i.e. x belongs to cluster i if  $\|x - z_i\|$  is smallest among all the distances of k.

### 2.4 Classification

It is the selection feature where classification is done on the data set. Learning must be done in standard way in order to obtain the best results of system. It classifies the gait data where a set of data is already given as crouch data. The training is done using K-Mean algorithm then new push recovery data collected from humanoid robot is tested on the same system to predict the behavior of humanoid robot learning.

Different joint angles (knee, hip, and ankle) were captured and control reverse torques can be calculated for the joints in a bipedal humanoid using the following equation,

$$\tau = W(\Phi) \ddot{\theta} + C(\Phi, \dot{\Phi}) + G(\Phi)$$

where  $W(\Phi)$  -inertial torque,  $C(\Phi)$  - centripetal and coriolis forces,  $G(\Phi)$  - gravitational force.

Once this is done, now the trajectories of push data is classified in order to check whether the humanoid robot lies in safe region or the humanoid robot cannot recover from the push.

### 3. EXPERIMENTAL RESULTS

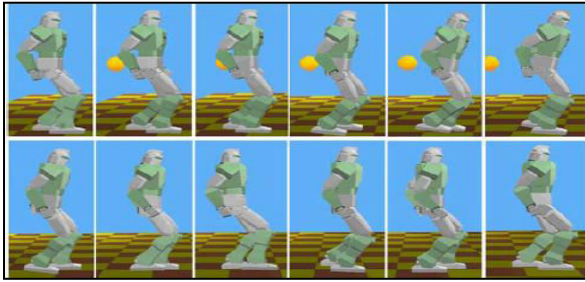
In simulated environment after training of push recovery controller we validate walking of humanoid robot.

#### 3.1 Experimental setup

The simulation is done in WEBOTS, where the humanoid robot is equipped with accelerometer in the pelvic. In the experiment, similar motion of the simulation is produced to intensify the push force by modifying the COM position. We keep changing the pushing force and push data is analysed using classifier. As the data is collected by humanoid robot it is passed through the proposed algorithm and it shows the learning pattern of push recovery.

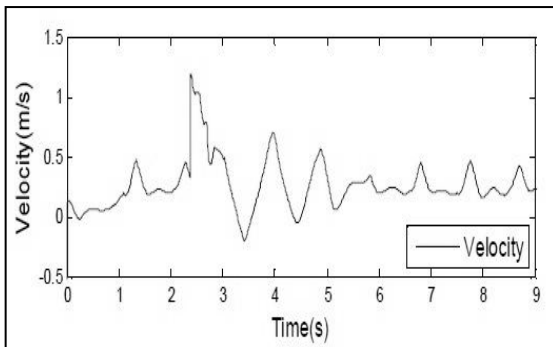
#### 3.2 Push simulation

Here a rigid ball is used to push the humanoid robot from behind shown in figure 8[24].



**Fig.8 Simulation results as humanoid robot manages to walk normally by recovering from push strategy.**

Figure 9[24] shows the influence inducement as an abrupt rise in the velocity of ball. To recover the humanoid robot the controller controls it and the unsupervised learning is done through K-Mean and manages it to its normal walking position.



**Fig.9 Velocity vs. Time diagram**

Once the push recovery strategy is applied, now the push recovery walking strategy is produced by controller and thus it will return to normal walking trajectory. Then ZMP controller generates centre of mass trajectory that will keep track of ZMP. The different type of data set is collected when the robot is pushed from behind from different four strategies i.e. low,

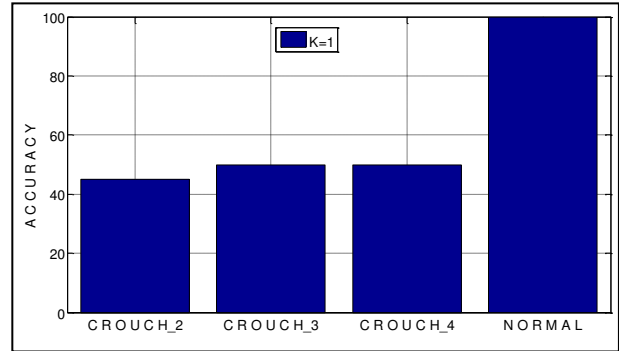
medium, medium high and high. When the classification of data is done the overall accuracy obtained is 60.275% shown in figure 10.

### 4. DISCUSSION

#### 4.1 Performance of system

Webots is chosen for the simulation as it uses an accurate physics engine. It simulates gravity and frictional forces that influence the humanoid robot's motion. Webots can easily simulate every position while applying gravity and external forces.

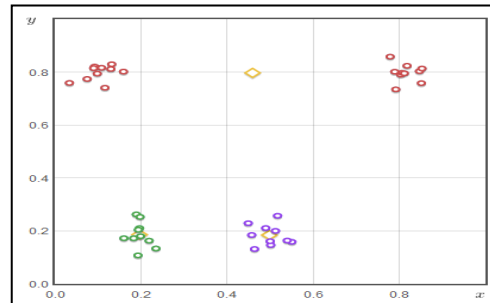
Therefore we get the applied force by simulation program. When the push is applied then the performance of proposed system is best and its impact lies near COM. As the pushes occur away from COM the angular momentum will be greater and the performance of system will degrade and it becomes difficult for humanoid robot to recover. Therefore as soon as the pushes occur away from COM we classify the data and try to bring it under COM.



**Fig.10 shows overall accuracy for training data set**

#### 4.2 Conclusion

Figure 11 shows that different clusters could be shown according to COM that falls in classification in 4 different points while doing classification of Figure 1.



**Fig.11 data points clustered at 4 distinct positions**

When the humanoid robot is tested with respect to proposed system, it shows the better push recovery strategy. Therefore we used the K-Mean classification technique which gives accuracy rate up to 60%. Hence the push recovery data is classified



properly and push recovery strategy is efficiently implied for humanoid robot.

So by the experimental comparisons, the proposed method can reduce perturbations up to large extend and it indicates that this method is an effective approach and can be use in the motion rules for humanoid robot walking and running.

## REFERENCES

- [1]. Aaron D. Ames, "Human-Inspired Control of Bipedal Walking Robots" IEEE TRANSACTIONS ON AUTOMATIC CONTROL, VOL. 59, NO. 5, MAY 2014 1115
- [2]. Wentao Mao, Jeong-Jung Kim and Ju-Jang Lee, "Continuous Steps toward Humanoid Push Recovery" Proceedings of the IEEE International Conference on Automation and Logistics Shenyang, China August 2009.
- [3] Vijay Bhaskar Semwal, Kaushik Mondal & G.C.Nandi "Robust and Accurate Feature Selection for Humanoid Push Recovery and Classification: Deep learning Approach" Neural Computer & Application DOI 10.1007/s005.21-015-2089-3.
- [4] Iamsa-at, S.; Horata, P., "Handwritten Character Recognition Using Histograms of Oriented Gradient Features in Deep Learning of Artificial Neural Network," IT Convergence and Security (ICITCS), 2013 International Conference on , vol., no., pp.1,5, 16-18 Dec. 2013.
- [5] Baptista, Darío, and Fernando Morgado-Dias. "A survey of artificial neural network training tools." *Neural Computing and Applications* 23.3-4 (2013): 609-615.
- [6] Chowdhury, Sujan, Brijesh Verma, and David Stockwell. "A novel texture feature based multiple classifier technique for roadside vegetation classification." *Expert Systems with Applications* 42.12 (2015): 5047-5055.
- [7] Taku Komura\*, Akinori Nagano, Howard Leung, and Yoshihisa Shinagawa. "Simulating Pathological Gait Using the Enhanced Linear Inverted Pendulum Model." IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING, VOL. 52, NO. 9, SEPTEMBER 2005.
- [8] S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Yokoi, and H.Hirukawa, "Biped Walking Pattern Generation by a Simple Threedimensional Inverted Pendulum Model," *Advanced Robotics*, vol. 17, pp. 131-147, 2003.
- [9] Albert, and W. Gerth, "Analytic Path Planning Algorithms for Bipedal Robots without a Trunk," *Journal of Intelligent and Robotic Systems*, vol. 36, pp. 109-127, 2003.
- [10] F. Miyazaki, S. Arimoto, "A Control Theoretic Study on Dynamical Biped Locomotion," *Journal of Dynamic Systems, Measurement, and Control*, vol. 102, pp. 233-239, 1980.
- [11] H. Miura, I. Shimoyama, "Dynamic Walk of A Biped," *Int. Journal of Robotics Research*, vol. 3, pp. 60-74, 1984.
- [12] F.L. Buzcek, K. M. Cooney, M. R. Walker, M. J. Rainbow, M. C. Concha and J. O. Sanders, "Performance of an inverted pendulum model directly applied to normal human gait," *Clinical Biomechanics*, vol. 21, no. 3, pp. 288-296, Mar. 2006.
- [13] A. Karlsson and T. Persson, "The ankle strategy for postural control: a comparison between a model-based and a marker-based method," *Computer Methods and Programs in Biomedicine*, vol 52, no. 3, pp. 165-173, Mar. 1997
- [14] M. A. Townsend, "Biped gait stabilization via foot placement," *Journal of Biomechanics*, vol. 18, no. 1, pp. 21-38, 1985.
- [15] M. A. Townsend, "Dynamics and coordination of torso motions in human locomotion," *Journal of Biomechanics*, vol 14, no. 11, pp. 727-738, 1981.
- [16] H. Hemami and C. L. Golliday, "The inverted pendulum and biped stability," *Mathematical Biosciences*, vol. 34, no. 1-2, pp. 95-110, 1977.
- [17] Akash, Shruti Chandra, Abba, G.C. Nandi, "Modeling a biped humanoid robot using inverted pendulum towards push recovery," *International Conference on Communication, Information, and Computing Technology*, Mumbai, India, 2012.
- [18] Awais Yasin, Qiang Huang, Qian Xu, M. Saad Sultan, "Humanoid Robot Push recovery through foot placement," *International Conference on Mechatronics and Automation*, Chengdu, China, 2012.
- [19] Semwal, Vijay Bhaskar, and Gora Chand Nandi. "Toward Developing a Computational Model for Bipedal Push Recovery—A Brief." *Sensors Journal*, IEEE 15.4
- [20] Semwal, Vijay Bhaskar, et al. "Biologically-inspired push recovery capable bipedal locomotion modeling through hybrid automata." *Robotics and Autonomous Systems* 70 (2015): 181-190.
- [21] Sharma, Pinki, and Abhishek Vaish. "Information-Theoretic Measures on Intrinsic Mode Function for the Individual Identification Using EEG Sensors."
- [22] Kumari, Pinki, and Abhishek Vaish. "Feature-level fusion of mental task's brain signal for an efficient identification system." *Neural Computing and Applications*: 1-11.
- [23] Kumari, Pinki, and Abhishek Vaish. "Brainwave's based user authentication system: A pilot study in robotic environment." *Robotics and Autonomous Systems*, Available online 11 (2014).
- [24] Albertus Hendrawan Adiwahono, Chee-Meng Chew, Weiwei Huang, and Yu Zheng, "Push Recovery Controller for Bipedal Robot Walking" *2009 IEEE/ASME International Conference on Advanced Intelligent Mechatronics Suntec Convention and Exhibition Center Singapore, July 14-17, 2009*

## Appendix – II

### Classifying Gait Data using different machine learning techniques and finding the optimum technique of classification

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**Abstract.** The Classification of Humanoid locomotion is a troublesome exercise because of non linearity associate with gait. The high dimension feature vector requires a high computational cost. The classification using the different machine learning technique leads for over fitting and under fitting. To select the correct feature is also the difficult task. The hand craft feature selection machine learning techniques performed poor. We have used the deep learning technique to get the trained feature and then classification we have used deep belief network based deep learning. Classification is utilized to see Gait pattern of different person and any upcoming disease can be detected earlier. So in this paper we first selected the feature and identify the principle feature then we classify gait data and use different machine learning technique (ANN, SVM, KNN, and Classifier fusion) and performance comparison is shown. Experimental result on real time datasets propose method is better than previous method as far as humanoid locomotion classification is concerned.

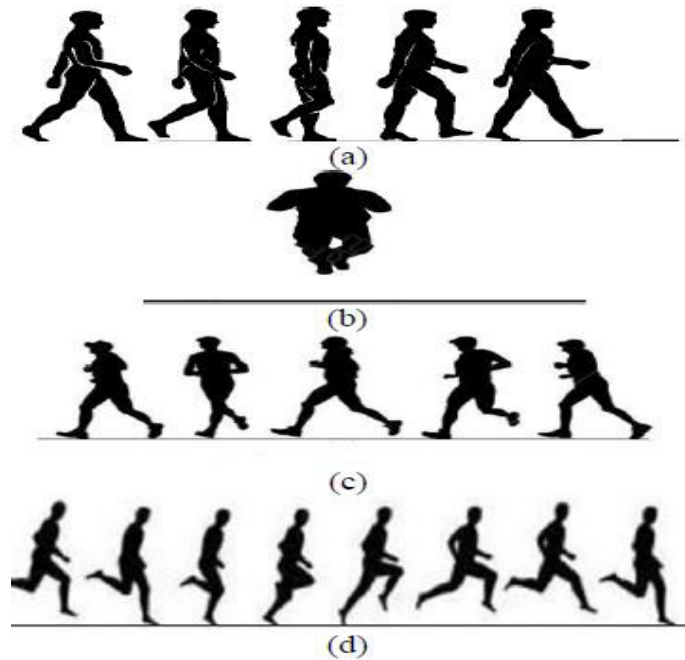
**Keywords:** GAIT, Classification, Biometric Identification, Feature selection.

## 1 Introduction

Human gait is supposed as unique biometric identification like thumb print. So in GAIT analysis to examine the human walking moment we collect and analyze the data [1]. Human gait can be utilized to identify people for diverse security reason and exercised before to detect many abnormalities. For disable person the classified data is used to implement the prosthetic leg.

The GAIT data can be useful to help in walking of old people and is used in pattern analysis and classification of different walk. By seeing the abnormality in GAIT

cycle, it is worth to do research for prognostication for many upcoming diseases. Differently abled and elderly people can be helped by any research regarding GAIT. During human movement, top and bottom limb of human body yield the periodic motion. Due to this particular motion human beings get a particular motion and a unique walking pattern. The biometric identity like gender, age and race based on human locomotion can be obtained by the use of GAIT analysis shown in figure 1.



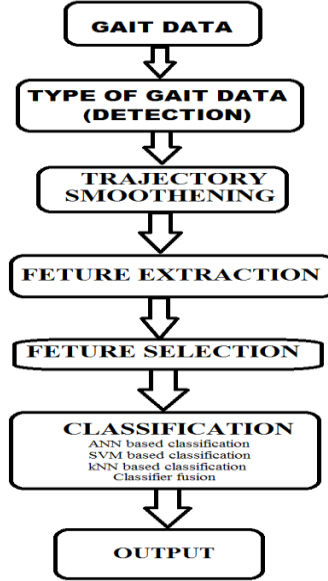
**Fig. 1.** Biomechanics of Locomotion: (a) Walk (b) Jump (c) Jog (d) Run

A lot of precursory research work about various machine learning techniques is gathered from the paper given [2-6] below. The classification of GAIT pattern is very accurate and unique due to rhythmic pattern. The new arena of biometric research is different category of GAIT classification. The GAIT can be helpful for security reason as it can identify from distance. In comparison with other biometric identification technique such as face reorganization i.e. iris, finger print which is considered as first generation biometric, the GAIT is divergent as it does not require any subject contact other than walking. This can be considered as behavioral authentication technique. Therefore identification using gait is becoming very popular in security. The development of highly sophisticated and more accurate humanoid can be made easy with the study of human GAIT data. So the foremost requirement is the feature reduction.

## 2 Proposed System



Proposed system consists of seven phases. They are collection of gait data, detection of gait data, trajectory smoothening, feature extraction, feature selection and classification. Fig. 2 tells systematic proposal. In the following subsections there are details of each phase.



**Fig. 2.** Architecture of the proposed system

$$w_i = \frac{1}{\sum_{k \neq i} ||x_k - x_i||_2}$$

According to the equation, if a feature point  $x_i$  is far from the other feature points then it has a small weight. Alternately, it is near to most of the other feature points then it will have a larger weight. After normalizing the weights as  $\sum_i w_i = 1$ , the centroid is calculated as:

$$x_c = \sum_i w_i x_i$$

The final step of compact criteria for a feature point  $x_i$  is expressed as  $||x_k - x_i||_2 < r_o$ , where  $r_o$  is a distance threshold. This criterion helps to eliminate the redundant feature points and ensures that the feature points are concentrated around the centroid.

### 3 Methodology

Firstly, we have collected data of different gait using accelerometer [8]. Secondly, features were selected on the basis of ANOVA and IFA (Incremental Feature

Analysis) to get optimal features from the existing features. Then different combinations of features are made by IFA using ANN, SVM, and KNN classifier. Lastly, classifier fusion technique is used.

### 3.1 Trajectory smoothening

After smoothening, we detect and remove the self co-articulated strokes. And as these strokes are not part of gesture therefore they need to be removed at this stage as they are just hand movements.

### 3.2 Feature Extraction

The features have been collected from [7].

Table.1. Crouch dataset description

| Feature Category | Feature Name    | Feature Category | Feature Name     |
|------------------|-----------------|------------------|------------------|
| F1               | Pelvis_tx       | F10              | knee_angle_r     |
| F2               | Pelvis_ty       | F11              | ankle_angle_r    |
| F3               | Pelvis_tz       | F12              | hip_flexion_l    |
| F4               | Pelvis_tilt     | F13              | hip_adduction_l  |
| F5               | Pelvis_list     | F14              | hip_rotation_l   |
| F6               | Pelvis_rotation | F15              | knee_angle_l     |
| F7               | hip_flexion_r   | F16              | ankle_angle_l    |
| F8               | hip_adduction_r | F17              | lumbar_extension |
| F9               | hip_rotation_r  |                  |                  |

### 3.3 Feature Selection

For selecting the final optimal features here we are using 2-stage process. In first stage we perform the ANOVA test to obtain feature [9], [10], [11]. In second stage after the significant features were obtained from Stage 1, the IFS technique is used to determine the optimal features.

## 4 Classification

Classification is done using following machine learning techniques.

### 4.1 Classification based on ANN

The network is trained many times till root mean squared error reaches an optimum level. To adjust the network weights, we train the system through back-propagation algorithm. Results of ANN classification is shown in Table 2.

Table.2. ANN Results

| Exp. No | Hidden units | Network structure | Iterations | Training accuracy (%) | Test accuracy (%) |
|---------|--------------|-------------------|------------|-----------------------|-------------------|
| 1       | 50           | 33-50-40          | 100        | 84                    | 80                |
|         |              |                   | 500        | 86                    | 80                |
| 2       | 52           | 33-52-40          | 100        | 86                    | 80                |
|         |              |                   | 500        | 90                    | 82                |
| 3       | 54           | 33-54-40          | 100        | 86                    | 82                |

#### 4.2 Classification based on SVM

SVM is supervised learning method. We optimize class separation hyper plane so we get the maximum distance between hyper plane and pattern separating the classes. Now dataset is trained with different kernel functions such as polynomial, quadratic, linear and radial basis function. Results of SVM classification is shown in Table 3.

Table.3. SVM Results

| Exp No. | Function Kernel       | Training accuracy(%) | Test accuracy (%) |
|---------|-----------------------|----------------------|-------------------|
| 1.      | Linear function       | 68                   | 71                |
| 2.      | Quadratic function    | 70                   | 76                |
| 3.      | Polynomial function   | 84                   | 84                |
| 4.      | Radial Basis Function | 88                   | 84                |

#### 4.3 Classification based on KNN

KNN classifier is based on relating the anonymous instances to the known according to some distance function. To train *KNN* we use different *k* values as it is an essential parameter. In training phase we used different number of *k* values such as 3, 5, 7 and 9. Results of KNN classification is shown in Table 4.

Table.4. KNN Results

| Exp No. | K | Training accuracy (%) | Testing accuracy (%) |
|---------|---|-----------------------|----------------------|
| 1       | 3 | 74                    | 70                   |
| 2       | 5 | 79                    | 80                   |
| 3       | 7 | 78                    | 74                   |
| 4       | 9 | 84                    | 81                   |

#### 4.4 Classifier fusion

Classifier fusion combines many classifiers and provides the result by combining the results of the individual classifiers. In this paper, we have combined the results of ANN, SVM and KNN models to achieve the classifier fusion result. The experiment of classifier fusion was done by combining the other classifiers. Thus, the performance of classifier fusion is likely superior than the individual classifiers used in isolation.

Once the training is done, testing is done by 5-fold cross validation shown in table 5.

Table.5. 5-fold cross validation result

| Expt# | ANN<br>classifier<br>Accuracy (%) | SVM classifier<br>Accuracy (%) | kNN with k=5<br>Accuracy (%) | Classifier<br>fusion<br>Accuracy (%) |
|-------|-----------------------------------|--------------------------------|------------------------------|--------------------------------------|
| 1     | 89.67                             | 91.62                          | 80.76                        | 89.67                                |
| 2     | 92.91                             | 91.92                          | 78.89                        | 90.75                                |
| 3     | 87.82                             | 88.52                          | 91.82                        | 87.82                                |
| 4     | 93.60                             | 85.60                          | 90.40                        | 89.60                                |
| 5     | 91.01                             | 89.21                          | 92.21                        | 86.75                                |

## 5 Experimental results and discussions

### 5.1 Data Set Used

In order to evaluate our proposed approach of human activity recognition, we have used crouch datasets. The data set have four classes. Each has three different data base so total 12 dataset for all four classes. It has total 17 features (Times series data). The dimension of data is 17\*300 for a particular data set, in which we choose one gait cycle for our experiment. So our dataset has 17\*100 elements each. Hence total 17\*4\*3\*100 element.

### 5.2 Performance of the classifiers

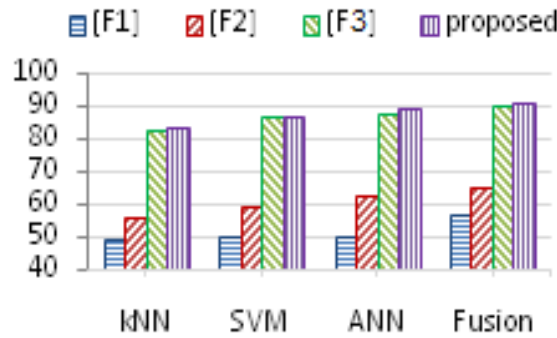
The train and test accuracies are evaluated for different parameters of ANN such as iterations and hidden units. The highest train and test accuracy was observed for network structure 40L54N40L (L, N corresponds to linear and nonlinear layers), iterations = 500. This network parameters were used for performing 5-fold cross validation to determine the performance and validity of the model. Results of cross validation process are shown in Table 6.

Table.6. Comparison of success rate by different classifier using 5–fold cross validation

| Classifier        | Rate of Success (%) |
|-------------------|---------------------|
| ANN               | 82.58               |
| SVM               | 80.11               |
| KNN               | 87.72               |
| Classifier fusion | 93.56               |

### 5.3 Comparative analysis

In our proposed system we perform comparative analysis by selecting the optimal features. Now the comparison is done with various set of gestures as seen in Fig. 3.



**Fig. 3.** Classification of Different GAIT pattern: comparison of proposed optimal features using ANN, SVM, KNN, and classifier fusion: (a) Crouch\_2 (b) Crouch\_3 (c) Crouch\_4 (d) Normal [10]

It was noted that from all the other individual classifiers for various feature set performed the performance of the classifier fusion is best. Therefore we get the highest overall accuracy of 93.56% by using the combination of optimal features and classifier fusion as compared to other feature and classifier combinations.

### 5.4 Conclusion

The proposed system can be used for developing a gesture controlled hexadecimal keyboard making human-computer interaction easier. From the existing literatures, a total of 10 features were selected. To check the statistical significance of the 10 features ANOVA test was performed. The 10 significant features were then arranged in the descending order of the F-static value which was then fed to the IFS to select

the optimal features. Total number of features or the optimal features was observed to be 6, 7 and 8 for ANN, SVM and KNN respectively. The results of the three individual classifiers were combining to provide the classifier fusion results. After this, 5-fold cross validation was used to provide overall accuracy of the system. Overall accuracy was observed to be 82.58, 8.11%, 87.72% and 93.56% using the ANN, SVM, KNN and classifier fusion respectively.

## References

1. Rattiya Thanasoontornerk, Chatkaew Pongmala, Areerat Suputtiada: Tree Induction for Diagnosis on Movement Disorders Using Gait Data. In: 2013 5th International Conference on Knowledge and smart technology (KST)
2. Kiichi\_Katoh,, *Makoto Katoh*.: Using FS in Tuning a Biped Locomotion Surrogate Robot for Walking Training. In: 2011 4th International Conference on Biomedical Engineering and Informatics (BMEI)
3. Mitsuru Yoneyama, Yosuke Kurihara, Kajiro Watanabe,: Accelerometry-Based Gait Analysis and Its Application to Parkinson's Disease Assessment—Part 2: IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING, VOL. 21, NO. 6, NOVEMBER 2013
4. M. Ogata1, T. Karato1, T. Tsuji1, H. Kobayashi1 and K. Irie2.: Development of Active Walker by using Hart Walker. In: SICE-ICASE International Joint Conference 2006 Oct. 18-21, 2006 in Bexco, Busan, Korea
5. Hiroshi Kobayashi, Takeo Karato, and Toshiaki Tsuji.: Development of an Active Walker as a New Orthosis. Proceedings of the 2007 IEEE International Conference on Mechatronics and Automation August 5 - 8, 2007, Harbin, China
6. Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami.: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. In Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
7. Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
8. Anubha Parashar, Somya Goyal, Apoorva Parashar, Bharat Sahjlan: Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Mean. ICTCS 2016 - CSI-Udaipur Chapter.
9. Semwal, Vijay Bhaskar, et al : Biped model based on human Gait pattern parameters for sagittal plane movement. Control, Automation, Robotics and Embedded Systems (CARE), 2013 International Conference on. IEEE, 2013.
10. Semwal, Vijay Bhaskar, et al.: Biologically-inspired push recovery capable bipedal locomotion modeling through hybrid automata. Robotics and Autonomous Systems 70 (2015): 181-190.
11. Semwal, Vijay Bhaskar, Manish Raj, and Gora Chand Nandi. :Biometric gait identification based on a multilayer perceptron. Robotics and Autonomous Systems 65 (2015): 65-75.

## Appendix - IV

### Clustering Gait Data using different machine learning techniques and finding the best technique

Anubha Parashar<sup>1</sup>; Deepak Goyal<sup>2</sup>

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**Abstract.** Clustering is done in order to group the entities which are alike in one group so that grouping of more similar objects can be done. The objects placed in one group are known as clusters. In this paper we are using clustering in order to identify the human locomotion and categories the dataset making clusters. We are using two clustering techniques i.e. SOM and K-Means. So we first selected the feature and identify the principle feature then we cluster gait data and use different machine learning technique (K-Means and SOM) and performance comparison is shown. Experimental result on real time datasets propose method is better than previous method as far as humanoid locomotion classification is concerned.

**Keywords:** Clustering, Biometric Identification, Feature selection, SOM, K-Means.

## 1 Introduction

In order to authenticate humans we use various verification and validation technologies. To know whether a person is authenticated or not we need verification. To authenticate some person and to check his/her identity various authentications techniques are used. First is *knowledge*; in this technique a person must have particular knowledge about of his PIN, password, or any other secret code. Second is *Possession*; in this technique a person must know particular token or item like identity card, smart card. Third is *Biometric*; in this technique a person's body part is taken to identify unique identity of person like fingerprint, retina, iris, facial

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expression. But there are various disadvantages of above techniques [1] [2] [3]. A person may forget its password or PIN. A person may lose its identity card or smart card. Biometric authentication may not be always user friendly and is expensive too.

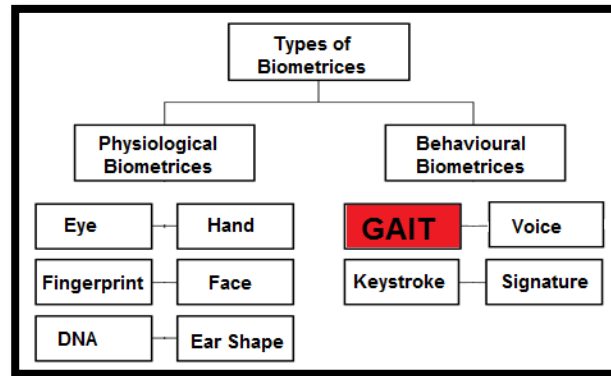


Fig. 1. Biometrics types

In contrast with such authentication techniques GAIT based biometric method outperforms all above disadvantages.

Gait means how a human move/walk and is very useful to identify person from a distance. It is very useful in detecting diseases and in surveillance system. In this paper we have taken four data sets namely croubh2, croubh3, croubh4, normal shown in figure 2.

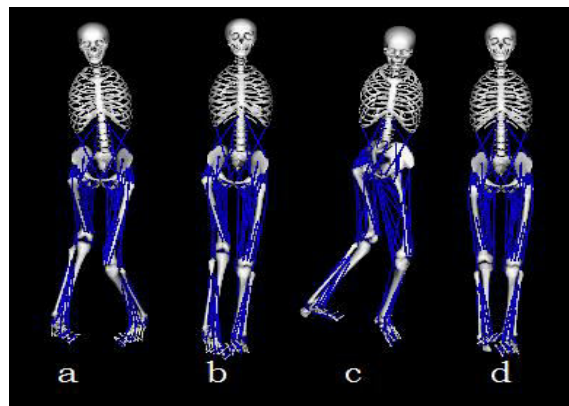


Fig. 2. Contrast enhanced images in gait sequences of Gait Database (Dataset - a (croubh2) b (croubh3) c (croubh4) d (normal)). Four samples from left to right show the gait of pathological walking (a,b,c) and normal walking (d).

## 2 Proposed System



Proposed system consists of eight phases. They are collection of Data Preprocessing, Gait Cycle Detection, Feature Extraction, Feature Reduction (PCA), Output Feature Vector, Clustering (K-Means) (SOM), and output.

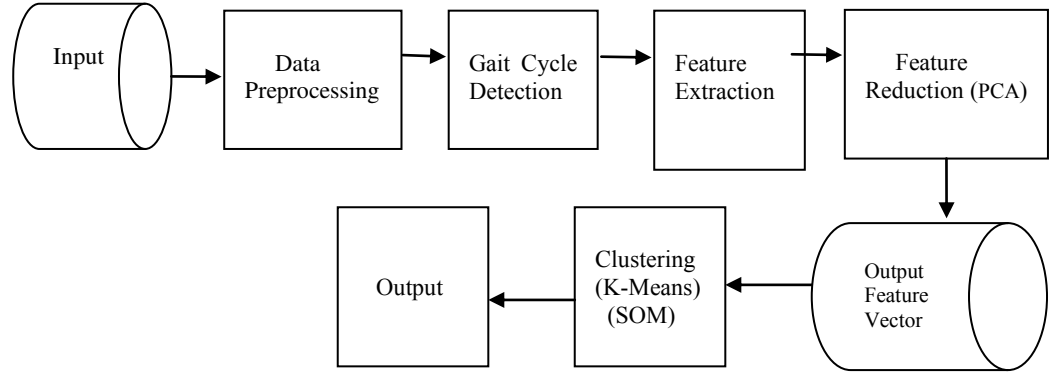


Fig. 3. Proposed System

$$w_i = \frac{1}{\sum_{k \neq i} ||x_k - x_i||_2}$$

According to the equation, if a feature point  $x_i$  is far from the other feature points then it has a small weight. Alternately, it is near to most of the other feature points then it will have a larger weight. After normalizing the weights as  $\sum_i w_i = 1$ , the centroid is calculated as:

$$x_c = \sum_i w_i x_i$$

The final step of compact criteria for a feature point  $x_i$  is expressed as  $||x_k - x_i||_2 < r_o$ , where  $r_o$  is a distance threshold. This criterion helps to eliminate the redundant feature points and ensures that the feature points are concentrated around the centroid.

### 3 Methodology

Firstly, we have collected data of different gait using HMDC [8], and then features were selected, then different combinations of features are made using K-Means and SOM clustering. Lastly, classifier fusion technique is used.

#### 3.1 Trajectory smoothening

After smoothening, we detect and remove the self co-articulated strokes. And as these strokes are not part of gesture therefore they need to be removed at this stage as they are just hand movements.

### 3.2 Feature Extraction

The features have been collected from [7].

Table.1. Crouch dataset description

| Feature Category | Feature Name    | Feature Category | Feature Name     |
|------------------|-----------------|------------------|------------------|
| F1               | Pelvis_tx       | F10              | knee_angle_r     |
| F2               | Pelvis_ty       | F11              | ankle_angle_r    |
| F3               | Pelvis_tz       | F12              | hip_flexion_l    |
| F4               | Pelvis_tilt     | F13              | hip_adduction_l  |
| F5               | Pelvis_list     | F14              | hip_rotation_l   |
| F6               | Pelvis_rotation | F15              | knee_angle_l     |
| F7               | hip_flexion_r   | F16              | ankle_angle_l    |
| F8               | hip_adduction_r | F17              | lumbar_extension |
| F9               | hip_rotation_r  |                  |                  |

### 3.3 Feature Selection

For selecting the final optimal features here we are using 2-stage process. In first stage we perform the ANOVA test to obtain feature [9], [10], [11]. In second stage after the significant features were obtained from Stage 1, the IFS technique is used to determine the optimal features.

## 4 Clustering

Clustering is done using following machine learning techniques.

### 4.1 K-Means

The K-Means algorithm is generally used for compact clustering. It is very much sensitive for outlier and noise. It is applicable only for numerical data.

Table.2. K-Means Results

| Exp. No | Training accuracy (%) | Test accuracy (%) |
|---------|-----------------------|-------------------|
| 1       | 100                   | 80                |

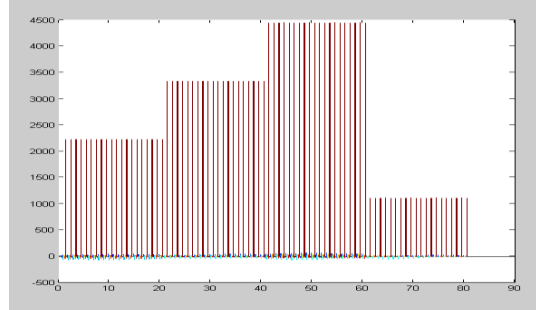


Fig. 4. Overall clustering using different no of majority vote

## 4.2 Clustering based on SOM

SOM is supervised learning method. We optimize class separation hyper plane so we get the maximum distance between hyper plane and pattern separating the classes. Now dataset is trained with different kernel functions such as polynomial, quadratic, linear and radial basis function. Results of SOM classification is shown in Table 3.

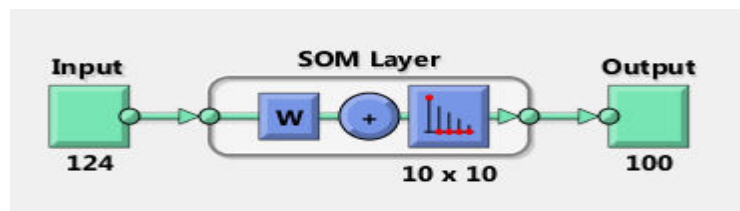


Fig. 5. SOM network architecture

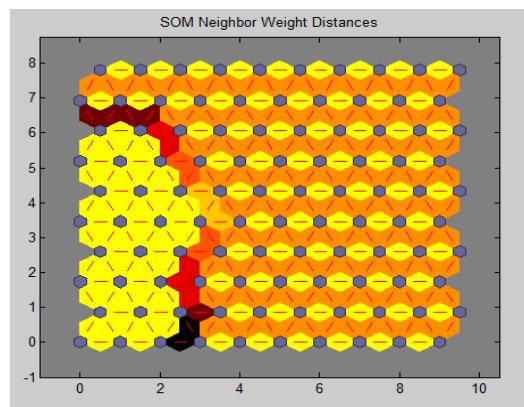


Fig. 6. SOM neighbouring distances

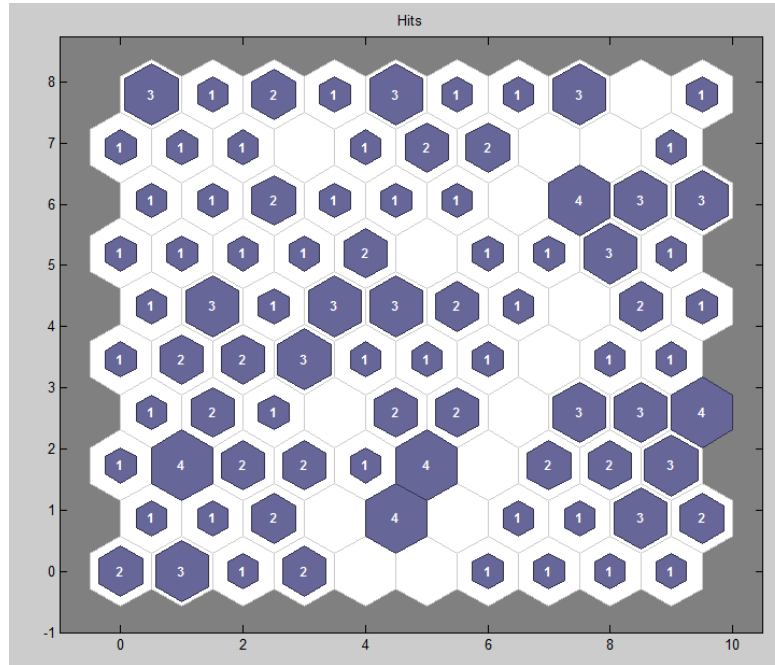


Fig. 7. Biometrics types (in this thesis GAIT biometric classification is used)

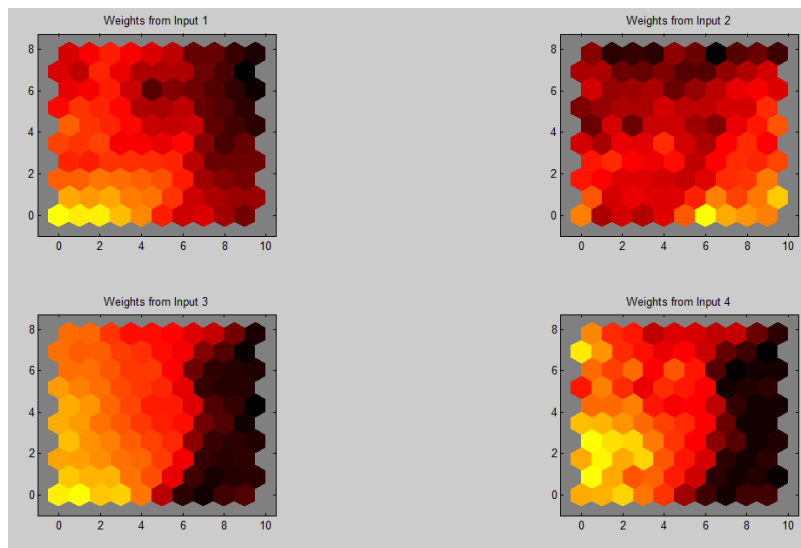


Fig. 8. SOM weight Planes

Table.3. SOM Results

| Exp No. | Training accuracy(%) | Test accuracy (%) |
|---------|----------------------|-------------------|
| 1.      | 100                  | 71                |

## 5 Experimental results and discussions

### 5.1 Data Set Used

In order to evaluate our proposed approach of human activity recognition, we have used crouch datasets. The data set have four classes. Each has three different data base so total 12 dataset for all four classes. It has total 17 features (Times series data). The dimension of data is  $17 \times 300$  for a particular data set, in which we choose one gait cycle for our experiment. So our dataset has  $17 \times 100$  elements each. Hence total  $17 \times 4 \times 3 \times 100$  element.

### 5.2 Performance of the classifiers

10-fold cross validation is used to determine the performance and validity of the model. Results of cross validation process are shown in Table 6.

Table.4. Comparison of success rate by different classifier using 5-fold cross validation

| Classifier | Rate of Success (%) |
|------------|---------------------|
| K-Means    | 80                  |
| SOM        | 71                  |

It is noted that from all the other individual classifiers for various feature set performed the performance of the K-Means is best. Therefore we get the highest overall accuracy of 90%.

### 5.4 Conclusion

The proposed system can be used for developing a gesture controlled hexadecimal keyboard making human-computer interaction easier. From the existing literatures, a total of 10 features were selected. To check the statistical significance of the 10 features ANOVA test was performed. The 10 significant features were then arranged in the descending order of the F-static value which was then fed to the IFS to select

the optimal features. Total number of features or the optimal features was observed to be 6, 7 and 8 for K-Means and SOM respectively. The results of the three individual classifiers were combining to provide the classifier fusion results. After this, 10-fold cross validation was used to provide overall accuracy of the system. Overall accuracy was observed to be 80, 71.1%, using the K-Means and SOM respectively.

## References

1. Rattiya Thanasoontornerk, Chatkaew Pongmala, Areerat Suputtiada: Tree Induction for Diagnosis on Movement Disorders Using Gait Data. In: 2013 5th International Conference on Knowledge and smart technology (KST)
2. Kiichi\_Katoh,, *Makoto Katoh*.: Using FS in Tuning a Biped Locomotion Surrogate Robot for Walking Training. In: 2011 4th International Conference on Biomedical Engineering and Informatics (BMEI)
3. Mitsuru Yoneyama, Yosuke Kurihara, Kajiro Watanabe,: Accelerometry-Based Gait Analysis and Its Application to Parkinson's Disease Assessment—Part 2: IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING, VOL. 21, NO. 6, NOVEMBER 2013
4. M. Ogata1, T. Karato1, T. Tsuji1, H. Kobayashi1 and K. Irie2.: Development of Active Walker by using Hart Walker. In: SICE-ICASE International Joint Conference 2006 Oct. 18-21, 2006 in Bexco, Busan, Korea
5. Hiroshi Kobayashi, Takeo Karato, and Toshiaki Tsuji.: Development of an Active Walker as a New Orthosis. Proceedings of the 2007 IEEE International Conference on Mechatronics and Automation August 5 - 8, 2007, Harbin, China
6. Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami.: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. In Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
7. Daniel T.H. Lai, Ahsan Khandoker, Rezaul K. Begg and M.Palaniswami: A hybrid Support Vector Machine and autoregressive model for detecting gait disorders in the elderly. Proceedings of International Joint Conference on Neural Networks, Orlando, Florida, USA, August 12-17, 2007.
8. Anubha Parashar, Somya Goyal, Apoorva Parashar, Bharat Sahjlan: Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Means. ICTCS 2016 - CSI-Udaipur Chapter.
9. Semwal, Vijay Bhaskar, et al : Biped model based on human Gait pattern parameters for sagittal plane movement. Control, Automation, Robotics and Embedded Systems (CARE), 2013 International Conference on. IEEE, 2013.
10. Semwal, Vijay Bhaskar, et al.: Biologically-inspired push recovery capable bipedal locomotion modeling through hybrid automata. Robotics and Autonomous Systems 70 (2015): 181-190.
11. Semwal, Vijay Bhaskar, Manish Raj, and Gora Chand Nandi. :Biometric gait identification based on a multilayer perceptron. Robotics and Autonomous Systems 65 (2015): 65-75.